

TCV Core | Database and Development Guide

Source review date: 6 October 2026 (Asia/Calcutta). Prepared from the current repository, without connecting to or modifying a database. Audience: frontend/backend developers, maintainers and technical onboarding.

Scope and evidence

This guide inventories 71 table names, 27 mounted backend routers, 84 lazy frontend routes, 302 router endpoint declarations and 164 ID relationships. Duplicate legacy API aliases are shown explicitly. The diagram appendix follows the development guide and includes declared foreign keys and separately labelled inferred ID links.

Schema baseline: backend/migrations/production_schema.sql, an August 2026 repository snapshot, augmented with table definitions found in migrations and runtime JavaScript. The live deployed schema has not been verified. CREATE IF NOT EXISTS does not update existing tables. Alternative CREATE definitions and ALTER statements are documented separately rather than being silently merged into a fictitious final schema. Inferred links are candidates, not verified database constraints. No customer records, database dumps containing rows, credentials or environment secrets are included.

Reading map

Section	What to use it for
1 Architecture and setup	Run the app and follow requests through frontend/API/database.
2 Business flows and ID rules	Understand billing, approvals and identity boundaries.
3 Module guide	Purpose, active screens, service files, controller tables and every endpoint.
4 Frontend navigation and HTTP inventory	Locate each lazy route and inspect service request expressions.
5 Schema dictionary	Table primary keys, columns, declared constraints and ID mappings.
6 Schema evolution and verification	Runtime variants, migrations, tests and deployment procedure.
Diagram appendix	Vector relationship diagrams grouped by business domain; use the dictionary for exact types and sources.

1. Architecture and local development

Angular 20 standalone/lazy components -> injected HttpClient services -> customInterceptor Bearer token -> Express 5 index.js route mounts and permission middleware -> routers -> controllers/schema helpers -> mysql2 connection pool -> MySQL. API responses update Angular views. Finance, stock and approval actions can write multiple related tables.

Layer	Repository entry point	Responsibility
Frontend bootstrap	frontend/src/main.ts; frontend/src/app/app.config.ts	Angular providers, router and HTTP setup.
Navigation	frontend/src/app/app.routes.ts; layout/	Lazy components and application navigation.
API configuration	frontend/src/app/app-config.ts	Runtime window.APP_CONFIG override; localhost defaults to http://localhost:8080/api; deployed default uses the configured public API.
Authorization	services/route-guard.ts; permission.service.ts; interceptor/custom-interceptor.ts	Authentication, refreshed route permissions, Bearer header.
Backend startup	backend/server.js -> backend/index.js	dotenv load, HTTP listener on PORT, CORS, JSON/urlencoded parsers, /uploads static files and router mounts.

Database access	backend/connection.js	mysql2 pool and promise facade. beginTransaction pins a connection until commit/rollback releases it.
Schema helpers	backend/utis/*Schema.js and controller ensure* functions	CREATE/ALTER compatibility logic; schema may evolve on request paths.

Setup procedure

1. Use Node.js compatible with the installed Angular 20 tooling. In backend and frontend separately run npm ci when their lockfiles are present, otherwise npm install. Package versions below are repository declarations, not claims about latest releases.
2. Create a dedicated development MySQL database. Inspect production_schema.sql and required migrations before importing. Schema files and consolidated dumps can contain DROP TABLE statements: select a fresh database explicitly. Do not execute the production data dump or one-off VPS repair scripts as a routine setup step.
3. Create backend/.env with your own development values. Set PORT=8080 to match local frontend defaults. Required database names are DB_HOST, DB_USERNAME and DB_NAME; DB_PASSWORD and DB_PORT configure credentials/port. Set ACCESS_TOKEN to a development JWT secret. This guide lists variable names only.
4. In backend run npm run dev (nodemon) or npm start. In frontend run npm start. Use npm start -- --port 4500 if that is the intended local UI port. Open the URL printed by Angular and verify login and /api/permissions before editing a domain module.
5. Build frontend with npm run build; the configured default is production. npm run build:tcverp sets /tcverp/ as base href. Check app-config.ts and the host runtime APP_CONFIG before assuming environment.ts controls every service. Database pool startup performs SELECT 1.

Variable group	Names and use
Backend listener/auth	PORT, ACCESS_TOKEN, CAPTCHA_SECRET
Database	DB_HOST, DB_USERNAME, DB_NAME, DB_PASSWORD, DB_PORT (default 3306)
Pool tuning	DB_CONNECTION_LIMIT, DB_MAX_IDLE, DB_IDLE_TIMEOUT_MS, DB_KEEPAIVE_DELAY_MS, DB_CONNECT_TIMEOUT_MS
Legacy account email	EMAIL, PASSWORD in userController; inspect intended SMTP behavior before use.
Invoice email	Inspect backend/INVOICE_EMAIL.md and internetSubscriptionEmail.js for INVOICE_EMAIL_USER, INVOICE_EMAIL_PASSWORD, INVOICE_SMTP_HOST, INVOICE_SMTP_PORT and other supported options.

Package	Declared version
frontend: @angular/animations	^20.3.16
frontend: @angular/cdk	^20.2.14
frontend: @angular/common	^20.3.0
frontend: @angular/compiler	^20.3.0
frontend: @angular/core	^20.3.0
frontend: @angular/forms	^20.3.0
frontend: @angular/material	^20.2.14
frontend: @angular/platform-browser	^20.3.0
frontend: @angular/router	^20.3.0
frontend: @popperjs/core	^2.11.8

frontend: @primeng/themes	^21.0.4
frontend: ag-grid-angular	^35.1.0
frontend: ag-grid-community	^35.1.0
frontend: bootstrap	^5.3.8
frontend: bootstrap-icons	^1.13.1
frontend: jwt-decode	^4.0.0
frontend: ngx-ui-loader	^19.0.0
frontend: primeng	^20.4.0
frontend: rxjs	~7.8.0
frontend: tslib	^2.3.0
frontend: zone.js	~0.15.0
backend: cors	^2.8.6
backend: crypto	^1.0.1
backend: dotenv	^17.2.4
backend: express	^5.2.1
backend: jsonwebtoken	^9.0.3
backend: multer	^2.1.1
backend: mysql	^2.18.1
backend: mysql2	^3.17.1
backend: nodemailer	^8.0.1

2. Business flows and ID mapping rules

Use database numeric IDs for joins and child actions. Customer numbers, NET IDs, STB numbers, quotation numbers and invoice numbers are business/display identifiers and may have independent uniqueness rules. Three customer families exist: customers.customer_id, cable_tv_customers.cable_customer_id and internet_customers.internet_customer_id. Never substitute one family for another because the numbers happen to match.

Flow	How to follow it	Critical identities
Purchase to stock	Supplier -> purchase header -> purchase items -> stock/ledger -> supplier payment	supplier_id, purchase_id, product_id; verify exact line PK in dictionary.
Quote to installation/ invoice	Customer -> quotation/items -> submit/review/ customer response -> work order -> material issues/ returns -> invoice -> receipts	customer_id, quotation_id, work_order_id, sales_id; use source link columns in schema.
Cable customer lifecycle	Customer -> connection/STB/package history -> pending review -> approved action -> account/stock effects -> monthly subscriptions/collection	cable_customer_id plus connection/STB/package/subscription row IDs; never pass an STB master ID where a customer STB row ID is required.
Internet lifecycle	Customer -> enrollment connection/router/package rows -> workflow review -> accounts and router stock -> subscription append/renewal -> payment/report/invoice email	internet_customer_id, internet_connection_id, internet_router_id, internet_customer_package_id, internet_subscription_id.

Technician material sales	Office stock -> technician stock/movement -> issue/sale batch -> customer mapping -> sale/payment/adjustment review	employee_id, product_id and material movement/sale IDs; trace materialSalesController.
Complaints	Customer family selection -> complaint -> assignment/attempts -> outcome/report	Complaint row ID, attempt row ID and customer family reference; inspect complainant_type.
Human resources	Employee profile -> dated attendance -> salary; user link supports identity	employee_id is separate from users.user_id.
Approval/audit references	Workflow dispatch or audit lookup selects target by discriminator and reference	workflow_approvals.module_name + reference_id; audit_log.table_name + record_id.

Nested ID example (illustrative IDs only)

GET /api/v1/internet/customers/887 selects internet_customer_id=887. PATCH /api/v1/internet/customers/887/connections/10 targets internet_connection_id=10 owned by that customer. Subscription operations take internet_subscription_id, and package operations take internet_customer_package_id. The connection-actions tests explicitly verify customer-scoped child updates. Illustrative values here are not live database records.

Permission behavior developers must preserve

Do not infer backend grants from expectedRole arrays in app.routes.ts. RouteGuard uses refreshed PermissionService decisions. index.js protects catalogued API aliases; Cable TV and Internet routers apply their own per-route grants. Some Internet subscription update/email routes intentionally check can_view, while connection edits and cash correction require ADMIN. transactionRouter authenticates all requests and restricts approve/delete to ADMIN. JWT role claims are used by current backend middleware; a role change may require a fresh login. Confirm current source semantics before altering any check.

3. Frontend and backend module guide

3.1 Role permissions

The permission catalog defines routes and module keys; role_permissions stores action grants. RouteGuard refreshes permissions and calls canRoute; expectedRole metadata alone does not enforce current route access. Backend permission middleware maps GET/POST/PUT/PATCH/DELETE to view/create/update/delete, with route-specific overrides. ADMIN bypasses generic grants.

Development path: Adding SENIOR_MANAGER requires add_senior_manager_role.sql for both role enums before deploying the UI/backend. Appending preserves existing enum indexes; this migration adds no users or grants. Review each endpoint middleware rather than assuming every PATCH requires can_update.

Router: backend/routes/permissionRouter.js. API mount aliases: /api/permissions.

Controllers/helpers imported by router: backend/controller/permissionController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): role_permissions

Frontend service files matched by API mount text: frontend/src/app/services/permission.service.ts

Frontend route	Active component
/dashboard	frontend/src/app/dashboard/dashboard.ts
/users	frontend/src/app/user/user-list/user-list.ts
/productList	frontend/src/app/products/product-list/product-list.ts
/role-permissions	frontend/src/app/administration/role-permissions/role-permissions.ts

/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts
/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts
/assign-net-collector	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-pending	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts
/net-subscription-report	frontend/src/app/administration/net-subscription-report/net-subscription-report.ts
/transactions	frontend/src/app/transactions/transaction-list/transaction-list.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts
/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/employee-salary	frontend/src/app/employee-salary/employee-salary.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): permissions, role

Method	Path appended to each mount	Handler / route middleware
GET	/	controller.getPermissions
PUT	/:role	controller.updatePermissions

3.2 Users and authentication

Login, captcha, account lifecycle and role administration. The controller signs an eight-hour JWT. The interceptor reads token from localStorage and supplies Authorization: Bearer. Users have user_id; employee references are a separate identity.

Development path: Trace login.ts -> user/auth services -> userRouter -> userController. Test login, expired token, a non-admin operation, and user-role changes.

Router: backend/routes/userRouter.js. API mount aliases: /api/user.

Controllers/helpers imported by router: backend/controller/userController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): employees, role_permissions, users

Frontend service files matched by API mount text: frontend/src/app/services/permission.service.ts, frontend/src/app/services/user-services.ts

Frontend route	Active component
/login	frontend/src/app/login/login.ts
/dashboard	frontend/src/app/dashboard/dashboard.ts
/users	frontend/src/app/user/user-list/user-list.ts

/productList	frontend/src/app/products/product-list/product-list.ts
/role-permissions	frontend/src/app/administration/role-permissions/role-permissions.ts
/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts
/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts
/assign-net-collector	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-pending	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts
/net-subscription-report	frontend/src/app/administration/net-subscription-report/net-subscription-report.ts
/transactions	frontend/src/app/transactions/transaction-list/transaction-list.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts
/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/employee-salary	frontend/src/app/employee-salary/employee-salary.ts
/**	frontend/src/app/login/login.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): `userId`, `user_id`

Method	Path appended to each mount	Handler / route middleware
POST	/login	login
GET	/captcha	getCaptcha
POST	/forgotPassword	forgotPassword
POST	/changePassword	auth.authenticateToken, changePassword
GET	/my-permissions	auth.authenticateToken, getMyPermissions
POST	/signup	auth.authenticateToken, auth.requireAdmin, signup
GET	/get	auth.authenticateToken, auth.requireAdmin, getAllUser
PATCH	/editUser	auth.authenticateToken, auth.requireAdmin, editUser
DELETE	/deleteUser	auth.authenticateToken, auth.requireAdmin, deleteUser

3.3 Brands

Product brand master and lookup maintenance. `brand_id` is referenced by products.

Development path: Edit brand service, brand list/dialog and brandController together; preserve unique names/codes and existing product references.

Router: backend/routes/brandRouter.js. API mount aliases: /api/brand, /api/v1/brands.

Controllers/helpers imported by router: backend/controller/brandController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): brands

Frontend service files matched by API mount text: frontend/src/app/services/brand-services.ts

Frontend route	Active component
/brands	frontend/src/app/products/brands-list/brands-list.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): brand_code, brand_id, brand_name, description, status

Method	Path appended to each mount	Handler / route middleware
POST	/	auth.authenticateToken, addBrand
POST	/add	auth.authenticateToken, addBrand
GET	/	auth.authenticateToken, getBrands
GET	/get	auth.authenticateToken, getBrands
DELETE	/delete/:brand_id	deleteBrand
PATCH	/	auth.authenticateToken, editBrand
PATCH	/edit	auth.authenticateToken, editBrand
PATCH	/:brand_id	auth.authenticateToken, (req, res) => { req.body = req.body {}; req.body.brand_id = req.body.brand_id req.params.brand_id; return editBrand(req, res); }
DELETE	/	auth.authenticateToken, deleteBrand
DELETE	/delete	auth.authenticateToken, deleteBrand
DELETE	/delete/:brand_id	auth.authenticateToken, (req, res) => { req.body = req.body {}; req.body.brand_id = req.body.brand_id req.params.brand_id; return deleteBrand(req, res); }
DELETE	/:brand_id	auth.authenticateToken, (req, res) => { req.body = req.body {}; req.body.brand_id = req.body.brand_id req.params.brand_id; return deleteBrand(req, res); }

3.4 Categories

Hierarchical product classification. parent_id identifies another category; product category_id identifies its classification.

Development path: Trace the active categories-list component and category services. Keep tree levels, slugs and parent relationships consistent; do not use no-need-trash components as the active implementation.

Router: backend/routes/categoryRouter.js. API mount aliases: /api/category, /api/v1/categories.

Controllers/helpers imported by router: backend/controller/categoryController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): categories

Frontend service files matched by API mount text: frontend/src/app/services/category-services.ts

Frontend route	Active component
/categoriesLists	frontend/src/app/products/categories-list/categories-list.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): category_id, id

Method	Path appended to each mount	Handler / route middleware
POST	/	addCategory
POST	/add	addCategory
POST	/create	createCategoryBulk
GET	/	getCategoriesTree
GET	/get	getCategoriesTree
PATCH	/	updateCategory
PATCH	/update	updateCategory
PATCH	/:id	(req, res) => { req.body.category_id = req.body.category_id req.params.id; return updateCategory(req, res); }
DELETE	/delete/:id	deleteCategory
DELETE	/:id	deleteCategory
GET	/get/:id	getCatById
GET	/:id	getCatById

3.5 Products

SKU, classification, pricing, tax, units and product type. product_id is reused throughout purchases, stock, sales and installation materials.

Development path: Update product forms, service payloads and controller validation together. Confirm product type enums through migration files; distinguish master pricing from transaction item pricing.

Router: backend/routes/productRouter.js. API mount aliases: /api/product, /api/v1/products.

Controllers/helpers imported by router: backend/controller/productController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): brands, categories, products

Frontend service files matched by API mount text: frontend/src/app/services/product-service.ts

Frontend route	Active component
/productList	frontend/src/app/products/product-list/product-list.ts
/quotation-templates	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/preview/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/work-orders/add	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/edit/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/preview/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/service-tickets	frontend/src/app/service-tickets/service-tickets.ts
/warranty-master	frontend/src/app/warranty-master/warranty-master.ts
/stock	frontend/src/app/stock/stock-list/stock-list.ts

/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/review/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/purchases/add	frontend/src/app/purchases/purchase-form/purchase-form.ts
/purchases/edit/:id	frontend/src/app/purchases/purchase-form/purchase-form.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): id, product_id

Method	Path appended to each mount	Handler / route middleware
GET	/	getProducts
GET	/get	getProducts
POST	/	addProduct
POST	/add	addProduct
PATCH	/	updateProduct
PATCH	/update	updateProduct
PATCH	/:product_id	(req, res) => { req.body.product_id = req.body.product_id req.params.product_id; return updateProduct(req, res); }
DELETE	/:product_id	deleteProduct
DELETE	/delete/:product_id	deleteProduct

3.6 Suppliers

Supplier contact and business details underpin purchases and supplier payments.

Development path: Use supplier_id as the relational key; names are display values. Trace supplier forms/services and supplierController; verify linked purchase and payment screens.

Router: backend/routes/supplierRouter.js. API mount aliases: /api/supplier, /api/v1/suppliers.

Controllers/helpers imported by router: backend/controller/supplierController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): suppliers

Frontend service files matched by API mount text: frontend/src/app/services/supplier-payment-services.ts, frontend/src/app/services/supplier-services.ts

Frontend route	Active component
/supplier-payments	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/add	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/edit/:id	frontend/src/app/supplier-payments/supplier-payments.ts
/purchases/add	frontend/src/app/purchases/purchase-form/purchase-form.ts
/purchases/edit/:id	frontend/src/app/purchases/purchase-form/purchase-form.ts
/suppliers	frontend/src/app/suppliers/supplier-list/supplier-list.ts
/suppliers/add	frontend/src/app/suppliers/add-supplier/add-supplier.ts

/suppliers/edit/:id

frontend/src/app/suppliers/add-supplier/add-supplier.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): supplier_id

Method	Path appended to each mount	Handler / route middleware
POST	/	addSupplier
POST	/add	addSupplier
GET	/	getSuppliers
GET	/get	getSuppliers
GET	/get/:supplier_id	getSupplierById
GET	/:supplier_id	getSupplierById
PATCH	/	updateSupplier
PATCH	/update	updateSupplier
PATCH	/:supplier_id	(req, res) => { req.body.supplier_id = req.body.supplier_id req.params.supplier_id; return updateSupplier(req, res); }
DELETE	/delete	deleteSupplier
DELETE	/:supplier_id	(req, res) => { req.body.supplier_id = req.body.supplier_id req.params.supplier_id; return deleteSupplier(req, res); }

3.7 General customers

General customer master for quotation, sales, work orders and service. Runtime schema adds salutation, marketing/referral and approval fields. This customer_id is distinct from cable_customer_id and internet_customer_id.

Development path: Trace customer form/list and customerSchema.js. Preserve approval state and existing references; confirm which customer family the calling screen expects.

Router: backend/routes/customerRouter.js. API mount aliases: /api/customer, /api/v1/customers.

Controllers/helpers imported by router: backend/controller/customerController.js, backend/controller/salesController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): customers, employees, products, quotation_master, sales_items, sales_master, work_orders, workflow_approvals

Frontend service files matched by API mount text: frontend/src/app/services/cable-tv-services.ts, frontend/src/app/services/customer-payment-services.ts, frontend/src/app/services/customer-services.ts, frontend/src/app/services/internet-customer-services.ts, frontend/src/app/services/quotation-services.ts, frontend/src/app/services/sales-services.ts

Frontend route	Active component
/quotation-templates	frontend/src/app/quotations/quotation-form/quotation-form.ts
/cable-tv-masters	frontend/src/app/administration/cable-tv-masters/cable-tv-masters.ts
/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts
/cable-tv-stbs	frontend/src/app/administration/cable-tv-stbs/cable-tv-stbs.ts
/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/lo-accounts	frontend/src/app/administration/lo-accounts/lo-accounts.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts

/cable-tv-subscription-append	frontend/src/app/administration/cable-tv-subscription-append/cable-tv-subscription-append.ts
/assign-net-collector	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-pending	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-append	frontend/src/app/administration/net-subscription-append/net-subscription-append.ts
/net-cash-admin-correction	frontend/src/app/administration/net-cash-admin-correction/net-cash-admin-correction.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts
/catv-customer-list-report	frontend/src/app/administration/catv-customer-list-report/catv-customer-list-report.ts
/net-subscription-report	frontend/src/app/administration/net-subscription-report/net-subscription-report.ts
/stb-payment-report	frontend/src/app/administration/stb-payment-report/stb-payment-report.ts
/material-sales-report	frontend/src/app/administration/material-sales-report/material-sales-report.ts
/complaint-report	frontend/src/app/administration/complaint-report/complaint-report.ts
/quotations	frontend/src/app/quotations/quotation-list/quotation-list.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/preview/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/work-orders/add	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/edit/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/preview/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/sales	frontend/src/app/sales/sales.ts
/customer-payments	frontend/src/app/customer-payments/customer-payments.ts
/service-tickets	frontend/src/app/service-tickets/service-tickets.ts
/warranty-master	frontend/src/app/warranty-master/warranty-master.ts
/quotations	frontend/src/app/quotations/quotation-list/quotation-list.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/review/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/workflow	frontend/src/app/workflow/workflow-list/workflow-list.ts
/customers	frontend/src/app/customers/customer-list/customer-list.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/internet/customers	frontend/src/app/internet/internet-customer-list/internet-customer-list.ts
/internet/customers/add	frontend/src/app/internet/internet-customer-form/internet-customer-form.ts
/internet/customers/edit/:id	frontend/src/app/internet/internet-customer-form/internet-customer-form.ts
/internet/customers/view/:id	frontend/src/app/internet/internet-customer-view/internet-customer-view.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts

/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/view/:id	frontend/src/app/cable-tv/cable-tv-customer-view/cable-tv-customer-view.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/customers/:customerId/invoice	frontend/src/app/sales/sales.ts
/customers/add	frontend/src/app/customers/customer-form/customer-form.ts
/customers/edit/:id	frontend/src/app/customers/customer-form/customer-form.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): sales_id

Method	Path appended to each mount	Handler / route middleware
POST	/	addCustomer
POST	/add	addCustomer
GET	/	getCustomers
GET	/get	getCustomers
GET	/marketing-employees	getMarketingEmployees
POST	/:customer_id/invoice	(req, res) => { req.body.customer_id = req.params.customer_id; return addSale(req, res); }
GET	/get/:customer_id	getCustomerById
GET	/:customer_id	getCustomerById
PATCH	/	updateCustomer
PATCH	/update	updateCustomer
PATCH	/:customer_id	(req, res) => { req.body.customer_id = req.body.customer_id req.params.customer_id; return updateCustomer(req, res); }
DELETE	/delete	deleteCustomer
DELETE	/:customer_id	(req, res) => { req.body.customer_id = req.body.customer_id req.params.customer_id; return deleteCustomer(req, res); }

3.8 Purchases

Supplier purchase header and line items connect supplier_id and product_id to quantities, tax, pricing and stock.

Development path: Follow purchase form/service into purchaseController. Review stock master and ledger writes in the same operation; verify quantity changes, payment balance and rollback behavior.

Router: backend/routes/purchaseRouter.js. API mount aliases: /api/purchase, /api/v1/purchase.

Controllers/helpers imported by router: backend/controller/purchaseController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): products, purchase_items, purchase_master, stock_ledger, stock_master, suppliers

Frontend service files matched by API mount text: frontend/src/app/services/purchase-services.ts

Frontend route

Active component

/purchases	frontend/src/app/purchases/purchase-list/purchase-list.ts
/supplier-payments	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/add	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/edit/:id	frontend/src/app/supplier-payments/supplier-payments.ts
/purchases/add	frontend/src/app/purchases/purchase-form/purchase-form.ts
/purchases/edit/:id	frontend/src/app/purchases/purchase-form/purchase-form.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): approved_by_employee_id, purchase_id

Method	Path appended to each mount	Handler / route middleware
GET	/	getPurchases
GET	/get	getPurchases
GET	/next-no	getNextPurchaseNo
GET	/get/:purchase_id	getPurchaseById
GET	/:purchase_id	getPurchaseById
POST	/	addPurchase
POST	/add	addPurchase
PATCH	/	updatePurchase
PATCH	/update	updatePurchase
PATCH	/:purchase_id	(req, res) => { req.body.purchase_id = req.body.purchase_id req.params.purchase_id; return updatePurchase(req, res); }
PATCH	/:purchase_id/approve	approvePurchase
POST	/:purchase_id/approve	approvePurchase
DELETE	/delete	deletePurchase
DELETE	/:purchase_id	(req, res) => { req.body.purchase_id = req.body.purchase_id req.params.purchase_id; return deletePurchase(req, res); }

3.9 Stock and inventory

Stock master represents balances; ledger records movement history. Installation, purchases, STBs, routers and technician material flows also touch inventory.

Development path: Use stockController plus the domain controller that performs the movement. Avoid editing balance without the corresponding ledger operation. Verify one receipt and one issue path.

Router: backend/routes/stockRouter.js. API mount aliases: /api/stock, /api/v1/stock.

Controllers/helpers imported by router: backend/controller/stockController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): brands, products, stock_ledger, stock_master

Frontend service files matched by API mount text: frontend/src/app/services/cable-tv-services.ts, frontend/src/app/services/stock-services.ts

Frontend route

Active component

/cable-tv-masters	frontend/src/app/administration/cable-tv-masters/cable-tv-masters.ts
/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts
/cable-tv-stbs	frontend/src/app/administration/cable-tv-stbs/cable-tv-stbs.ts
/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/lo-accounts	frontend/src/app/administration/lo-accounts/lo-accounts.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts
/cable-tv-subscription-append	frontend/src/app/administration/cable-tv-subscription-append/cable-tv-subscription-append.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts
/catv-customer-list-report	frontend/src/app/administration/catv-customer-list-report/catv-customer-list-report.ts
/stb-payment-report	frontend/src/app/administration/stb-payment-report/stb-payment-report.ts
/material-sales-report	frontend/src/app/administration/material-sales-report/material-sales-report.ts
/complaint-report	frontend/src/app/administration/complaint-report/complaint-report.ts
/stock	frontend/src/app/stock/stock-list/stock-list.ts
/stock-quick-view	frontend/src/app/stock/stock-quick-view/stock-quick-view.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts
/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/view/:id	frontend/src/app/cable-tv/cable-tv-customer-view/cable-tv-customer-view.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): Inspect destructuring and handler validation in the controller.

Method	Path appended to each mount	Handler / route middleware
GET	/	getStock
GET	/get	getStock
GET	/ledger	getLedger
PATCH	/settings	upsertStockSettings
POST	/adjust	adjustStock

3.10 Quotations and templates

Quotation headers/items, submission, approval and customer response. Template CRUD is in quotationTemplateController; template writes are admin protected.

Development path: Trace quotation list/form, quotation-services and quotationRouter. Keep route-specific operations such as submit, approve and customer-response distinct from header edits. Check preview and linked work-order behavior.

Router: backend/routes/quotationRouter.js. API mount aliases: /api/quotation, /api/v1/quotations.

Controllers/helpers imported by router: backend/controller/quotationController.js, backend/controller/quotationTemplateController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): customers, employees, products, quotation_items, quotation_master, quotation_templates, workflow_approvals

Frontend service files matched by API mount text: frontend/src/app/services/permission.service.ts, frontend/src/app/services/quotation-services.ts

Frontend route	Active component
/dashboard	frontend/src/app/dashboard/dashboard.ts
/users	frontend/src/app/user/user-list/user-list.ts
/productList	frontend/src/app/products/product-list/product-list.ts
/quotation-templates	frontend/src/app/quotations/quotation-form/quotation-form.ts
/role-permissions	frontend/src/app/administration/role-permissions/role-permissions.ts
/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts
/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts
/assign-net-collector	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-pending	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts
/net-subscription-report	frontend/src/app/administration/net-subscription-report/net-subscription-report.ts
/transactions	frontend/src/app/transactions/transaction-list/transaction-list.ts
/quotations	frontend/src/app/quotations/quotation-list/quotation-list.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/preview/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/work-orders/add	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/edit/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/preview/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/quotations	frontend/src/app/quotations/quotation-list/quotation-list.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/review/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/workflow	frontend/src/app/workflow/workflow-list/workflow-list.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts
/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts

/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/employee-salary	frontend/src/app/employee-salary/employee-salary.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): approved_by_employee_id, id, quotation_id, remarks, status

Method	Path appended to each mount	Handler / route middleware
GET	/templates	templates.list
GET	/templates/:id	templates.get
POST	/templates	requireAdmin, templates.save
PUT	/templates/:id	requireAdmin, templates.save
GET	/	getQuotations
GET	/get	getQuotations
GET	/next-no	getNextQuotationNo
GET	/get/:quotation_id	getQuotationById
GET	/:quotation_id	getQuotationById
POST	/	addQuotation
POST	/add	addQuotation
PATCH	/	updateQuotation
PATCH	/update	updateQuotation
PATCH	/:quotation_id	(req, res) => { req.body.quotation_id = req.body.quotation_id req.params.quotation_id; return updateQuotation(req, res); }
PATCH	/:quotation_id/approve	approveQuotation
PATCH	/:quotation_id/submit	submitQuotation
PATCH	/:quotation_id/customer-response	updateCustomerResponse
DELETE	/	deleteQuotation
DELETE	/delete	deleteQuotation
DELETE	/:quotation_id	deleteQuotation

3.11 Workflow approval

Central workflow_approvals uses module_name plus reference_id as a polymorphic link. The controller dispatches to domain-specific approval/rejection logic and stock/account synchronization. reference_id alone is insufficient to identify a table.

Development path: Review the matching module_name branch in workflowController before changing approval behavior. Cable/Internet/stock/account effects may occur at approval time. Test approve, reject and repeat-review handling.

Router: backend/routes/workflowRouter.js. API mount aliases: /api/workflow, /api/v1/workflow, /api/v1/workflows.

Controllers/helpers imported by router: backend/controller/workflowController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables):
 cable_approval_groups, cable_connections, cable_customer_accounts, cable_customer_packages,
 cable_customer_stb_accessories, cable_customer_stbs, cable_stb_issue_master, cable_stb_master, cable_subscriptions,
 cable_tv_customers, categories, customers, finance_transactions, internet_connections, internet_customer_accounts,
 internet_customer_packages, internet_customer_routers, internet_customers, products, purchase_master, quotation_master,
 stock_ledger, stock_master, suppliers, technician_material_stock, work_order_material_issues, work_orders,
 workflow_approvals

Frontend service files matched by API mount text: frontend/src/app/services/quotation-services.ts, frontend/src/app/services/workflow-services.ts

Frontend route	Active component
/quotation-templates	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations	frontend/src/app/quotations/quotation-list/quotation-list.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/preview/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/workflow-approval	frontend/src/app/workflows/workflow-list/workflow-list.ts
/work-orders/add	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/edit/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/preview/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/quotations	frontend/src/app/quotations/quotation-list/quotation-list.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/review/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/workflow	frontend/src/app/workflow/workflow-list/workflow-list.ts
/internet/customers/view/:id	frontend/src/app/internet/internet-customer-view/internet-customer-view.ts
/cable-tv/customers/view/:id	frontend/src/app/cable-tv/cable-tv-customer-view/cable-tv-customer-view.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/customers/add	frontend/src/app/customers/customer-form/customer-form.ts
/customers/edit/:id	frontend/src/app/customers/customer-form/customer-form.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): approved_by_employee_id, remarks, workflow_id

Method	Path appended to each mount	Handler / route middleware
GET	/	getWorkflowApprovals
POST	/:workflow_id/approve	approveWorkflow
GET	/get	getWorkflowApprovals

3.12 Work orders and installation

Work order create/edit/review, material issue submission/review, returns and invoice generation. Linked employee

assignments and quotation/customer references connect the job to people and commercial documents.

Development path: Trace work-order form/material screens and workOrderController. Preserve issue approvals, return quantity rules and invoice linkage. Verify a material issue, return and invoice path with current IDs.

Router: backend/routes/workOrderRouter.js. API mount aliases: /api/work-order, /api/v1/work-orders.

Controllers/helpers imported by router: backend/controller/workOrderController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): customers, employees, material_master, products, quotation_items, quotation_master, sales_items, sales_master, stock_ledger, stock_master, work_order_employees, work_order_items, work_order_material_issues, work_order_material_returns, work_orders, workflow_approvals

Frontend service files matched by API mount text: frontend/src/app/services/work-order-services.ts

Frontend route	Active component
/work-orders	frontend/src/app/work-orders/work-order-list/work-order-list.ts
/work-orders/add	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/edit/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/material-issue/:id	frontend/src/app/work-orders/work-order-material/work-order-material.ts
/work-orders/preview/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): action, created_by_employee_id, due_date, invoice_date, invoice_no, issued_by_employee_id, payment_mode, remarks, reviewed_by_employee_id, rows, work_order_id

Method	Path appended to each mount	Handler / route middleware
GET	/materials	getMaterials
GET	/next-no	getNextWorkOrderNo
GET	/	getWorkOrders
GET	/get	getWorkOrders
GET	/get/:work_order_id	getWorkOrderById
GET	/:work_order_id	getWorkOrderById
POST	/	addWorkOrder
POST	/add	addWorkOrder
PATCH	/	updateWorkOrder
PATCH	/update	updateWorkOrder
PATCH	/:work_order_id	(req, res) => { req.body.work_order_id = req.body.work_order_id req.params.work_order_id; return updateWorkOrder(req, res); }
POST	/:work_order_id/review	reviewWorkOrder
DELETE	/:work_order_id	deleteWorkOrder
POST	/:work_order_id/material-issue	addMaterialIssue
POST	/:work_order_id/material-issues/submit	submitMaterialIssueList
POST	/:work_order_id/material-issues/review	reviewMaterialIssueList

POST	/:work_order_id/material-return	addMaterialReturn
POST	/:work_order_id/create-invoice	createInvoiceFromWorkOrder

3.13 Employees

Employee profiles, departments, user links and uploaded photos; employee_id is used for attendance, salary, installers, collectors and assignees.

Development path: Review multer handling and backend/uploads/employees. Do not confuse employee_id with user_id. Apply department enum migration only after inspecting intended database DDL.

Router: backend/routes/employeeRouter.js. API mount aliases: /api/employee.

Controllers/helpers imported by router: backend/controller/employeeController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): employees

Frontend service files matched by API mount text: frontend/src/app/services/employee-attendance-services.ts, frontend/src/app/services/employee-salary-services.ts, frontend/src/app/services/employee-services.ts

Frontend route	Active component
/quotation-templates	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/preview/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/work-orders/add	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/edit/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/material-issue/:id	frontend/src/app/work-orders/work-order-material/work-order-material.ts
/work-orders/preview/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/customer-payments	frontend/src/app/customer-payments/customer-payments.ts
/supplier-payments	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/add	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/edit/:id	frontend/src/app/supplier-payments/supplier-payments.ts
/service-tickets	frontend/src/app/service-tickets/service-tickets.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/review/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/employees	frontend/src/app/employees/employee-list/employee-list.ts
/employees/add	frontend/src/app/employees/employee-form/employee-form.ts
/employees/edit/:id	frontend/src/app/employees/employee-form/employee-form.ts
/employee-salary	frontend/src/app/employee-salary/employee-salary.ts
/employee-attendance	frontend/src/app/employee-attendance/employee-attendance.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): Inspect destructuring and handler validation in the controller.

Method	Path appended to each mount	Handler / route middleware
GET	/get	auth.authenticateToken, getEmployees
GET	/next-code	auth.authenticateToken, getNextEmployeeCode
GET	/get/:employee_id	auth.authenticateToken, getEmployeeById
POST	/upload-photo	auth.authenticateToken, uploadEmployeePhoto
POST	/add	auth.authenticateToken, addEmployee
PATCH	/update	auth.authenticateToken, updateEmployee
DELETE	/delete	auth.authenticateToken, deleteEmployee

3.14 Employee salary

Salary-related reads and writes by employee. Permission behavior is covered by salaryReadAccess.test.js.

Development path: Trace employee-salary.ts/services into employeeSalaryRouter/controller. Verify permitted salary reads and denied writes independently.

Router: backend/routes/employeeSalaryRouter.js. API mount aliases: /api/employee-salary.

Controllers/helpers imported by router: backend/controller/employeeSalaryController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): employee_salary, employee_salary_items, employees, users

Frontend service files matched by API mount text: frontend/src/app/services/employee-salary-services.ts

Frontend route	Active component
/employee-salary	frontend/src/app/employee-salary/employee-salary.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): Inspect destructuring and handler validation in the controller.

Method	Path appended to each mount	Handler / route middleware
GET	/get	auth.authenticateToken, getSalaries
GET	/get/:salary_id	auth.authenticateToken, getSalaryById
GET	/employee/:employee_id	auth.authenticateToken, getSalaryByEmployee
POST	/add	auth.authenticateToken, addSalary
PATCH	/update	auth.authenticateToken, updateSalary
DELETE	/delete	auth.authenticateToken, deleteSalary

3.15 Audit logs

Records user, module, action and affected record. table_name plus record_id identifies the changed business record; user_id identifies the actor. JSON old/new values support change inspection.

Development path: Trace audit list/form and auditLogController; preserve actor and record identity when changing write paths. The polymorphic record reference is not one fixed FK.

Router: backend/routes/auditLogRouter.js. API mount aliases: /api/audit-log.

Controllers/helpers imported by router: backend/controller/auditLogController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): audit_log

Frontend service files matched by API mount text: frontend/src/app/services/audit-log-services.ts

Frontend route	Active component
/audit-logs	frontend/src/app/audit-logs/audit-log-list/audit-log-list.ts
/audit-logs/add	frontend/src/app/audit-logs/audit-log-form/audit-log-form.ts
/audit-logs/edit/:id	frontend/src/app/audit-logs/audit-log-form/audit-log-form.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): new_values, old_values

Method	Path appended to each mount	Handler / route middleware
GET	/get	auth.authenticateToken, getAuditLogs
GET	/get/:audit_id	auth.authenticateToken, getAuditLogById
POST	/add	auth.authenticateToken, addAuditLog
PATCH	/update	auth.authenticateToken, updateAuditLog
DELETE	/delete	auth.authenticateToken, deleteAuditLog

3.16 Locations

Location endpoints and Cable TV masters support geographic classification. Cable location/area/street relationships also appear in the Cable TV router.

Development path: Check which location family the screen uses; inspect locationRouter and cableTvRouter lookups before reusing an ID across modules.

Router: backend/routes/locationRouter.js. API mount aliases: /api/location.

Controllers/helpers imported by router: backend/controller/locationController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): No direct table matches found.

Frontend service files matched by API mount text: frontend/src/app/services/cable-tv-services.ts, frontend/src/app/services/customer-services.ts, frontend/src/app/services/employee-services.ts, frontend/src/app/services/supplier-services.ts

Frontend route	Active component
/quotation-templates	frontend/src/app/quotations/quotation-form/quotation-form.ts
/cable-tv-masters	frontend/src/app/administration/cable-tv-masters/cable-tv-masters.ts
/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts
/cable-tv-stbs	frontend/src/app/administration/cable-tv-stbs/cable-tv-stbs.ts
/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/lo-accounts	frontend/src/app/administration/lo-accounts/lo-accounts.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts
/cable-tv-subscription-append	frontend/src/app/administration/cable-tv-subscription-append/cable-tv-subscription-append.ts
/assign-net-collector	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-pending	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-append	frontend/src/app/administration/net-subscription-append/net-subscription-append.ts

/net-cash-admin-correction	frontend/src/app/administration/net-cash-admin-correction/net-cash-admin-correction.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts
/catv-customer-list-report	frontend/src/app/administration/catv-customer-list-report/catv-customer-list-report.ts
/net-subscription-report	frontend/src/app/administration/net-subscription-report/net-subscription-report.ts
/stb-payment-report	frontend/src/app/administration/stb-payment-report/stb-payment-report.ts
/material-sales-report	frontend/src/app/administration/material-sales-report/material-sales-report.ts
/complaint-report	frontend/src/app/administration/complaint-report/complaint-report.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/preview/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/work-orders/add	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/edit/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/material-issue/:id	frontend/src/app/work-orders/work-order-material/work-order-material.ts
/work-orders/preview/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/sales	frontend/src/app/sales/sales.ts
/customer-payments	frontend/src/app/customer-payments/customer-payments.ts
/supplier-payments	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/add	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/edit/:id	frontend/src/app/supplier-payments/supplier-payments.ts
/service-tickets	frontend/src/app/service-tickets/service-tickets.ts
/warranty-master	frontend/src/app/warranty-master/warranty-master.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/review/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/purchases/add	frontend/src/app/purchases/purchase-form/purchase-form.ts
/purchases/edit/:id	frontend/src/app/purchases/purchase-form/purchase-form.ts
/suppliers	frontend/src/app/suppliers/supplier-list/supplier-list.ts
/customers	frontend/src/app/customers/customer-list/customer-list.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/internet/customers	frontend/src/app/internet/internet-customer-list/internet-customer-list.ts
/internet/customers/add	frontend/src/app/internet/internet-customer-form/internet-customer-form.ts
/internet/customers/edit/:id	frontend/src/app/internet/internet-customer-form/internet-customer-form.ts
/internet/customers/view/:id	frontend/src/app/internet/internet-customer-view/internet-customer-view.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts

/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/view/:id	frontend/src/app/cable-tv/cable-tv-customer-view/cable-tv-customer-view.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/customers/:customerId/invoice	frontend/src/app/sales/sales.ts
/customers/add	frontend/src/app/customers/customer-form/customer-form.ts
/customers/edit/:id	frontend/src/app/customers/customer-form/customer-form.ts
/suppliers/add	frontend/src/app/suppliers/add-supplier/add-supplier.ts
/suppliers/edit/:id	frontend/src/app/suppliers/add-supplier/add-supplier.ts
/employees	frontend/src/app/employees/employee-list/employee-list.ts
/employees/add	frontend/src/app/employees/employee-form/employee-form.ts
/employees/edit/:id	frontend/src/app/employees/employee-form/employee-form.ts
/employee-salary	frontend/src/app/employee-salary/employee-salary.ts
/employee-attendance	frontend/src/app/employee-attendance/employee-attendance.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): Inspect destructuring and handler validation in the controller.

Method	Path appended to each mount	Handler / route middleware
GET	/states	auth.authenticateToken, getStates
GET	/districts/:state_id	auth.authenticateToken, getDistricts

3.17 Attendance

Attendance records join employee identity to dated attendance data; runtime initialization is in businessModuleSchema.js.

Development path: Trace employee-attendance component/service/controller and verify date filters, employee selection and duplicate-date behavior against source rules.

Router: backend/routes/employeeAttendanceRouter.js. API mount aliases: /api/employee-attendance, /api/v1/employee-attendance.

Controllers/helpers imported by router: backend/controller/employeeAttendanceController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): employee_attendance, employees

Frontend service files matched by API mount text: frontend/src/app/services/employee-attendance-services.ts

Frontend route	Active component
/employee-attendance	frontend/src/app/employee-attendance/employee-attendance.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): Inspect destructuring and handler validation in the controller.

Method	Path appended to each mount	Handler / route middleware
GET	/	getAttendance

GET	/get	getAttendance
POST	/	addAttendance
POST	/add	addAttendance
PATCH	/	updateAttendance
PATCH	/update	updateAttendance
DELETE	/	deleteAttendance
DELETE	/delete	deleteAttendance

3.18 Customer payments

Customer receipts connect to invoice/customer balances. payment IDs identify receipts, not sales headers.

Development path: Trace customer-payments screen/services and paymentController. Review amount, date, reference and outstanding balance rules; verify partial and fully paid examples.

Router: backend/routes/customerPaymentRouter.js. API mount aliases: /api/customer-payment, /api/v1/customer-payments.

Controllers/helpers imported by router: backend/controller/paymentController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): customer_payments, customers, employees, purchase_master, sales_master, supplier_payments, suppliers

Frontend service files matched by API mount text: frontend/src/app/services/customer-payment-services.ts

Frontend route	Active component
/customer-payments	frontend/src/app/customer-payments/customer-payments.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): payment_id

Method	Path appended to each mount	Handler / route middleware
GET	/	getCustomerPayments
GET	/get	getCustomerPayments
POST	/	addCustomerPayment
POST	/add	addCustomerPayment
PATCH	/	updateCustomerPayment
PATCH	/update	updateCustomerPayment
DELETE	/	deleteCustomerPayment
DELETE	/delete	deleteCustomerPayment

3.19 Supplier payments

Supplier disbursements link purchase and supplier balances.

Development path: Trace supplier-payment services and paymentController. Check payment allocation and balance logic without changing purchase stock calculations.

Router: backend/routes/supplierPaymentRouter.js. API mount aliases: /api/supplier-payment, /api/v1/supplier-payments.

Controllers/helpers imported by router: backend/controller/paymentController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables):

customer_payments, customers, employees, purchase_master, sales_master, supplier_payments, suppliers

Frontend service files matched by API mount text: frontend/src/app/services/supplier-payment-services.ts

Frontend route	Active component
/supplier-payments	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/add	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/edit/:id	frontend/src/app/supplier-payments/supplier-payments.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): payment_id

Method	Path appended to each mount	Handler / route middleware
GET	/	getSupplierPayments
GET	/get	getSupplierPayments
POST	/	addSupplierPayment
POST	/add	addSupplierPayment
PATCH	/	updateSupplierPayment
PATCH	/update	updateSupplierPayment
DELETE	/	deleteSupplierPayment
DELETE	/delete	deleteSupplierPayment

3.20 Sales and invoices

Sales headers/items store customer, source documents, taxes and payment metadata. Runtime schema adds paid_date and payment_reference. Customer invoice routes reuse sales.ts.

Development path: Trace sales-services and salesController; preserve source work-order/quotation identity, line totals and payment behavior. The salesBalance test provides focused balance coverage.

Router: backend/routes/salesRouter.js. API mount aliases: /api/sales, /api/v1/sales.

Controllers/helpers imported by router: backend/controller/salesController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): customers, products, quotation_master, sales_items, sales_master, work_orders

Frontend service files matched by API mount text: frontend/src/app/services/cable-tv-services.ts, frontend/src/app/services/sales-services.ts

Frontend route	Active component
/cable-tv-masters	frontend/src/app/administration/cable-tv-masters/cable-tv-masters.ts
/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts
/cable-tv-stbs	frontend/src/app/administration/cable-tv-stbs/cable-tv-stbs.ts
/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/lo-accounts	frontend/src/app/administration/lo-accounts/lo-accounts.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts
/cable-tv-subscription-append	frontend/src/app/administration/cable-tv-subscription-append/cable-tv-subscription-append.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts

/catv-customer-list-report	frontend/src/app/administration/catv-customer-list-report/catv-customer-list-report.ts
/stb-payment-report	frontend/src/app/administration/stb-payment-report/stb-payment-report.ts
/material-sales-report	frontend/src/app/administration/material-sales-report/material-sales-report.ts
/complaint-report	frontend/src/app/administration/complaint-report/complaint-report.ts
/sales	frontend/src/app/sales/sales.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts
/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/view/:id	frontend/src/app/cable-tv/cable-tv-customer-view/cable-tv-customer-view.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/customers/:customerId/invoice	frontend/src/app/sales/sales.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): sales_id

Method	Path appended to each mount	Handler / route middleware
GET	/	getSales
GET	/get	getSales
GET	/:sales_id	getSaleById
POST	/	addSale
POST	/add	addSale
PATCH	/	updateSale
PATCH	/update	updateSale
DELETE	/	deleteSale
DELETE	/delete	deleteSale

3.21 Service tickets

General service tickets use customer/product/employee references, status and assignment fields. This API is separate from Cable TV and Internet complaint APIs.

Development path: Trace service-ticket screen/services and serviceTicketController. Check assignment, priority and status validation in the controller before adding UI options.

Router: backend/routes/serviceTicketRouter.js. API mount aliases: /api/service-ticket, /api/v1/service-tickets.

Controllers/helpers imported by router: backend/controller/serviceTicketController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): customers, employees, products, sales_master, service_tickets

Frontend service files matched by API mount text: frontend/src/app/services/service-ticket-services.ts

Frontend route**Active component**

/service-tickets frontend/src/app/service-tickets/service-tickets.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): service_ticket_id

Method	Path appended to each mount	Handler / route middleware
GET	/	getServiceTickets
GET	/get	getServiceTickets
POST	/	addServiceTicket
POST	/add	addServiceTicket
PATCH	/	updateServiceTicket
PATCH	/update	updateServiceTicket
DELETE	/	deleteServiceTicket
DELETE	/delete	deleteServiceTicket

3.22 Warranty

Warranty records connect customer/product and serial number to validity dates and status.

Development path: Trace warranty services/controller and active route components. Verify product/customer linkage and validity dates together.

Router: backend/routes/warrantyRouter.js. API mount aliases: /api/warranty, /api/v1/warranties.

Controllers/helpers imported by router: backend/controller/warrantyController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): customers, products, sales_master, warranty_master

Frontend service files matched by API mount text: frontend/src/app/services/warranty-services.ts

Frontend route**Active component**

/warranty-master frontend/src/app/warranty-master/warranty-master.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): warranty_id

Method	Path appended to each mount	Handler / route middleware
GET	/	getWarranties
GET	/get	getWarranties
POST	/	addWarranty
POST	/add	addWarranty
PATCH	/	updateWarranty
PATCH	/update	updateWarranty
DELETE	/	deleteWarranty
DELETE	/delete	deleteWarranty

3.23 Notifications

Notification list and state operations provide workflow visibility for users.

Development path: Trace notification services/router/controller; check intended recipient identity and read-state updates. Review create_notifications.sql when deploying to an older schema.

Router: backend/routes/notificationRouter.js. API mount aliases: /api/notifications, /api/v1/notifications.

Controllers/helpers imported by router: backend/controller/notificationController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): cable_customer_accounts, cable_tv_customers, customers, notifications, products, purchase_master, quotation_master, sales_master, stock_master, suppliers, workflow_approvals

Frontend service files matched by API mount text: frontend/src/app/services/notification-services.ts

Frontend route	Active component
/notifications	frontend/src/app/notifications/notification-list/notification-list.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): limit, notification_id, unread

Method	Path appended to each mount	Handler / route middleware
GET	/	getNotifications
PATCH	/read-all	markAllNotificationsRead
PATCH	/:notification_id/read	markNotificationRead

3.24 Cable TV, masters, billing, complaints and materials

Cable customer onboarding, connections, STBs/accessories, packages and subscriptions use separate child-row IDs under cable_customer_id. Accounts and office receipt flows are distinct from subscription collection. This router also hosts technician material movements/sales/adjustments, complaint attempts and reports.

Development path: For customer actions trace cable-tv-services and the relevant list/form/history/view screen into cableTvController. For materials use materialSalesController; for complaints use cableTvComplaintController. Inspect per-route grants, approval groups, deferred stock and account effects; verify the existing cable tests before altering these paths.

Router: backend/routes/cableTvRouter.js. API mount aliases: /api/cable-tv, /api/v1/cable-tv.

Controllers/helpers imported by router: backend/controller/materialSalesController.js, backend/controller/cableTvComplaintController.js, backend/controller/cableTvController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): cable_approval_groups, cable_areas, cable_connection_materials, cable_connection_sources, cable_connections, cable_customer_account_payments, cable_customer_accounts, cable_customer_packages, cable_customer_stb_accessories, cable_customer_stbs, cable_locations, cable_mso_master, cable_network_master, cable_package_master, cable_stb_issue_master, cable_stb_master, cable_streets, cable_subscription_payments, cable_subscriptions, cable_tv_complaint_attempts, cable_tv_complaints, cable_tv_customers, categories, customers, employees, finance_transactions, internet_customer_account_payments, internet_customer_accounts, internet_customers, internet_subscriptions, products, stock_ledger, stock_master, technician_material_movements, technician_material_sale_adjustments, technician_material_sale_payments, technician_material_stock, users

Frontend service files matched by API mount text: frontend/src/app/services/cable-tv-services.ts

Frontend route	Active component
/cable-tv-masters	frontend/src/app/administration/cable-tv-masters/cable-tv-masters.ts
/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts
/cable-tv-stbs	frontend/src/app/administration/cable-tv-stbs/cable-tv-stbs.ts

/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/lo-accounts	frontend/src/app/administration/lo-accounts/lo-accounts.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts
/cable-tv-subscription-append	frontend/src/app/administration/cable-tv-subscription-append/cable-tv-subscription-append.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts
/catv-customer-list-report	frontend/src/app/administration/catv-customer-list-report/catv-customer-list-report.ts
/stb-payment-report	frontend/src/app/administration/stb-payment-report/stb-payment-report.ts
/material-sales-report	frontend/src/app/administration/material-sales-report/material-sales-report.ts
/complaint-report	frontend/src/app/administration/complaint-report/complaint-report.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts
/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/view/:id	frontend/src/app/cable-tv/cable-tv-customer-view/cable-tv-customer-view.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): accountId, action, adjustmentId, adjustment_type, amount, approval_status, area_id, area_name, assigned_employee_id, billing_basis, cable_customer_id, cash_amount, collect_date, collected_by_employee_id, collected_date, comments, complaintId, connectionId, connection_charge, connection_date, connection_discount, connection_type, customer_id, customer_ids, customer_name, customer_no, customer_paid_amount, customer_type, days_in_month, description, disconnection_date, due_date, employee_id, end_date, expiry_date, id, installed_by_employee_id, installed_date, internet_network_type, is_active, issue_mode, issued_by_employee_id, items, labour_service_charge, location_id, movementId, movement_type, name, network_id, number_of_days_or_months, old_customer_no, online_amount, overall_discount, packageId, package_id, package_name, package_price, package_type, paid_amount, paid_date, paid_from_account, payment_mapped_employee_id, payment_mode, payment_reference, payment_status, payment_type, price, qty, reason, reason_remarks, received_amount, received_by_employee_id, received_count, received_date, received_id, received_name, remarks, review_remarks, search, service_category, service_customer_id, start_date, status, stbId, stbMasterId, stb_amount, stb_discount, streetId, street_id, street_name, subscriptionId, subscription_month, subscription_year, type, update_reason, updated_by_employee_id, updated_date

Method	Path appended to each mount	Handler / route middleware
GET	/lookups	getLookups
GET	/material-sales/lookups	auth.requirePermission('MATERIAL_SALES'), getMaterialSalesLookups
GET	/material-sales/stock	auth.requirePermission('MATERIAL_SALES'), getTechnicianStock
GET	/material-sales/movements	auth.requirePermission('MATERIAL_SALES'), getMaterialMovements
POST	/material-sales/movements	auth.requirePermissionAction('MATERIAL_SALES', 'can_create'), addMaterialMovement
POST	/material-sales/issues/batch	auth.requirePermissionAction('MATERIAL_SALES', 'can_create'), addMaterialIssueBatch
POST	/material-sales/sales/batch	auth.requirePermissionAction('MATERIAL_SALES', 'can_create'), addMaterialSaleBatch

GET	/material-sales/issued	auth.requirePermission('MATERIAL_SALES'), getIssuedMaterialSales
PATCH	/material-sales/issued/:movementId	auth.requireAdmin, correctIssuedMaterial
DELETE	/material-sales/issued/:movementId	auth.requireAdmin, correctIssuedMaterial
GET	/material-sales/adjustments	auth.requirePermission('MATERIAL_SALES'), getMaterialSaleAdjustments
POST	/material-sales/movements/:movementId/adjustments	auth.requirePermissionAction('MATERIAL_SALES', 'can_update'), requestMaterialSaleAdjustment
PATCH	/material-sales/adjustments/:adjustmentId/review	auth.requireAdmin, reviewMaterialSaleAdjustment
PATCH	/material-sales/movements/:movementId/customer	auth.requirePermissionAction('MATERIAL_SALES', 'can_update'), mapMaterialSaleCustomer
PATCH	/material-sales/movements/:movementId/sold	auth.requirePermissionAction('MATERIAL_SALES', 'can_update'), markMaterialSaleSold
GET	/complaints	auth.requirePermission('CABLE_TV_COMPLAINTS'), getComplaints
GET	/reports/complaints	auth.requirePermission('CABLE_TV_COMPLAINT_REPORT'), getComplaintReport
GET	/complaints/customers/lookup	auth.requirePermission('CABLE_TV_COMPLAINTS'), getComplaintCustomers
GET	/complaints/:complaintId	auth.requirePermission('CABLE_TV_COMPLAINTS'), getComplaintById
POST	/complaints	auth.requirePermissionAction('CABLE_TV_COMPLAINTS', 'can_create'), addComplaint
POST	/complaints/:complaintId/attempts	addComplaintAttempt
GET	/masters	auth.requireAnyPermission(['CABLE_TV_MASTERS', 'CABLE_TV_PACKAGES', 'CABLE_TV_STBS']), getMasters
POST	/masters/locations	auth.requirePermission('CABLE_TV_MASTERS'), addLocation
POST	/masters/areas	auth.requirePermission('CABLE_TV_MASTERS'), addArea
POST	/masters/streets	auth.requirePermission('CABLE_TV_MASTERS'), addStreet
POST	/masters/location-info	auth.requirePermission('CABLE_TV_MASTERS'), addLocationInfo
PATCH	/masters/location-info/:streetId	auth.requirePermission('CABLE_TV_MASTERS'), updateLocationInfo
DELETE	/masters/location-info/:streetId	auth.requirePermission('CABLE_TV_MASTERS'), deleteLocationInfo
POST	/masters/packages	auth.requirePermission('CABLE_TV_PACKAGES'), addPackage
PATCH	/masters/packages/:packageId	auth.requirePermission('CABLE_TV_PACKAGES'), updatePackage
POST	/masters/stbs	auth.requirePermission('CABLE_TV_STBS'), addStbMaster
PATCH	/masters/stbs/:stbMasterId	auth.requirePermission('CABLE_TV_STBS'), updateStbMaster
DELETE	/masters/stbs/:stbMasterId	auth.requirePermission('CABLE_TV_STBS'), deleteStbMaster
PATCH	/masters/stbs/:stbMasterId/assign	auth.requirePermission('CABLE_TV_STBS'), assignStbMaster
GET	/accounts/pending	auth.requirePermission('CABLE_TV_ACCOUNTS'), getPendingAccounts
GET	/accounts/lo-customers	auth.requirePermission('LO_ACCOUNTS'), getLoAccounts
GET	/accounts/:accountId/payments	auth.requirePermission('CABLE_TV_ACCOUNTS'), getAccountPayments
PATCH	/accounts/:accountId/receive	auth.requirePermission('CABLE_TV_ACCOUNTS'), receiveAccount
PATCH	/accounts/:accountId/revert-pending	auth.requireAdmin, revertAccountToPending

GET	/subscriptions/pending	auth.requirePermission('CABLE_TV_SUBSCRIPTION_DUES'), getPendingSubscriptions
GET	/subscriptions/generation-preview	auth.requireAdmin, previewSubscriptionGeneration
POST	/subscriptions/generate	auth.requireAdmin, generateMonthlySubscriptions
PATCH	/subscriptions/:subscriptionId/receive	auth.requirePermissionAction('CABLE_TV_SUBSCRIPTION_DUES', 'can_view'), receiveSubscriptionPayment
GET	/reports/subscriptions	auth.requirePermission('CABLE_TV_SUBSCRIPTION_REPORT'), getCableSubscriptionReport
GET	/reports/customers	auth.requirePermission('CATV_CUSTOMER_LIST_REPORT'), getCableCustomerListReport
GET	/reports/stb-payments	auth.requirePermission('STB_PAYMENT_REPORT'), getStbPaymentReport
GET	/customers	auth.requirePermission('CABLE_TV_CUSTOMERS'), getCableCustomers
GET	/customers/:id	auth.requirePermission('CABLE_TV_CUSTOMERS'), getCableCustomerById
POST	/customers	auth.requirePermission('CABLE_TV_CUSTOMERS'), addCableCustomer
PATCH	/customers/:id	auth.requirePermission('CABLE_TV_CUSTOMERS'), updateCableCustomer
PATCH	/customers/:id/information	auth.requireAdmin, updateCableCustomerInformation
POST	/customers/:id/connections	auth.requirePermission('CABLE_TV_CONNECTIONS'), addCustomerConnection
PATCH	/customers/:id/ connections/:connectionId	auth.requireAdmin, updateCustomerConnection
DELETE	/customers/:id/ connections/:connectionId	auth.requireAdmin, deleteCustomerConnection
POST	/customers/:id/stbs	auth.requirePermission('CABLE_TV_CUSTOMER_STBS'), addCustomerStb
PATCH	/customers/:id/stbs/:stbId	auth.requireAdmin, updateCustomerStb
DELETE	/customers/:id/stbs/:stbId	auth.requireAdmin, deleteCustomerStb
POST	/customers/:id/packages	auth.requirePermission('CABLE_TV_CUSTOMER_PACKAGES'), addCustomerPackage
POST	/customers/:id/packages/:packageId/ remove	auth.requirePermission('CABLE_TV_CUSTOMER_PACKAGES'), removeCustomerPackage
PATCH	/customers/:id/packages/:packageId	auth.requireAdmin, updateCustomerPackage
DELETE	/customers/:id/packages/:packageId	auth.requireAdmin, deleteCustomerPackage
POST	/customers/:id/subscriptions	auth.requirePermission('CABLE_TV_SUBSCRIPTIONS'), addCustomerSubscription
PATCH	/customers/:id/ subscriptions/:subscriptionId	auth.requirePermissionAction('CABLE_TV_SUBSCRIPTIONS', 'can_create'), updateCustomerSubscription
DELETE	/customers/:id/ subscriptions/:subscriptionId	auth.requireAdmin, deleteCustomerSubscription

3.25 Finance transactions

finance_transactions records transaction operations; approval and delete routes require ADMIN, while list/create use authenticated router access. Inspect controller checks for any further restrictions.

Development path: Trace transaction-list and transaction-services into transactionController. Confirm transaction type/reference identity, approval and balance impact in the concrete handler.

Router: backend/routes/transactionRouter.js. API mount aliases: /api/transactions, /api/v1/transactions.

Controllers/helpers imported by router: backend/controller/transactionController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): employees, finance_transactions, users

Frontend service files matched by API mount text: frontend/src/app/services/transaction-services.ts

Frontend route	Active component
/transactions	frontend/src/app/transactions/transaction-list/transaction-list.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): amount, bill_copy_available, category, created_by_user_id, description, end_date, instructed_by_user_id, item_list, payment_mode, received_by, received_date, reference_no, start_date, transactionId, transaction_date, transaction_type

Method	Path appended to each mount	Handler / route middleware
GET	/	getTransactions
POST	/	addTransaction
PATCH	/:transactionId/approve	auth.requireAdmin, approveTransaction
DELETE	/:transactionId	auth.requireAdmin, deleteTransaction

3.26 Internet customers, routers, billing and collection

Internet onboarding uses internet_customer_id, with separate router, connection, package, subscription and account IDs. Includes collector assignment, renewal/append preview, receive-payment, reports, admin cash correction and invoice email.

Development path: Trace internet-customer-services and Internet/administration components into internetCustomerController and its helper controllers. Enrollment, router stock, connection actions and account synchronization have focused tests. Match both customer and child IDs; preserve approved history and payment allocations.

Router: backend/routes/internetCustomerRouter.js. API mount aliases: /api/internet, /api/v1/internet.

Controllers/helpers imported by router: backend/controller/internetCustomerController.js, backend/controller/internetSubscriptionEmail.js, backend/controller/internetCustomerController.js, backend/controller/internetPackageController.js, backend/controller/internetConnectionActions.js, backend/controller/internetCustomerController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): cable_areas, cable_locations, cable_streets, categories, employees, finance_transactions, internet_connection_materials, internet_connections, internet_customer_account_payments, internet_customer_accounts, internet_customer_complaints, internet_customer_packages, internet_customer_routers, internet_customers, internet_package_master, internet_router_material_issues, internet_subscriptions, products, technician_material_movements, technician_material_stock, users, workflow_approvals

Frontend service files matched by API mount text: frontend/src/app/services/internet-customer-services.ts, frontend/src/app/services/permission.service.ts

Frontend route	Active component
/dashboard	frontend/src/app/dashboard/dashboard.ts
/users	frontend/src/app/user/user-list/user-list.ts
/productList	frontend/src/app/products/product-list/product-list.ts
/role-permissions	frontend/src/app/administration/role-permissions/role-permissions.ts
/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts

/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts
/assign-net-collector	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-pending	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-append	frontend/src/app/administration/net-subscription-append/net-subscription-append.ts
/net-cash-admin-correction	frontend/src/app/administration/net-cash-admin-correction/net-cash-admin-correction.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts
/net-subscription-report	frontend/src/app/administration/net-subscription-report/net-subscription-report.ts
/transactions	frontend/src/app/transactions/transaction-list/transaction-list.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/internet/customers	frontend/src/app/internet/internet-customer-list/internet-customer-list.ts
/internet/customers/add	frontend/src/app/internet/internet-customer-form/internet-customer-form.ts
/internet/customers/edit/:id	frontend/src/app/internet/internet-customer-form/internet-customer-form.ts
/internet/customers/view/:id	frontend/src/app/internet/internet-customer-view/internet-customer-view.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts
/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/employee-salary	frontend/src/app/employee-salary/employee-salary.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): amount, area_id, collect_date, collected_by_employee_id, complaint_date, connectionId, customer_ids, customer_name, customer_no, description, end_date, free_period_unit, free_period_value, id, net_id, net_ids, network_type, packageRowId, payment_mapped_employee_id, payment_mode, payment_reference, period_count, period_unit, period_value, received_amount, renewed_by_employee_id, routerId, section, start_date, street_id, subject, subscriptionId, subscription_month, subscription_year

Method	Path appended to each mount	Handler / route middleware
GET	/customers/:id/subscriptions/:subscriptionId/email-preview	auth.requirePermissionAction('INTERNET_CUSTOMERS', 'can_view'), invoiceEmail.preview
POST	/customers/:id/subscriptions/:subscriptionId/email	auth.requirePermissionAction('INTERNET_CUSTOMERS', 'can_view'), (req, res, next) => { invoiceU res.status(400).json({ message: 'Attach one PDF invoice under 4 MB' }) : next(); }, invoiceEmail.send
GET	/packages	auth.requireAdmin,internetPackages.listPackages
POST	/packages	auth.requireAdmin,internetPackages.savePackage
PUT	/packages/:id	auth.requireAdmin,internetPackages.savePackage
GET	/assign-net-collector	auth.requirePermission('NET_SUBSCRIPTION'),getUnassignedNetCollectors

PATCH	/assign-net-collector/:id	auth.requirePermissionAction('NET_SUBSCRIPTION','can_update'),assignNetCollector
GET	/subscription-dues/lookups	auth.requirePermission('NET_SUBSCRIPTION'),internetLookups
GET	/subscription-dues	auth.requirePermission('NET_SUBSCRIPTION'),getPendingInternetSubscriptions
PATCH	/subscription-dues/:id/ receive	auth.requirePermissionAction('NET_SUBSCRIPTION','can_update'),receiveInternetSubscriptionPayr
GET	/subscription-report	auth.requirePermission('NET_SUBSCRIPTION_REPORT'),getInternetSubscriptionReport
GET	/subscriptions/append- preview	auth.requirePermission('NET_SUBSCRIPTION_APPEND'),allowPermissionGrantedAdminOperation
POST	/subscriptions/append	auth.requirePermissionAction('NET_SUBSCRIPTION_APPEND','can_create'),allowPermissionGrant
POST	/subscriptions/cash-admin- correction/preview	auth.requireAdmin,previewCashAdminCorrection
POST	/subscriptions/cash-admin- correction/apply	auth.requireAdmin,applyCashAdminCorrection
PATCH	/customers/:id/ subscriptions/:subscriptionId	auth.requirePermissionAction('INTERNET_CUSTOMERS','can_view'),updateInternetSubscription
GET	/lookups	internetLookups
GET	/customers	getInternetCustomers
GET	/customers/:id	getInternetCustomer
POST	/customers	saveInternetCustomer
PUT	/customers/:id	saveInternetCustomer
PATCH	/customers/:id/information	updateInternetCustomerInformation
DELETE	/customers/:id/ subscriptions/:subscriptionId	deleteInternetSubscription
PATCH	/customers/:id/ packages/:packageRowId	updateInternetCustomerPackage
DELETE	/customers/:id/ packages/:packageRowId	deleteInternetCustomerPackage
GET	/customers/:id/ connections/:connectionId	auth.requireAdmin,connectionActions.get
PATCH	/customers/:id/ connections/:connectionId	auth.requireAdmin,connectionActions.update
DELETE	/customers/:id/ connections/:connectionId	auth.requireAdmin,connectionActions.remove
PATCH	/customers/:id/ routers/:routerId	updateInternetCustomerRouter
DELETE	/customers/:id/ routers/:routerId	deleteInternetCustomerRouter
GET	/customers/:id/complaints	getInternetComplaints
POST	/customers/:id/complaints	addInternetComplaint
POST	/customers/:id/:section	addInternetCustomerHistory

3.27 Dashboard

Aggregated dashboard reads present cross-module summaries.

Development path: Trace dashboard.ts and its HTTP calls into dashboardRouter/controller. Confirm each source query, time range and status filter before changing a tile or total.

Router: backend/routes/dashboardRouter.js. API mount aliases: /api/dashboard, /api/v1/dashboard.

Controllers/helpers imported by router: backend/controller/dashboardController.js

Database tables named by directly imported controller SQL (static inventory; helper/dynamic SQL can add tables): cable_approval_groups, cable_stb_master, cable_tv_complaints, cable_tv_customers, internet_customers, purchase_master, workflow_approvals

Frontend service files matched by API mount text: frontend/src/app/services/route-guard.ts

Direct req.body/params/query field accesses found across imported controllers (not a required-field contract; destructured fields are not included): Inspect destructuring and handler validation in the controller.

Method	Path appended to each mount	Handler / route middleware
GET	/summary	getDashboardSummary

4. Frontend navigation and HTTP service reference

This is the active lazy-route inventory from app.routes.ts. Child routes resolve under the layout. Some components serve more than one screen, and administrative reports/actions can share a domain service. Root redirects and wildcard navigation are defined in app.routes.ts; this inventory focuses on loadComponent declarations.

Route	Component source
/login	frontend/src/app/login/login.ts
/	frontend/src/app/layout/layout.ts
/dashboard	frontend/src/app/dashboard/dashboard.ts
/users	frontend/src/app/user/user-list/user-list.ts
/brands	frontend/src/app/products/brands-list/brands-list.ts
/categoriesLists	frontend/src/app/products/categories-list/categories-list.ts
/productList	frontend/src/app/products/product-list/product-list.ts
/quotation-templates	frontend/src/app/quotations/quotation-form/quotation-form.ts
/role-permissions	frontend/src/app/administration/role-permissions/role-permissions.ts
/cable-tv-masters	frontend/src/app/administration/cable-tv-masters/cable-tv-masters.ts
/cable-tv-packages	frontend/src/app/administration/cable-tv-packages/cable-tv-packages.ts
/internet-packages	frontend/src/app/administration/internet-packages/internet-packages.ts
/cable-tv-stbs	frontend/src/app/administration/cable-tv-stbs/cable-tv-stbs.ts
/cable-tv-account-pending	frontend/src/app/administration/cable-tv-account-pending/cable-tv-account-pending.ts
/lo-accounts	frontend/src/app/administration/lo-accounts/lo-accounts.ts
/cable-tv-subscription-pending	frontend/src/app/administration/cable-tv-subscription-pending/cable-tv-subscription-pending.ts
/cable-tv-subscription-append	frontend/src/app/administration/cable-tv-subscription-append/cable-tv-subscription-append.ts
/assign-net-collector	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts
/net-subscription-pending	frontend/src/app/administration/net-subscription-pending/net-subscription-pending.ts

/net-subscription-append	frontend/src/app/administration/net-subscription-append/net-subscription-append.ts
/net-cash-admin-correction	frontend/src/app/administration/net-cash-admin-correction/net-cash-admin-correction.ts
/cable-tv-subscription-report	frontend/src/app/administration/cable-tv-subscription-report/cable-tv-subscription-report.ts
/catv-customer-list-report	frontend/src/app/administration/catv-customer-list-report/catv-customer-list-report.ts
/net-subscription-report	frontend/src/app/administration/net-subscription-report/net-subscription-report.ts
/stb-payment-report	frontend/src/app/administration/stb-payment-report/stb-payment-report.ts
/material-sales-report	frontend/src/app/administration/material-sales-report/material-sales-report.ts
/complaint-report	frontend/src/app/administration/complaint-report/complaint-report.ts
/transactions	frontend/src/app/transactions/transaction-list/transaction-list.ts
/purchases	frontend/src/app/purchases/purchase-list/purchase-list.ts
/quotations	frontend/src/app/quotations/quotation-list/quotation-list.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/preview/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/workflow-approval	frontend/src/app/workflows/workflow-list/workflow-list.ts
/notifications	frontend/src/app/notifications/notification-list/notification-list.ts
/work-orders	frontend/src/app/work-orders/work-order-list/work-order-list.ts
/work-orders/add	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/edit/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/work-orders/material-issue/:id	frontend/src/app/work-orders/work-order-material/work-order-material.ts
/work-orders/preview/:id	frontend/src/app/work-orders/work-order-form/work-order-form.ts
/sales	frontend/src/app/sales/sales.ts
/customer-payments	frontend/src/app/customer-payments/customer-payments.ts
/supplier-payments	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/add	frontend/src/app/supplier-payments/supplier-payments.ts
/supplier-payments/edit/:id	frontend/src/app/supplier-payments/supplier-payments.ts
/service-tickets	frontend/src/app/service-tickets/service-tickets.ts
/warranty-master	frontend/src/app/warranty-master/warranty-master.ts
/stock	frontend/src/app/stock/stock-list/stock-list.ts
/stock-quick-view	frontend/src/app/stock/stock-quick-view/stock-quick-view.ts
/quotations	frontend/src/app/quotations/quotation-list/quotation-list.ts
/quotations/add	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/edit/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/quotations/review/:id	frontend/src/app/quotations/quotation-form/quotation-form.ts
/workflow	frontend/src/app/workflow/workflow-list/workflow-list.ts

/purchases/add	frontend/src/app/purchases/purchase-form/purchase-form.ts
/purchases/edit/:id	frontend/src/app/purchases/purchase-form/purchase-form.ts
/suppliers	frontend/src/app/suppliers/supplier-list/supplier-list.ts
/customers	frontend/src/app/customers/customer-list/customer-list.ts
/material-sales	frontend/src/app/cable-tv/material-sales/material-sales.ts
/cable-tv/complaints	frontend/src/app/cable-tv/cable-tv-complaints/cable-tv-complaints.ts
/internet/customers	frontend/src/app/internet/internet-customer-list/internet-customer-list.ts
/internet/customers/add	frontend/src/app/internet/internet-customer-form/internet-customer-form.ts
/internet/customers/edit/:id	frontend/src/app/internet/internet-customer-form/internet-customer-form.ts
/internet/customers/view/:id	frontend/src/app/internet/internet-customer-view/internet-customer-view.ts
/cable-tv/customers	frontend/src/app/cable-tv/cable-tv-customer-list/cable-tv-customer-list.ts
/cable-tv/customers/add	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/edit/:id	frontend/src/app/cable-tv/cable-tv-customer-form/cable-tv-customer-form.ts
/cable-tv/customers/view/:id	frontend/src/app/cable-tv/cable-tv-customer-view/cable-tv-customer-view.ts
/cable-tv/customers/:id	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/cable-tv/customers/:id/:section	frontend/src/app/cable-tv/cable-tv-customer-history/cable-tv-customer-history.ts
/customers/:customerId/invoice	frontend/src/app/sales/sales.ts
/customers/add	frontend/src/app/customers/customer-form/customer-form.ts
/customers/edit/:id	frontend/src/app/customers/customer-form/customer-form.ts
/suppliers/add	frontend/src/app/suppliers/add-supplier/add-supplier.ts
/suppliers/edit/:id	frontend/src/app/suppliers/add-supplier/add-supplier.ts
/employees	frontend/src/app/employees/employee-list/employee-list.ts
/employees/add	frontend/src/app/employees/employee-form/employee-form.ts
/employees/edit/:id	frontend/src/app/employees/employee-form/employee-form.ts
/employee-salary	frontend/src/app/employee-salary/employee-salary.ts
/employee-attendance	frontend/src/app/employee-attendance/employee-attendance.ts
/audit-logs	frontend/src/app/audit-logs/audit-log-list/audit-log-list.ts
/audit-logs/add	frontend/src/app/audit-logs/audit-log-form/audit-log-form.ts
/audit-logs/edit/:id	frontend/src/app/audit-logs/audit-log-form/audit-log-form.ts
/**	frontend/src/app/login/login.ts

frontend/src/app/services/audit-log-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	`\${this.url}/audit-log/get`);

GET	`\${this.url}/audit-log/get/\${auditId}`);
POST	`\${this.url}/audit-log/add`, data, {
PATCH	`\${this.url}/audit-log/update`, data, {
DELETE	`\${this.url}/audit-log/delete`, {

frontend/src/app/services/auth-service.ts

Shared frontend utility or auth/permission behavior; inspect its source for non-HttpClient responsibilities and indirect calls.

frontend/src/app/services/brand-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
POST	this.endpoint, data,{
PATCH	this.endpoint,data,{
DELETE	`\${this.endpoint}/\${brandId}`, {

frontend/src/app/services/cable-tv-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	`\${this.endpoint}/lookups`);
GET	`\${this.endpoint}/masters`);
GET	`\${this.endpoint}/customers`, {
GET	`\${this.endpoint}/customers/\${customerId}`);
POST	`\${this.endpoint}/customers`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/customers/\${customerId}`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/customers/\${customerId}/information`, data, { headers: this.jsonHeaders });
POST	`\${this.endpoint}/customers/\${customerId}/connections`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/customers/\${customerId}/connections/\${connectionId}`, data, { headers: this.jsonHeaders });
DELETE	`\${this.endpoint}/customers/\${customerId}/connections/\${connectionId}`);
POST	`\${this.endpoint}/customers/\${customerId}/stbs`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/customers/\${customerId}/stbs/\${stbId}`, data, { headers: this.jsonHeaders });
DELETE	`\${this.endpoint}/customers/\${customerId}/stbs/\${stbId}`);
POST	`\${this.endpoint}/customers/\${customerId}/packages`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/customers/\${customerId}/packages/\${packageId}`, data, { headers: this.jsonHeaders });
POST	`\${this.endpoint}/customers/\${customerId}/packages/\${packageId}/remove`, {}, { headers: this.jsonHeaders });
DELETE	`\${this.endpoint}/customers/\${customerId}/packages/\${packageId}`);

POST	`\${this.endpoint}/customers/\${customerId}/subscriptions`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/customers/\${customerId}/subscriptions/\${subscriptionId}`, data, { headers: this.jsonHeaders });
DELETE	`\${this.endpoint}/customers/\${customerId}/subscriptions/\${subscriptionId}`;
POST	`\${this.endpoint}/masters/locations`, data, { headers: this.jsonHeaders });
POST	`\${this.endpoint}/masters/areas`, data, { headers: this.jsonHeaders });
POST	`\${this.endpoint}/masters/streets`, data, { headers: this.jsonHeaders });
POST	`\${this.endpoint}/masters/location-info`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/masters/location-info/\${streetId}`, data, { headers: this.jsonHeaders });
DELETE	`\${this.endpoint}/masters/location-info/\${streetId}`;
POST	`\${this.endpoint}/masters/packages`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/masters/packages/\${packageId}`, data, { headers: this.jsonHeaders });
POST	`\${this.endpoint}/masters/stbs`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/masters/stbs/\${stbMasterId}`, data, { headers: this.jsonHeaders });
DELETE	`\${this.endpoint}/masters/stbs/\${stbMasterId}`;
PATCH	`\${this.endpoint}/masters/stbs/\${stbMasterId}/assign`, { assigned_employee_id: assignedEmployeeId }, { headers: this.jsonHeaders });
GET	`\${this.endpoint}/accounts/pending`, { params });
GET	`\${this.endpoint}/accounts/lo-customers`, { params });
GET	`\${this.endpoint}/accounts/\${accountId}/payments`;
PATCH	`\${this.endpoint}/accounts/\${accountId}/receive`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/accounts/\${accountId}/revert-pending`, {}, { headers: this.jsonHeaders });
GET	`\${this.endpoint}/subscriptions/pending`, { params });
PATCH	`\${this.endpoint}/subscriptions/\${subscriptionId}/receive`, data, { headers: this.jsonHeaders });
GET	`\${this.endpoint}/reports/subscriptions`, { params });
GET	`\${this.endpoint}/reports/customers`, { params });
GET	`\${this.endpoint}/reports/stb-payments`, {params:filters});
GET	`\${this.endpoint}/subscriptions/generation-preview`, {
POST	`\${this.endpoint}/subscriptions/generate`, data, { headers: this.jsonHeaders });
GET	`\${this.endpoint}/complaints`, { params });
GET	`\${this.endpoint}/reports/complaints`, { params });
GET	`\${this.endpoint}/complaints/\${complaintId}`;
GET	`\${this.endpoint}/complaints/customers/lookup`, { params });
POST	`\${this.endpoint}/complaints`, data, { headers: this.jsonHeaders });
POST	`\${this.endpoint}/complaints/\${complaintId}/attempts`, data, { headers: this.jsonHeaders });
GET	`\${this.endpoint}/material-sales/lookups`;

GET	`\${this.endpoint}/material-sales/stock`, {
GET	`\${this.endpoint}/material-sales/movements`, { params });
POST	`\${this.endpoint}/material-sales/movements`, data, { headers: this.jsonHeaders });
POST	`\${this.endpoint}/material-sales/issues/batch`, { items }, { headers: this.jsonHeaders });
POST	`\${this.endpoint}/material-sales/sales/batch`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/material-sales/issued/\${movementId}`, data, { headers: this.jsonHeaders });
DELETE	`\${this.endpoint}/material-sales/issued/\${movementId}`;
GET	`\${this.endpoint}/material-sales/issued`, {
GET	`\${this.endpoint}/material-sales/adjustments`, {
POST	`\${this.endpoint}/material-sales/movements/\${movementId}/adjustments`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/material-sales/adjustments/\${adjustmentId}/review`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/material-sales/movements/\${movementId}/sold`, data, { headers: this.jsonHeaders });
PATCH	`\${this.endpoint}/material-sales/movements/\${movementId}/customer`, data, { headers: this.jsonHeaders });

frontend/src/app/services/category-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
DELETE	`\${this.endpoint}/\${id}`;
POST	this.endpoint, data,{
PATCH	this.endpoint, data, {
GET	`\${this.endpoint}/\${parentId}`)
PUT	`\${this.apiUrl}/\${id}`, payload);
DELETE	`\${this.apiUrl}/\${id}`;

frontend/src/app/services/customer-payment-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
POST	this.endpoint, data, {
PATCH	this.endpoint, data, {
DELETE	this.endpoint, {

frontend/src/app/services/customer-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
GET	`\${this.endpoint}/\${customerId}`);
GET	`\${this.url}/location/states`);
GET	`\${this.url}/location/districts/\${stateId}`);
GET	`\${this.endpoint}/marketing-employees`);
POST	this.endpoint, data, {
PATCH	this.endpoint, data, {
DELETE	this.endpoint, {

frontend/src/app/services/employee-attendance-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
POST	this.endpoint, data, {
PATCH	this.endpoint, data, {
DELETE	this.endpoint, {

frontend/src/app/services/employee-salary-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	`\${this.url}/employee-salary/get`);
GET	`\${this.url}/employee-salary/get/\${salaryId}`);
POST	`\${this.url}/employee-salary/add`, data, {
PATCH	`\${this.url}/employee-salary/update`, data, {
DELETE	`\${this.url}/employee-salary/delete`, {

frontend/src/app/services/employee-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	`\${this.url}/employee/get`);
GET	`\${this.url}/employee/get/\${employeeId}`);
GET	`\${this.url}/employee/next-code`);
POST	`\${this.url}/employee/upload-photo`, formData);
GET	`\${this.url}/location/states`);

GET	`\${this.url}/location/districts/\${stateId}`);
POST	`\${this.url}/employee/add`, data, {
PATCH	`\${this.url}/employee/update`, data, {
DELETE	`\${this.url}/employee/delete`, {

frontend/src/app/services/global-constants.ts

Shared frontend utility or auth/permission behavior; inspect its source for non-HttpClient responsibilities and indirect calls.

frontend/src/app/services/idle-logout.service.ts

Shared frontend utility or auth/permission behavior; inspect its source for non-HttpClient responsibilities and indirect calls.

frontend/src/app/services/internet-customer-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	`\${this.endpoint}/assign-net-collector`, {params:filters}); }
PATCH	`\${this.endpoint}/assign-net-collector/\${id}`,data); }
GET	`\${this.endpoint}/lookups`); }
GET	`\${this.endpoint}/customers`); }
GET	`\${this.endpoint}/customers/\${id}`); }
POST	`\${this.endpoint}/customers`, data); }
PUT	`\${this.endpoint}/customers/\${id}`, data); }
PATCH	`\${this.endpoint}/customers/\${id}/information`, data); }
POST	`\${this.endpoint}/customers/\${id}/\${section}`, data); }
GET	`\${this.endpoint}/customers/\${id}/connections/\${rowId}`); }
PATCH	`\${this.endpoint}/customers/\${id}/connections/\${rowId}`,data); }
DELETE	`\${this.endpoint}/customers/\${id}/connections/\${rowId}`); }
GET	`\${this.endpoint}/customers/\${id}/complaints`); }
POST	`\${this.endpoint}/customers/\${id}/complaints`, data); }
GET	`\${this.endpoint}/subscription-dues/lookups`); }
GET	`\${this.endpoint}/subscription-dues`, { params: filters }); }
PATCH	`\${this.endpoint}/subscription-dues/\${id}/receive`, data); }
PATCH	`\${this.endpoint}/customers/\${customerId}/subscriptions/\${subscriptionId}`, data); }
GET	`\${this.endpoint}/customers/\${customerId}/subscriptions/\${subscriptionId}/email-preview`); }
POST	`\${this.endpoint}/customers/\${customerId}/subscriptions/\${subscriptionId}/email`, data); }
DELETE	`\${this.endpoint}/customers/\${customerId}/subscriptions/\${subscriptionId}`); }
PATCH	`\${this.endpoint}/customers/\${customerId}/packages/\${packageRowId}`, data); }

DELETE	`\${this.endpoint}/customers/\${customerId}/packages/\${packageRowId}`); }
PATCH	`\${this.endpoint}/customers/\${customerId}/routers/\${routerId}`, data); }
DELETE	`\${this.endpoint}/customers/\${customerId}/routers/\${routerId}`); }
GET	`\${this.endpoint}/subscription-report`, { params: filters }); }
GET	`\${this.endpoint}/subscriptions/append-preview`, {params:{subscription_month:month,subscription_year:year}}); }
POST	`\${this.endpoint}/subscriptions/append`, data); }
POST	`\${this.endpoint}/subscriptions/cash-admin-correction/preview`, {net_ids:netIds,subscription_month:month,subscription_year:year,...selection}); }
POST	`\${this.endpoint}/subscriptions/cash-admin-correction/apply`, {customer_ids:customerIds,net_ids:netIds,subscription_month:month,subscription_year:year,...selection}); }

frontend/src/app/services/ngx-loader.ts

Shared frontend utility or auth/permission behavior; inspect its source for non-HttpClient responsibilities and indirect calls.

frontend/src/app/services/notification-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	`\${this.endpoint}?limit=\${limit}&unread=\${unread}`);
PATCH	`\${this.endpoint}/\${notificationId}/read`, {});
PATCH	`\${this.endpoint}/read-all`, {});

frontend/src/app/services/permission.service.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	`\${appConfig.apiUrl}/user/my-permissions`).pipe(
GET	`\${appConfig.apiUrl}/permissions`); }
PUT	`\${appConfig.apiUrl}/permissions/\${role}`, { permissions });

frontend/src/app/services/product-service.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
POST	this.endpoint, data, {
PATCH	this.endpoint, data,{
DELETE	`\${this.endpoint}/\${productId}`);

frontend/src/app/services/purchase-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
GET	`\${this.endpoint}/\${purchaseId}`);
GET	`\${this.endpoint}/next-no`);
POST	this.endpoint, data, { headers: this.headers });
PATCH	this.endpoint, data, { headers: this.headers });
PATCH	`\${this.endpoint}/\${purchaseId}/approve`, data, { headers: this.headers });
DELETE	this.endpoint, {

frontend/src/app/services/quotation-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	`\${this.endpoint}/templates`); }
GET	`\${this.endpoint}/templates/\${id}`); }
PUT	`\${this.endpoint}/templates/\${id}`, data) : this.http.post(`\${this.endpoint}/templates`, data);
GET	this.endpoint);
GET	`\${this.endpoint}/\${quotationId}`);
GET	`\${this.endpoint}/next-no`);
GET	`\${this.url}/v1/workflow`);
POST	this.endpoint, data, { headers: this.headers });
PATCH	this.endpoint, data, { headers: this.headers });
PATCH	`\${this.endpoint}/\${quotationId}/approve`, data, { headers: this.headers });
PATCH	`\${this.endpoint}/\${quotationId}/submit`, {}, { headers: this.headers });
PATCH	`\${this.endpoint}/\${quotationId}/customer-response`, { status }, { headers: this.headers });
DELETE	this.endpoint, {

frontend/src/app/services/route-guard.ts

Shared frontend utility or auth/permission behavior; inspect its source for non-HttpClient responsibilities and indirect calls.

frontend/src/app/services/sales-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
GET	`\${this.endpoint}/\${salesId}`);

POST	this.endpoint, data, {
POST	`\${appConfig.apiUrl}/customer/\${customerId}/invoice`, data, {
PATCH	this.endpoint, data, {
DELETE	this.endpoint, {

frontend/src/app/services/service-ticket-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
POST	this.endpoint, data, {
PATCH	this.endpoint, data, {
DELETE	this.endpoint, {

frontend/src/app/services/snackbar.ts

Shared frontend utility or auth/permission behavior; inspect its source for non-HttpClient responsibilities and indirect calls.

frontend/src/app/services/stock-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
GET	`\${this.endpoint}/ledger\${suffix}`);
PATCH	`\${this.endpoint}/settings`, data, { headers: this.headers });
POST	`\${this.endpoint}/adjust`, data, { headers: this.headers });

frontend/src/app/services/supplier-payment-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
POST	this.endpoint, data, {
PATCH	this.endpoint, data, {
DELETE	this.endpoint, {

frontend/src/app/services/supplier-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
------	--------------------

GET	this.endpoint);
GET	`\${this.endpoint}/\${supplierId}`);
GET	`\${this.url}/location/states`);
GET	`\${this.url}/location/districts/\${stateId}`);
POST	this.endpoint, data, {
PATCH	this.endpoint, data, {
DELETE	this.endpoint, {

frontend/src/app/services/transaction-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint, { params });
POST	this.endpoint, data);
PATCH	`\${this.endpoint}/\${transactionId}/approve`, {});
DELETE	`\${this.endpoint}/\${transactionId}`);

frontend/src/app/services/user-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
POST	`\${this.url}/user/login`, data,{
POST	`\${this.url}/user/signup`, data,{
PATCH	`\${this.url}/user/editUser`, data,{
DELETE	`\${this.url}/user/deleteUser`, {
GET	`\${this.url}/user/get`);
POST	`\${this.url}/user/forgotPassword`, data,{
POST	`\${this.url}/user/changePassword`,data,{

frontend/src/app/services/warranty-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
POST	this.endpoint, data, {
PATCH	this.endpoint, data, {
DELETE	this.endpoint, {

frontend/src/app/services/work-order-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
GET	`\${this.endpoint}/\${workOrderId}`);
GET	`\${this.endpoint}/next-no`);
GET	`\${this.endpoint}/materials`);
POST	this.endpoint, data, { headers: this.headers });
PATCH	this.endpoint, data, { headers: this.headers });
DELETE	`\${this.endpoint}/\${workOrderId}`, { headers: this.headers });
POST	`\${this.endpoint}/\${workOrderId}/review`, { action }, { headers: this.headers });
POST	`\${this.endpoint}/\${workOrderId}/material-issue`, data, { headers: this.headers });
POST	`\${this.endpoint}/\${workOrderId}/material-issues/submit`, { rows }, { headers: this.headers });
POST	`\${this.endpoint}/\${workOrderId}/material-issues/review`, { action }, { headers: this.headers });
POST	`\${this.endpoint}/\${workOrderId}/material-return`, data, { headers: this.headers });
POST	`\${this.endpoint}/\${workOrderId}/create-invoice`, data, { headers: this.headers });

frontend/src/app/services/workflow-services.ts

Request expressions copied from current source. Local variable/template expressions are preserved; inspect the service definition for endpoint base and exact method signature.

HTTP	Request expression
GET	this.endpoint);
POST	`\${this.endpoint}/\${workflowId}/approve`, data);

Editing a frontend module

Find the active route, open its .ts/.html/.scss files, identify injected services and reactive/template form fields, then map each service request to the backend router and handler. Keep display labels separate from IDs. Preserve loading/error/snackbar behavior. For new CRUD fields, update form defaults, edit hydration, request payload, backend validation and SQL together. Avoid old copies, category experiments and no-need-trash folders unless the active import graph uses them.

5. Schema dictionary and exact ID relationships

Each table below lists its selected CREATE source, primary key, full column declarations and constraints. Selection uses the production schema baseline first; missing tables come from other migration/runtime definitions. Extra or alternative runtime columns are in Section 6. Refer to each relationship kind: Declared FK means found in a repository CREATE definition, not proof of a constraint on the live database; Inferred ID link means a candidate based on matching a unique PK name.

audit_log

Source: backend/migrations/production_schema.sql:9. Primary key: audit_id

Column	SQL declaration
--------	-----------------

audit_id	int NOT NULL AUTO_INCREMENT
user_id	int DEFAULT NULL
module	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
action	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL
table_name	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
record_id	int DEFAULT NULL
old_values	json DEFAULT NULL
new_values	json DEFAULT NULL
ip_address	varchar(45) COLLATE utf8mb4_unicode_ci DEFAULT NULL
browser_info	varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL
change_reason	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`audit_id`)
KEY `idx_user_id` (`user_id`)
KEY `idx_module` (`module`)
KEY `idx_table_name` (`table_name`)
KEY `idx_created_at` (`created_at`)
KEY `idx_record` (`table_name`,`record_id`)
CONSTRAINT `audit_log_ibfk_1` FOREIGN KEY (`user_id`) REFERENCES `users` (`user_id`) ON DELETE SET NULL

Child column -> parent column	Evidence	Delete/update rule or caveat
user_id -> users.user_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

brands

Source: backend/migrations/production_schema.sql:32. Primary key: brand_id

Column	SQL declaration
brand_id	int NOT NULL AUTO_INCREMENT
brand_name	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
brand_code	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
description	varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL
logo_url	varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`brand\_id`)

UNIQUE KEY `brand\_name` (`brand\_name`)

UNIQUE KEY `brand\_code` (`brand\_code`)

KEY `idx\_brand\_name` (`brand\_name`)

KEY `idx\_active` (`is\_active`)

cable_approval_groups

Source: backend/migrations/production_schema.sql:49. Primary key: approval_group_id

Column	SQL declaration
approval_group_id	bigint NOT NULL AUTO_INCREMENT
approval_group_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
group_type	enum('NEW_CUSTOMER_ONBOARDING','CUSTOMER_UPDATE','STB_UPDATE','CONNECTION_UPDATE','MATE' COLLATE utf8mb4_unicode_ci NOT NULL
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
requested_by_user_id	int DEFAULT NULL
approved_by_user_id	int DEFAULT NULL
requested_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
approved_at	timestamp NULL DEFAULT NULL
rejected_reason	text COLLATE utf8mb4_unicode_ci
remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`approval\_group\_id`)

UNIQUE KEY `uk\_cable\_approval\_group\_no` (`approval\_group\_no`)

KEY `fk\_cable\_approval\_groups\_requested\_by` (`requested\_by\_user\_id`)

KEY `fk\_cable\_approval\_groups\_approved\_by` (`approved\_by\_user\_id`)

KEY `idx\_cable\_approval\_group\_type` (`group\_type`)

KEY `idx\_cable\_approval\_group\_status` (`approval\_status`)

CONSTRAINT `fk\_cable\_approval\_groups\_approved\_by` FOREIGN KEY (`approved\_by\_user\_id`) REFERENCES `users` (`user\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_approval\_groups\_requested\_by` FOREIGN KEY (`requested\_by\_user\_id`) REFERENCES `users` (`user\_id`) ON DELETE SET NULL ON UPDATE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
-------------------------------	----------	------------------------------

approved_by_user_id -> users.user_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
requested_by_user_id -> users.user_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

cable_areas

Source: backend/migrations/production_schema.sql:73. Primary key: area_id

Column	SQL declaration
area_id	int NOT NULL AUTO_INCREMENT
network_id	int DEFAULT NULL
location_id	int NOT NULL
area_name	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`area\_id`)

UNIQUE KEY `uk\_cable\_network\_location\_area` (`network\_id`,`location\_id`,`area\_name`)

KEY `idx\_cable\_area\_location` (`location\_id`)

KEY `idx\_cable\_area\_active` (`is\_active`)

CONSTRAINT `fk\_cable\_areas\_location` FOREIGN KEY (`location\_id`) REFERENCES `cable\_locations` (`location\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_areas\_network` FOREIGN KEY (`network\_id`) REFERENCES `cable\_network\_master` (`network\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
location_id -> cable_locations.location_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE
network_id -> cable_network_master.network_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE

cable_connection_materials

Source: backend/migrations/production_schema.sql:90. Primary key: connection_material_id

Column	SQL declaration
connection_material_id	bigint NOT NULL AUTO_INCREMENT
approval_group_id	bigint DEFAULT NULL
connection_id	bigint NOT NULL
product_id	int DEFAULT NULL

material_id	int DEFAULT NULL
item_name	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL
qty	decimal(10,2) NOT NULL
unit	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PCS'
unit_rate	decimal(12,2) NOT NULL DEFAULT '0.00'
amount	decimal(12,2) NOT NULL DEFAULT '0.00'
issued_by_employee_id	int DEFAULT NULL
updated_by_employee_id	int DEFAULT NULL
updated_date	date DEFAULT NULL
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
created_by_user_id	int DEFAULT NULL
approved_by_user_id	int DEFAULT NULL
approved_at	timestamp NULL DEFAULT NULL
rejected_reason	text COLLATE utf8mb4_unicode_ci
remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`connection\_material\_id`)

KEY `fk\_cable\_materials\_issued\_by` (`issued\_by\_employee\_id`)

KEY `fk\_cable\_materials\_created\_by\_user` (`created\_by\_user\_id`)

KEY `fk\_cable\_materials\_approved\_by\_user` (`approved\_by\_user\_id`)

KEY `idx\_cable\_materials\_connection` (`connection\_id`)

KEY `idx\_cable\_materials\_approval\_group` (`approval\_group\_id`)

KEY `idx\_cable\_materials\_product` (`product\_id`)

KEY `idx\_cable\_materials\_material` (`material\_id`)

KEY `idx\_cable\_materials\_approval\_status` (`approval\_status`)

CONSTRAINT `fk\_cable\_materials\_approval\_group` FOREIGN KEY (`approval\_group\_id`) REFERENCES `cable\_approval\_groups` (`approval\_group\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_materials\_approved\_by\_user` FOREIGN KEY (`approved\_by\_user\_id`) REFERENCES `users` (`user\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_materials\_connection` FOREIGN KEY (`connection\_id`) REFERENCES `cable\_connections` (`connection\_id`) ON DELETE CASCADE ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_materials\_created\_by\_user` FOREIGN KEY (`created\_by\_user\_id`) REFERENCES `users` (`user\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_materials\_issued\_by` FOREIGN KEY (`issued\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_materials\_material\_master` FOREIGN KEY (`material\_id`) REFERENCES `material\_master` (`material\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_materials\_product` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `cable\_connection\_materials\_chk\_1` CHECK ((`qty` > 0))

CONSTRAINT `cable\_connection\_materials\_chk\_2` CHECK ((`unit\_rate` >= 0))

CONSTRAINT `cable\_connection\_materials\_chk\_3` CHECK ((`amount` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
approval_group_id -> cable_approval_groups.approval_group_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
approved_by_user_id -> users.user_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
connection_id -> cable_connections.connection_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
created_by_user_id -> users.user_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
issued_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
material_id -> material_master.material_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

cable_connection_sources

Source: backend/migrations/production_schema.sql:133. Primary key: source_id

Column	SQL declaration
source_id	int NOT NULL AUTO_INCREMENT
source_name	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`source\_id`)

UNIQUE KEY `uk\_cable\_source\_name` (`source\_name`)

KEY `idx\_cable\_source\_active` (`is\_active`)

cable_connections

Source: backend/migrations/production_schema.sql:145. Primary key: connection_id

Column	SQL declaration
connection_id	bigint NOT NULL AUTO_INCREMENT
approval_group_id	bigint DEFAULT NULL
cable_customer_id	bigint NOT NULL
connection_date	date NOT NULL DEFAULT (curdate())
disconnection_date	date DEFAULT NULL
connection_type	enum('NEW','RECONNECTION','SHIFTED','TRANSFERRED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'NEW'
old_network_id	int DEFAULT NULL
new_network_id	int DEFAULT NULL
old_door_no	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL
new_door_no	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL
old_location_id	int DEFAULT NULL
old_area_id	int DEFAULT NULL
old_street_id	int DEFAULT NULL
new_location_id	int DEFAULT NULL
new_area_id	int DEFAULT NULL
new_street_id	int DEFAULT NULL
old_address	varchar(500) COLLATE utf8mb4_unicode_ci DEFAULT NULL
new_address	varchar(500) COLLATE utf8mb4_unicode_ci DEFAULT NULL
connected_by_employee_id	int DEFAULT NULL
entered_by_employee_id	int DEFAULT NULL
connection_charge	decimal(12,2) NOT NULL DEFAULT '0.00'
connection_discount	decimal(12,2) NOT NULL DEFAULT '0.00'
labour_service_charge	decimal(12,2) NOT NULL DEFAULT '0.00'
status	enum('ACTIVE','DISCONNECTED','SHIFTED','TRANSFERRED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ACTIVE'
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
remarks	text COLLATE utf8mb4_unicode_ci
created_by_user_id	int DEFAULT NULL
approved_by_user_id	int DEFAULT NULL
approved_at	timestamp NULL DEFAULT NULL
rejected_reason	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`connection_id`)
KEY `fk_cable_connections_connected_by` (`connected_by_employee_id`)
KEY `fk_cable_connections_created_by_user` (`created_by_user_id`)
KEY `fk_cable_connections_approved_by_user` (`approved_by_user_id`)
KEY `idx_cable_connections_customer` (`cable_customer_id`)
KEY `idx_cable_connections_approval_group` (`approval_group_id`)
KEY `idx_cable_connections_date` (`connection_date`)
KEY `idx_cable_connections_type` (`connection_type`)
KEY `idx_cable_connections_status` (`status`)
KEY `idx_cable_connections_approval_status` (`approval_status`)
CONSTRAINT `fk_cable_connections_approval_group` FOREIGN KEY (`approval_group_id`) REFERENCES `cable_approval_groups` (`approval_group_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_connections_approved_by_user` FOREIGN KEY (`approved_by_user_id`) REFERENCES `users` (`user_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_connections_connected_by` FOREIGN KEY (`connected_by_employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_connections_created_by_user` FOREIGN KEY (`created_by_user_id`) REFERENCES `users` (`user_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_connections_customer` FOREIGN KEY (`cable_customer_id`) REFERENCES `cable_tv_customers` (`cable_customer_id`) ON DELETE CASCADE ON UPDATE CASCADE
CONSTRAINT `cable_connections_chk_1` CHECK ((`connection_charge` >= 0))
CONSTRAINT `cable_connections_chk_2` CHECK ((`labour_service_charge` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
approval_group_id -> cable_approval_groups.approval_group_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
approved_by_user_id -> users.user_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
connected_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
created_by_user_id -> users.user_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE

cable_customer_account_payments

Source: backend/migrations/production_schema.sql:198. Primary key: payment_id

Column	SQL declaration
payment_id	bigint NOT NULL AUTO_INCREMENT

account_id	bigint NOT NULL
cash_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
online_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
received_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
paid_date	date NOT NULL
received_date	date NOT NULL
due_date	date DEFAULT NULL
balance_after_payment	decimal(12,2) NOT NULL DEFAULT '0.00'
payment_status	enum('PARTIAL','PAID') NOT NULL
received_by_user_id	int NOT NULL
received_by_employee_id	int DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`payment\_id`)

KEY `idx\_cable\_account\_payments\_account` (`account\_id`)

KEY `idx\_cable\_account\_payments\_paid\_date` (`paid\_date`)

CONSTRAINT `fk\_cable\_account\_payments\_account` FOREIGN KEY (`account\_id`) REFERENCES `cable\_customer\_accounts` (`account\_id`) ON DELETE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
account_id -> cable_customer_accounts.account_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE

cable_customer_accounts

Source: backend/migrations/production_schema.sql:219. Primary key: account_id

Column	SQL declaration
account_id	bigint NOT NULL AUTO_INCREMENT
approval_group_id	bigint NOT NULL
cable_customer_id	bigint NOT NULL
stb_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
connection_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
labor_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
material_cost	decimal(12,2) NOT NULL DEFAULT '0.00'
material_discount	decimal(12,2) NOT NULL DEFAULT '0.00'
subscription_amount	decimal(12,2) NOT NULL DEFAULT '0.00'

sub_total	decimal(12,2) NOT NULL DEFAULT '0.00'
discount	decimal(12,2) NOT NULL DEFAULT '0.00'
overall_discount	decimal(12,2) NOT NULL DEFAULT '0.00'
grand_total	decimal(12,2) NOT NULL DEFAULT '0.00'
customer_paid_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
office_received_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
office_balance_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
balance_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
due_date	date DEFAULT NULL
account_status	enum('PENDING','PARTIAL','PAID','RECEIVED') NOT NULL DEFAULT 'PENDING'
received_by_user_id	int DEFAULT NULL
received_by_employee_id	int DEFAULT NULL
received_at	timestamp NULL DEFAULT NULL
approval_status	enum('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	int DEFAULT NULL
approved_by_user_id	int DEFAULT NULL
approved_at	timestamp NULL DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`account\_id`)

KEY `idx\_cable\_customer\_accounts\_customer` (`cable\_customer\_id`)

KEY `idx\_cable\_customer\_accounts\_status` (`account\_status`)

KEY `fk\_cable\_accounts\_approval\_group` (`approval\_group\_id`)

CONSTRAINT `fk\_cable\_accounts\_approval\_group` FOREIGN KEY (`approval\_group\_id`) REFERENCES `cable\_approval\_groups` (`approval\_group\_id`)

CONSTRAINT `fk\_cable\_accounts\_customer` FOREIGN KEY (`cable\_customer\_id`) REFERENCES `cable\_tv\_customers` (`cable\_customer\_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
approval_group_id -> cable_approval_groups.approval_group_id	Declared FK; backend/migrations/ production_schema.sql	No explicit rule in source definition.
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/ production_schema.sql	No explicit rule in source definition.

cable_customer_packages

Source: backend/migrations/production_schema.sql:257. Primary key: customer_package_id

Column	SQL declaration
customer_package_id	bigint NOT NULL AUTO_INCREMENT
approval_group_id	bigint DEFAULT NULL
cable_customer_id	bigint NOT NULL
package_id	int NOT NULL
package_type	enum('ADDON','ALACARTE','BROADCASTER') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ADDON'
package_price	decimal(12,2) NOT NULL DEFAULT '0.00'
start_date	date NOT NULL DEFAULT (curdate())
end_date	date DEFAULT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
updated_by_employee_id	int DEFAULT NULL
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
removal_status	enum('NONE','PENDING','APPROVED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'NONE'
removal_requested_at	datetime DEFAULT NULL
created_by_user_id	int DEFAULT NULL
approved_by_user_id	int DEFAULT NULL
approved_at	timestamp NULL DEFAULT NULL
rejected_reason	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`customer\_package\_id`)

KEY `fk\_cable\_customer\_packages\_created\_by\_user` (`created\_by\_user\_id`)

KEY `fk\_cable\_customer\_packages\_approved\_by\_user` (`approved\_by\_user\_id`)

KEY `idx\_cable\_customer\_packages\_customer` (`cable\_customer\_id`)

KEY `idx\_cable\_customer\_packages\_approval\_group` (`approval\_group\_id`)

KEY `idx\_cable\_customer\_packages\_package` (`package\_id`)

KEY `idx\_cable\_customer\_packages\_active` (`is\_active`)

KEY `idx\_cable\_customer\_packages\_approval\_status` (`approval\_status`)

KEY `idx\_cable\_customer\_packages\_type` (`package\_type`)

KEY `idx\_cable\_customer\_packages\_updated\_by` (`updated\_by\_employee\_id`)

CONSTRAINT `fk\_cable\_customer\_packages\_approval\_group` FOREIGN KEY (`approval\_group\_id`) REFERENCES `cable\_approval\_groups` (`approval\_group\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customer\_packages\_approved\_by\_user` FOREIGN KEY (`approved\_by\_user\_id`) REFERENCES `users` (`user\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customer\_packages\_created\_by\_user` FOREIGN KEY (`created\_by\_user\_id`) REFERENCES `users` (`user\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customer\_packages\_customer` FOREIGN KEY (`cable\_customer\_id`) REFERENCES `cable\_tv\_customers` (`cable\_customer\_id`) ON DELETE CASCADE ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customer\_packages\_package` FOREIGN KEY (`package\_id`) REFERENCES `cable\_package\_master` (`package\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customer\_packages\_updated\_by` FOREIGN KEY (`updated\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `cable\_customer\_packages\_chk\_1` CHECK ((`package\_price` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
approval_group_id -> cable_approval_groups.approval_group_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
approved_by_user_id -> users.user_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
created_by_user_id -> users.user_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
package_id -> cable_package_master.package_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE
updated_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

cable_customer_stb_accessories

Source: backend/migrations/production_schema.sql:297. Primary key: stb_accessory_id

Column	SQL declaration
stb_accessory_id	bigint NOT NULL AUTO_INCREMENT
approval_group_id	bigint DEFAULT NULL
cable_customer_id	bigint NOT NULL
customer_stb_id	bigint NOT NULL
product_id	int NOT NULL
movement_type	enum('ISSUE','RETURN') NOT NULL DEFAULT 'ISSUE'
accessory_name	varchar(200) NOT NULL
qty	decimal(10,2) NOT NULL DEFAULT '1.00'
unit	varchar(20) NOT NULL DEFAULT 'PCS'
issued_by_employee_id	int DEFAULT NULL
issued_date	date DEFAULT NULL
approval_status	enum('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'

created_by_user_id	int DEFAULT NULL
created_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`stb_accessory_id`)
KEY `idx_cable_stb_accessories_customer` (`cable_customer_id`)
KEY `idx_cable_stb_accessories_stb` (`customer_stb_id`)
KEY `idx_cable_stb_accessories_product` (`product_id`)
KEY `fk_cable_stb_accessories_group` (`approval_group_id`)
KEY `fk_cable_stb_accessories_employee` (`issued_by_employee_id`)
CONSTRAINT `fk_cable_stb_accessories_customer` FOREIGN KEY (`cable_customer_id`) REFERENCES `cable_tv_customers` (`cable_customer_id`)
CONSTRAINT `fk_cable_stb_accessories_employee` FOREIGN KEY (`issued_by_employee_id`) REFERENCES `employees` (`employee_id`)
CONSTRAINT `fk_cable_stb_accessories_group` FOREIGN KEY (`approval_group_id`) REFERENCES `cable_approval_groups` (`approval_group_id`)
CONSTRAINT `fk_cable_stb_accessories_product` FOREIGN KEY (`product_id`) REFERENCES `products` (`product_id`)
CONSTRAINT `fk_cable_stb_accessories_stb` FOREIGN KEY (`customer_stb_id`) REFERENCES `cable_customer_stbs` (`customer_stb_id`) ON DELETE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
issued_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
approval_group_id -> cable_approval_groups.approval_group_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
customer_stb_id -> cable_customer_stbs.customer_stb_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE

cable_customer_stbs

Source: backend/migrations/production_schema.sql:327. Primary key: customer_stb_id

Column	SQL declaration
customer_stb_id	bigint NOT NULL AUTO_INCREMENT
approval_group_id	bigint DEFAULT NULL
cable_customer_id	bigint NOT NULL
stb_master_id	bigint DEFAULT NULL

stb_type	enum('NEW','SERVICED','RETURNED','FAULT','DAMAGED','UPGRADE','REPLACED','EXCHANGE','CUSTOMER') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'NEW'
issue_mode	enum('FULL_SET','BOX_ONLY') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'BOX_ONLY'
installed_mso_id	int DEFAULT NULL
exchange_original_mso_id	int DEFAULT NULL
stb_no	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
stb_image_path	varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL
stb_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
stb_discount	decimal(12,2) NOT NULL DEFAULT '0.00'
labour_service_charge	decimal(12,2) NOT NULL DEFAULT '0.00'
refund_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
refund_payment_mode	enum('CASH','ONLINE','BANK','UPI','OTHER') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'CASH'
installed_by_employee_id	int DEFAULT NULL
entered_by_employee_id	int DEFAULT NULL
installed_date	date NOT NULL DEFAULT (curdate())
updated_date	date DEFAULT NULL
update_reason	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL
reason_remarks	varchar(500) COLLATE utf8mb4_unicode_ci DEFAULT NULL
status	enum('ACTIVE','RETRIEVED','FAULT','DISCONNECTED','UPGRADE','RETURNED','FAULTY','REPLACED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ACTIVE'
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
created_by_user_id	int DEFAULT NULL
approved_by_user_id	int DEFAULT NULL
approved_at	timestamp NULL DEFAULT NULL
rejected_reason	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`customer\_stb\_id`)

KEY `fk\_cable\_stbs\_installed\_mso` (`installed\_mso\_id`)

KEY `fk\_cable\_stbs\_exchange\_mso` (`exchange\_original\_mso\_id`)

KEY `fk\_cable\_stbs\_installed\_by` (`installed\_by\_employee\_id`)

KEY `fk\_cable\_stbs\_created\_by\_user` (`created\_by\_user\_id`)

KEY `fk\_cable\_stbs\_approved\_by\_user` (`approved\_by\_user\_id`)

KEY `idx\_cable\_stbs\_approval\_group` (`approval\_group\_id`)

KEY `idx_cable_stbs_customer` (`cable_customer_id`)
KEY `idx_cable_stbs_type` (`stb_type`)
KEY `idx_cable_stbs_status` (`status`)
KEY `idx_cable_stbs_approval_status` (`approval_status`)
KEY `idx_cable_stbs_master` (`stb_master_id`)
CONSTRAINT `fk_cable_stbs_approval_group` FOREIGN KEY (`approval_group_id`) REFERENCES `cable_approval_groups` (`approval_group_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_stbs_approved_by_user` FOREIGN KEY (`approved_by_user_id`) REFERENCES `users` (`user_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_stbs_created_by_user` FOREIGN KEY (`created_by_user_id`) REFERENCES `users` (`user_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_stbs_customer` FOREIGN KEY (`cable_customer_id`) REFERENCES `cable_tv_customers` (`cable_customer_id`) ON DELETE CASCADE ON UPDATE CASCADE
CONSTRAINT `fk_cable_stbs_exchange_mso` FOREIGN KEY (`exchange_original_mso_id`) REFERENCES `cable_mso_master` (`mso_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_stbs_installed_by` FOREIGN KEY (`installed_by_employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_stbs_installed_mso` FOREIGN KEY (`installed_mso_id`) REFERENCES `cable_mso_master` (`mso_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `fk_cable_stbs_master` FOREIGN KEY (`stb_master_id`) REFERENCES `cable_stb_master` (`stb_master_id`)
CONSTRAINT `cable_customer_stbs_chk_1` CHECK ((`stb_amount` >= 0))
CONSTRAINT `cable_customer_stbs_chk_2` CHECK ((`stb_discount` >= 0))
CONSTRAINT `cable_customer_stbs_chk_3` CHECK ((`labour_service_charge` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
approval_group_id -> cable_approval_groups.approval_group_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
approved_by_user_id -> users.user_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
created_by_user_id -> users.user_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
exchange_original_mso_id -> cable_mso_master.mso_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
installed_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
installed_mso_id -> cable_mso_master.mso_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
stb_master_id -> cable_stb_master.stb_master_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.

cable_locations

Source: backend/migrations/production_schema.sql:383. Primary key: location_id

Column	SQL declaration
location_id	int NOT NULL AUTO_INCREMENT
location_name	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL
post_short_code	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
city	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
pincode	varchar(10) COLLATE utf8mb4_unicode_ci DEFAULT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`location_id`)
UNIQUE KEY `uk_cable_location_city` (`location_name`,`city`)
UNIQUE KEY `uk_cable_location_short_code` (`post_short_code`)
UNIQUE KEY `uk_cable_location_pincode` (`pincode`)
KEY `idx_cable_location_city` (`city`)
KEY `idx_cable_location_active` (`is_active`)

cable_mso_master

Source: backend/migrations/production_schema.sql:401. Primary key: mso_id

Column	SQL declaration
mso_id	int NOT NULL AUTO_INCREMENT
mso_name	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`mso_id`)
UNIQUE KEY `uk_cable_mso_name` (`mso_name`)
KEY `idx_cable_mso_active` (`is_active`)

cable_network_master

Source: backend/migrations/production_schema.sql:413. Primary key: network_id

Column	SQL declaration
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network_id	int NOT NULL AUTO_INCREMENT
network_code	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL
network_name	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
customer_no_start	int DEFAULT NULL
customer_no_end	int DEFAULT NULL
remarks	varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`network\_id`)

UNIQUE KEY `uk\_cable\_network\_code` (`network\_code`)

KEY `idx\_cable\_network\_active` (`is\_active`)

cable_package_master

Source: backend/migrations/production_schema.sql:429. Primary key: package_id

Column	SQL declaration
package_id	int NOT NULL AUTO_INCREMENT
package_name	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL
package_type	enum('MSO_PACKAGE','ADDON','ALACARTE','BROADCAST') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'MSO_PACKAGE'
service_category	enum('CATV','INTERNET') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'CATV'
internet_network_type	enum('KRISHI','RAILWIRE') COLLATE utf8mb4_unicode_ci DEFAULT NULL
price	decimal(12,2) NOT NULL DEFAULT '0.00'
gst_percent	decimal(5,2) NOT NULL DEFAULT '0.00'
price_including_gst	decimal(12,2) NOT NULL DEFAULT '0.00'
description	text COLLATE utf8mb4_unicode_ci
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`package\_id`)

UNIQUE KEY `uk\_cable\_package\_name\_type` (`package\_name`,`package\_type`)

KEY `idx\_cable\_package\_type` (`package\_type`)

KEY `idx\_cable\_package\_active` (`is\_active`)

CONSTRAINT `cable\_package\_master\_chk\_1` CHECK ((`price` >= 0))

cable_stb_issue_master

Source: backend/migrations/production_schema.sql:450. Primary key: stb_issue_id

Column	SQL declaration
stb_issue_id	bigint NOT NULL AUTO_INCREMENT
stb_master_id	bigint NOT NULL
cable_customer_id	bigint NOT NULL
customer_stb_id	bigint DEFAULT NULL
issued_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
issued_by_employee_id	int DEFAULT NULL
issue_status	enum('ISSUED','RETURNED','CANCELLED') NOT NULL DEFAULT 'ISSUED'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`stb\_issue\_id`)

KEY `idx\_cable\_stb\_issue\_customer` (`cable\_customer\_id`)

KEY `idx\_cable\_stb\_issue\_stb` (`stb\_master\_id`)

KEY `fk\_cable\_stb\_issue\_customer\_stb` (`customer\_stb\_id`)

KEY `fk\_cable\_stb\_issue\_employee` (`issued\_by\_employee\_id`)

CONSTRAINT `fk\_cable\_stb\_issue\_customer` FOREIGN KEY (`cable\_customer\_id`) REFERENCES `cable\_tv\_customers` (`cable\_customer\_id`)

CONSTRAINT `fk\_cable\_stb\_issue\_customer\_stb` FOREIGN KEY (`customer\_stb\_id`) REFERENCES `cable\_customer\_stbs` (`customer\_stb\_id`)

CONSTRAINT `fk\_cable\_stb\_issue\_employee` FOREIGN KEY (`issued\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`)

CONSTRAINT `fk\_cable\_stb\_issue\_master` FOREIGN KEY (`stb\_master\_id`) REFERENCES `cable\_stb\_master` (`stb\_master\_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
customer_stb_id -> cable_customer_stbs.customer_stb_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
issued_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
stb_master_id -> cable_stb_master.stb_master_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.

cable_stb_master

Source: backend/migrations/production_schema.sql:471. Primary key: stb_master_id

Column	SQL declaration
stb_master_id	bigint NOT NULL AUTO_INCREMENT
stb_number	varchar(100) NOT NULL
box_type	enum('HD','SD') NOT NULL DEFAULT 'HD'
stock_type	enum('NEW','SERVICED','RETURNED','FAULT','DAMAGED','BURNT','NOT_SERVICEABLE') NOT NULL DEFAULT 'NEW'
mso_id	int DEFAULT NULL
stb_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
full_set_amount	decimal(12,2) NOT NULL DEFAULT '800.00'
assigned_employee_id	int DEFAULT NULL
status	enum('AVAILABLE','IN_SERVICE','NOT_SERVICEABLE','NOT_AVAILABLE') NOT NULL DEFAULT 'AVAILABLE'
updated_date	date DEFAULT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`stb_master_id`)
KEY `idx_cable_stb_master_status` (`status`)
KEY `idx_cable_stb_master_stock_type` (`stock_type`)
KEY `fk_cable_stb_master_mso` (`mso_id`)
KEY `idx_cable_stb_master_number` (`stb_number`)
CONSTRAINT `fk_cable_stb_master_mso` FOREIGN KEY (`mso_id`) REFERENCES `cable_mso_master` (`mso_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
mso_id -> cable_mso_master.mso_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
assigned_employee_id -> employees.employee_id	Declared FK; backend/migrations/create_cable_tv_module.sql	ON DELETE SET NULL ON UPDATE CASCADE

cable_streets

Source: backend/migrations/production_schema.sql:494. Primary key: street_id

Column	SQL declaration
street_id	int NOT NULL AUTO_INCREMENT
area_id	int NOT NULL
street_name	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL

is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`street\_id`)

UNIQUE KEY `uk\_cable\_area\_street` (`area\_id`, `street\_name`)

KEY `idx\_cable\_street\_area` (`area\_id`)

KEY `idx\_cable\_street\_active` (`is\_active`)

CONSTRAINT `fk\_cable\_streets\_area` FOREIGN KEY (`area\_id`) REFERENCES `cable\_areas` (`area\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
area_id -> cable_areas.area_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE

cable_subscription_payments

Source: backend/migrations/production_schema.sql:509. Primary key: subscription_payment_id

Column	SQL declaration
subscription_payment_id	bigint NOT NULL AUTO_INCREMENT
subscription_id	bigint NOT NULL
cable_customer_id	bigint NOT NULL
received_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
collected_date	date NOT NULL
payment_mode	varchar(30) DEFAULT NULL
payment_reference	varchar(150) DEFAULT NULL
received_by_employee_id	int DEFAULT NULL
comments	varchar(500) DEFAULT NULL
balance_after_payment	decimal(12,2) NOT NULL DEFAULT '0.00'
payment_status	enum('PARTIAL','PAID') NOT NULL
created_by_user_id	int DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`subscription\_payment\_id`)

KEY `idx\_cable\_subscription\_payment\_subscription` (`subscription\_id`)

KEY `idx\_cable\_subscription\_payment\_customer` (`cable\_customer\_id`)

KEY `idx\_cable\_subscription\_payment\_date` (`collected\_date`)

CONSTRAINT `fk\_cable\_subscription\_payment\_customer` FOREIGN KEY (`cable\_customer\_id`) REFERENCES `cable\_tv\_customers` (`cable\_customer\_id`) ON DELETE CASCADE

CONSTRAINT `fk\_cable\_subscription\_payment\_subscription` FOREIGN KEY (`subscription\_id`) REFERENCES `cable\_subscriptions` (`subscription\_id`) ON DELETE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE
subscription_id -> cable_subscriptions.subscription_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE

cable_subscriptions

Source: backend/migrations/production_schema.sql:532. Primary key: subscription_id

Column	SQL declaration
subscription_id	bigint NOT NULL AUTO_INCREMENT
approval_group_id	bigint DEFAULT NULL
cable_customer_id	bigint NOT NULL
customer_package_id	bigint NOT NULL
subscription_month	tinyint NOT NULL
subscription_year	smallint NOT NULL
days_in_month	int NOT NULL
billing_basis	enum('DAY','MONTH','YEAR') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'MONTH'
number_of_days_or_months	decimal(8,2) NOT NULL DEFAULT '1.00'
received_count	decimal(8,2) NOT NULL DEFAULT '1.00'
amount	decimal(12,2) NOT NULL DEFAULT '0.00'
paid_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
balance_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
collect_date	date DEFAULT NULL
start_date	date DEFAULT NULL
expiry_date	date DEFAULT NULL
collected_by_employee_id	int DEFAULT NULL
payment_mapped_employee_id	int DEFAULT NULL
payment_mode	enum('CASH','ONLINE','OFFICE','UPI','CARD','BANK','CHEQUE') COLLATE utf8mb4_unicode_ci DEFAULT NULL
payment_reference	varchar(150) COLLATE utf8mb4_unicode_ci DEFAULT NULL
payment_status	enum('PENDING','PARTIAL','PAID','CANCELLED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'

approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
remarks	text COLLATE utf8mb4_unicode_ci
created_by_user_id	int DEFAULT NULL
approved_by_user_id	int DEFAULT NULL
approved_at	timestamp NULL DEFAULT NULL
rejected_reason	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`subscription\_id`)

UNIQUE KEY `uk\_cable\_subscription\_customer\_package\_month`
(`cable\_customer\_id`,`customer\_package\_id`,`subscription\_month`,`subscription\_year`)

KEY `fk\_cable\_subscriptions\_collected\_by` (`collected\_by\_employee\_id`)

KEY `fk\_cable\_subscriptions\_created\_by\_user` (`created\_by\_user\_id`)

KEY `fk\_cable\_subscriptions\_approved\_by\_user` (`approved\_by\_user\_id`)

KEY `idx\_cable\_subscriptions\_customer` (`cable\_customer\_id`)

KEY `idx\_cable\_subscriptions\_approval\_group` (`approval\_group\_id`)

KEY `idx\_cable\_subscriptions\_package` (`customer\_package\_id`)

KEY `idx\_cable\_subscriptions\_month\_year` (`subscription\_month`,`subscription\_year`)

KEY `idx\_cable\_subscriptions\_collect\_date` (`collect\_date`)

KEY `idx\_cable\_subscriptions\_payment\_status` (`payment\_status`)

KEY `idx\_cable\_subscriptions\_approval\_status` (`approval\_status`)

CONSTRAINT `fk\_cable\_subscriptions\_approval\_group` FOREIGN KEY (`approval\_group\_id`) REFERENCES `cable\_approval\_groups`
(`approval\_group\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_subscriptions\_approved\_by\_user` FOREIGN KEY (`approved\_by\_user\_id`) REFERENCES `users` (`user\_id`)
ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_subscriptions\_collected\_by` FOREIGN KEY (`collected\_by\_employee\_id`) REFERENCES `employees`
(`employee\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_subscriptions\_created\_by\_user` FOREIGN KEY (`created\_by\_user\_id`) REFERENCES `users` (`user\_id`) ON
DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_subscriptions\_customer` FOREIGN KEY (`cable\_customer\_id`) REFERENCES `cable\_tv\_customers`
(`cable\_customer\_id`) ON DELETE CASCADE ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_subscriptions\_customer\_package` FOREIGN KEY (`customer\_package\_id`) REFERENCES
`cable\_customer\_packages` (`customer\_package\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

CONSTRAINT `cable\_subscriptions\_chk\_1` CHECK ((`subscription\_month` between 1 and 12))

CONSTRAINT `cable\_subscriptions\_chk\_2` CHECK ((`subscription\_year` >= 2000))

CONSTRAINT `cable\_subscriptions\_chk\_3` CHECK ((`days\_in\_month` between 28 and 31))

CONSTRAINT `cable\_subscriptions\_chk\_4` CHECK ((`number\_of\_days\_or\_months` > 0))

CONSTRAINT `cable\_subscriptions\_chk\_5` CHECK ((`amount` >= 0))

CONSTRAINT `cable\_subscriptions\_chk\_6` CHECK ((`paid\_amount` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
approval_group_id -> cable_approval_groups.approval_group_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
approved_by_user_id -> users.user_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
collected_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
created_by_user_id -> users.user_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
customer_package_id -> cable_customer_packages.customer_package_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE
payment_mapped_employee_id -> employees.employee_id	Declared FK; backend/migrations/ create_cable_tv_module.sql	ON DELETE SET NULL ON UPDATE CASCADE

cable_tv_complaint_attempts

Source: backend/migrations/production_schema.sql:589. Primary key: complaint_attempt_id

Column	SQL declaration
complaint_attempt_id	bigint NOT NULL AUTO_INCREMENT
complaint_id	bigint NOT NULL
attempt_no	int NOT NULL
status	enum('OPEN','IN_PROGRESS','HOLD','PENDING','COMPLETED') NOT NULL
assigned_employee_id	int DEFAULT NULL
start_time	datetime DEFAULT NULL
end_time	datetime DEFAULT NULL
reason	text
remedy	text
notes	text
entered_by_user_id	int DEFAULT NULL
entered_by_employee_id	int DEFAULT NULL
entered_at	datetime NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`complaint\_attempt\_id`)

KEY `idx\_cable\_tv\_attempt\_complaint` (`complaint\_id`)

KEY `fk\_cable\_tv\_attempt\_employee` (`assigned\_employee\_id`)

CONSTRAINT `fk\_cable\_tv\_attempt\_complaint` FOREIGN KEY (`complaint\_id`) REFERENCES `cable\_tv\_complaints` (`complaint\_id`)

CONSTRAINT `fk\_cable\_tv\_attempt\_employee` FOREIGN KEY (`assigned\_employee\_id`) REFERENCES `employees` (`employee\_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
complaint_id -> cable_tv_complaints.complaint_id	Declared FK; backend/migrations/ production_schema.sql	No explicit rule in source definition.
assigned_employee_id -> employees.employee_id	Declared FK; backend/migrations/ production_schema.sql	No explicit rule in source definition.

cable_tv_complaints

Source: backend/migrations/production_schema.sql:611. Primary key: complaint_id

Column	SQL declaration
complaint_id	bigint NOT NULL AUTO_INCREMENT
complaint_no	varchar(30) NOT NULL
complainant_type	enum('CATV','NET','CCTV','ANONYMOUS') NOT NULL DEFAULT 'CATV'
cable_customer_id	bigint DEFAULT NULL
service_customer_id	int DEFAULT NULL
anonymous_name	varchar(150) DEFAULT NULL
anonymous_mobile	varchar(20) DEFAULT NULL
reported_mobile	varchar(20) DEFAULT NULL
anonymous_address	varchar(500) DEFAULT NULL
nature_of_complaint	varchar(250) NOT NULL
complaint_description	text
status	enum('OPEN','IN_PROGRESS','HOLD','PENDING','COMPLETED') NOT NULL DEFAULT 'OPEN'
assigned_employee_id	int DEFAULT NULL
registered_by_user_id	int DEFAULT NULL
registered_by_employee_id	int DEFAULT NULL
registered_at	datetime NOT NULL DEFAULT CURRENT_TIMESTAMP
completed_at	datetime DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`complaint\_id`)

UNIQUE KEY `uk\_cable\_tv\_complaint\_no` (`complaint\_no`)

KEY `idx_cable_tv_complaint_customer` (`cable_customer_id`)
KEY `idx_cable_tv_complaint_status` (`status`)
KEY `idx_cable_tv_complaint_assigned` (`assigned_employee_id`)
KEY `idx_cable_tv_complaint_service_customer` (`service_customer_id`)
CONSTRAINT `fk_cable_tv_complaint_assigned` FOREIGN KEY (`assigned_employee_id`) REFERENCES `employees` (`employee_id`)
CONSTRAINT `fk_cable_tv_complaint_customer` FOREIGN KEY (`cable_customer_id`) REFERENCES `cable_tv_customers` (`cable_customer_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
assigned_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
service_customer_id -> customers.customer_id	Declared FK; backend/migrations/create_cable_tv_module.sql	No explicit rule in source definition.
internet_customer_id -> internet_customers.internet_customer_id	Inferred ID link; backend/migrations/production_schema.sql	Candidate only: same ID name as a unique primary key; confirm SQL join and live DDL.

cable_tv_customers

Source: backend/migrations/production_schema.sql:642. Primary key: cable_customer_id

Column	SQL declaration
cable_customer_id	bigint NOT NULL AUTO_INCREMENT
approval_group_id	bigint DEFAULT NULL
erp_customer_id	int DEFAULT NULL
network_id	int NOT NULL
network_type	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
customer_type	enum('REGULAR','BUSINESS','FREE','LEASE_LINE') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'REG'
topup_base_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
topup_gst_percent	decimal(5,2) NOT NULL DEFAULT '18.00'
recharge_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
legacy_customer_no	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL
network_customer_no	varchar(30) COLLATE utf8mb4_unicode_ci DEFAULT NULL
customer_code	int NOT NULL
full_name	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL
door_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
location_id	int NOT NULL
area_id	int NOT NULL

street_id	int NOT NULL
city	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
pincode	varchar(10) COLLATE utf8mb4_unicode_ci DEFAULT NULL
mobile_no	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL
aadhaar_no	varchar(12) COLLATE utf8mb4_unicode_ci DEFAULT NULL
alternate_mobile_no	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
source_id	int DEFAULT NULL
installed_by_employee_id	int DEFAULT NULL
labour_service_charge	decimal(12,2) NOT NULL DEFAULT '0.00'
status	enum('ACTIVE','INACTIVE','DISCONNECTED','SHIFTED','TRANSFERRED','RETRIEVED','FAULT','UPGRADE','F') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ACTIVE'
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
created_by_user_id	int DEFAULT NULL
approved_by_user_id	int DEFAULT NULL
approved_at	timestamp NULL DEFAULT NULL
rejected_reason	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`cable\_customer\_id`)

UNIQUE KEY `uk\_cable\_customer\_code` (`customer\_code`)

UNIQUE KEY `uk\_cable\_network\_legacy\_customer` (`network\_id`,`legacy\_customer\_no`)

KEY `fk\_cable\_customers\_erp\_customer` (`erp\_customer\_id`)

KEY `fk\_cable\_customers\_area` (`area\_id`)

KEY `fk\_cable\_customers\_street` (`street\_id`)

KEY `fk\_cable\_customers\_source` (`source\_id`)

KEY `fk\_cable\_customers\_installed\_by` (`installed\_by\_employee\_id`)

KEY `fk\_cable\_customers\_created\_by\_user` (`created\_by\_user\_id`)

KEY `fk\_cable\_customers\_approved\_by\_user` (`approved\_by\_user\_id`)

KEY `idx\_cable\_customer\_approval\_group` (`approval\_group\_id`)

KEY `idx\_cable\_customer\_name` (`full\_name`)

KEY `idx\_cable\_customer\_mobile` (`mobile\_no`)

KEY `idx\_cable\_customer\_network` (`network\_id`)

KEY `idx\_cable\_customer\_location\_area\_street` (`location\_id`,`area\_id`,`street\_id`)

KEY `idx\_cable\_customer\_status` (`status`)

KEY `idx\_cable\_customer\_approval\_status` (`approval\_status`)

CONSTRAINT `fk\_cable\_customers\_approval\_group` FOREIGN KEY (`approval\_group\_id`) REFERENCES `cable\_approval\_groups` (`approval\_group\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customers\_approved\_by\_user` FOREIGN KEY (`approved\_by\_user\_id`) REFERENCES `users` (`user\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customers\_area` FOREIGN KEY (`area\_id`) REFERENCES `cable\_areas` (`area\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customers\_created\_by\_user` FOREIGN KEY (`created\_by\_user\_id`) REFERENCES `users` (`user\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customers\_erp\_customer` FOREIGN KEY (`erp\_customer\_id`) REFERENCES `customers` (`customer\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customers\_installed\_by` FOREIGN KEY (`installed\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customers\_location` FOREIGN KEY (`location\_id`) REFERENCES `cable\_locations` (`location\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customers\_network` FOREIGN KEY (`network\_id`) REFERENCES `cable\_network\_master` (`network\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customers\_source` FOREIGN KEY (`source\_id`) REFERENCES `cable\_connection\_sources` (`source\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `fk\_cable\_customers\_street` FOREIGN KEY (`street\_id`) REFERENCES `cable\_streets` (`street\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

CONSTRAINT `cable\_tv\_customers\_chk\_1` CHECK ((`labour\_service\_charge` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
approval_group_id -> cable_approval_groups.approval_group_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
approved_by_user_id -> users.user_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
area_id -> cable_areas.area_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE
created_by_user_id -> users.user_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
erp_customer_id -> customers.customer_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
installed_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
location_id -> cable_locations.location_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE
network_id -> cable_network_master.network_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE
source_id -> cable_connection_sources.source_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
street_id -> cable_streets.street_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE

categories

Source: backend/migrations/production_schema.sql:707. Primary key: category_id

Column	SQL declaration
category_id	int NOT NULL AUTO_INCREMENT
category_name	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL
parent_id	int DEFAULT NULL COMMENT 'For hierarchical categories (NULL for root)'
level	int NOT NULL DEFAULT '1' COMMENT '1=Root, 2=Sub-category, etc.'
slug	varchar(200) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'URL-friendly identifier'
sort_order	smallint NOT NULL DEFAULT '0'
description	text COLLATE utf8mb4_unicode_ci
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`category_id`)
UNIQUE KEY `slug` (`slug`)
KEY `idx_parent_id` (`parent_id`)
KEY `idx_level` (`level`)
KEY `idx_slug` (`slug`)
KEY `idx_active` (`is_active`)
KEY `idx_parent_active` (`parent_id`, `is_active`)
CONSTRAINT `categories_ibfk_1` FOREIGN KEY (`parent_id`) REFERENCES `categories` (`category_id`) ON DELETE SET NULL ON UPDATE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
parent_id -> categories.category_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

customer_payments

Source: backend/migrations/production_schema.sql:729. Primary key: payment_id

Column	SQL declaration
payment_id	int NOT NULL AUTO_INCREMENT
customer_id	int NOT NULL
sales_id	int DEFAULT NULL COMMENT 'Against which invoice'
amount	decimal(12,2) NOT NULL
payment_date	date NOT NULL DEFAULT (curdate())
payment_mode	enum('CASH','CARD','UPI','BANK','CHEQUE','ONLINE') COLLATE utf8mb4_unicode_ci NOT NULL

reference_no	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Cheque/Receipt/Transaction number'
payment_against	enum('INVOICE','ADVANCE','ADJUSTMENT') COLLATE utf8mb4_unicode_ci DEFAULT 'INVOICE'
narration	text COLLATE utf8mb4_unicode_ci
received_by_employee_id	int DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`payment_id`)
KEY `received_by_employee_id` (`received_by_employee_id`)
KEY `idx_customer_id` (`customer_id`)
KEY `idx_sales_id` (`sales_id`)
KEY `idx_payment_date` (`payment_date`)
KEY `idx_reference_no` (`reference_no`)
CONSTRAINT `customer_payments_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customers` (`customer_id`) ON UPDATE CASCADE
CONSTRAINT `customer_payments_ibfk_2` FOREIGN KEY (`sales_id`) REFERENCES `sales_master` (`sales_id`) ON DELETE SET NULL
CONSTRAINT `customer_payments_ibfk_3` FOREIGN KEY (`received_by_employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE SET NULL
CONSTRAINT `customer_payments_chk_1` CHECK ((`amount` > 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
customer_id -> customers.customer_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE
sales_id -> sales_master.sales_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
received_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

customers

Source: backend/migrations/production_schema.sql:754. Primary key: customer_id

Column	SQL declaration
customer_id	int NOT NULL AUTO_INCREMENT
salutation	enum('Mr/Mrs/Ms','Mr.','Mrs.','Ms.','M/S') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'Mr/Mrs/Ms'
customer_name	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL
contact_person	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
phone	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL
alternate_phone	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL

email	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
gst_no	varchar(15) COLLATE utf8mb4_unicode_ci DEFAULT NULL
customer_type	enum('RETAIL','WHOLESALE','DEALER','CORPORATE','SERVICE') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'RETAIL'
marketing_employee_id	int DEFAULT NULL
referral_details	varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL
address	text COLLATE utf8mb4_unicode_ci NOT NULL
city_district	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
state	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
pincode	varchar(10) COLLATE utf8mb4_unicode_ci DEFAULT NULL
credit_limit	decimal(12,2) NOT NULL DEFAULT '0.00'
opening_balance	decimal(12,2) NOT NULL DEFAULT '0.00' COMMENT 'Customer credit balance'
outstanding_balance	decimal(12,2) NOT NULL DEFAULT '0.00'
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`customer_id`)
KEY `idx_customer_name` (`customer_name`)
KEY `idx_customer_type` (`customer_type`)
KEY `idx_city` (`city_district`)
KEY `idx_active` (`is_active`)
KEY `idx_email` (`email`)
KEY `idx_phone` (`phone`)
KEY `idx_salutation` (`salutation`)
KEY `idx_marketing_employee_id` (`marketing_employee_id`)
CONSTRAINT `fk_customers_marketing_employee` FOREIGN KEY (`marketing_employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `customers_chk_1` CHECK ((`credit_limit` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
marketing_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

employee_attendance

Source: backend/migrations/production_schema.sql:790. Primary key: attendance_id

Column	SQL declaration
attendance_id	int NOT NULL AUTO_INCREMENT
employee_id	int NOT NULL
attendance_date	date NOT NULL
status	enum('PRESENT','ABSENT','HALF_DAY','LEAVE','HOLIDAY') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PRESENT'
check_in	time DEFAULT NULL
check_out	time DEFAULT NULL
remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`attendance\_id`)

UNIQUE KEY `uk\_employee\_attendance\_date` (`employee\_id`, `attendance\_date`)

KEY `idx\_attendance\_date` (`attendance\_date`)

KEY `idx\_attendance\_status` (`status`)

CONSTRAINT `employee\_attendance\_ibfk\_1` FOREIGN KEY (`employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE CASCADE ON UPDATE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
employee_id -> employees.employee_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE

employee_salary

Source: backend/migrations/production_schema.sql:808. Primary key: salary_id

Column	SQL declaration
salary_id	int NOT NULL AUTO_INCREMENT
employee_id	int NOT NULL
company_name	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'TCV'
salary_month	tinyint NOT NULL
salary_year	smallint NOT NULL
period_start_date	date NOT NULL
period_end_date	date NOT NULL
salary_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
earnings_total	decimal(12,2) NOT NULL DEFAULT '0.00'
deductions_total	decimal(12,2) NOT NULL DEFAULT '0.00'
net_salary	decimal(12,2) NOT NULL DEFAULT '0.00'

status	enum('DRAFT','FINAL') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'FINAL'
remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`salary_id`)
UNIQUE KEY `uk_employee_salary_month` (`employee_id`,`salary_month`,`salary_year`)
KEY `idx_employee_id` (`employee_id`)
KEY `idx_salary_period` (`salary_year`,`salary_month`)
KEY `idx_period_start_date` (`period_start_date`)
CONSTRAINT `employee_salary_ibfk_1` FOREIGN KEY (`employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE CASCADE ON UPDATE CASCADE
CONSTRAINT `employee_salary_chk_1` CHECK ((`salary_month` between 1 and 12))
CONSTRAINT `employee_salary_chk_2` CHECK ((`salary_year` >= 2000))
CONSTRAINT `employee_salary_chk_3` CHECK ((`salary_amount` >= 0))
CONSTRAINT `employee_salary_chk_4` CHECK ((`earnings_total` >= 0))
CONSTRAINT `employee_salary_chk_5` CHECK ((`deductions_total` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
employee_id -> employees.employee_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE

employee_salary_items

Source: backend/migrations/production_schema.sql:838. Primary key: salary_item_id

Column	SQL declaration
salary_item_id	int NOT NULL AUTO_INCREMENT
salary_id	int NOT NULL
item_type	enum('EARNING','DEDUCTION') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'EARNING'
line_no	smallint NOT NULL
description	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL
qty	decimal(10,2) NOT NULL DEFAULT '1.00'
price	decimal(12,2) NOT NULL DEFAULT '0.00'
total	decimal(12,2) NOT NULL DEFAULT '0.00'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`salary_item_id`)
KEY `idx_salary_id` (`salary_id`)
KEY `idx_item_type` (`item_type`)
CONSTRAINT `employee_salary_items_ibfk_1` FOREIGN KEY (`salary_id`) REFERENCES `employee_salary` (`salary_id`) ON DELETE CASCADE ON UPDATE CASCADE
CONSTRAINT `employee_salary_items_chk_1` CHECK ((`qty` >= 0))
CONSTRAINT `employee_salary_items_chk_2` CHECK ((`price` >= 0))
CONSTRAINT `employee_salary_items_chk_3` CHECK ((`total` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
salary_id -> employee_salary.salary_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE

employees

Source: backend/migrations/production_schema.sql:858. Primary key: employee_id

Column	
employee_id	int NOT NULL AUTO_INCREMENT
employee_code	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL COMMENT 'Auto generated employee code, e.g. TCV1'
first_name	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
last_name	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
phone	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL
alternate_phone	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
email	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
designation	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
department	enum('ADMIN','ENGINEER','TECHNICAL','STAFF','SALES','PURCHASE','STORE','INSTALLATION','SERVICE','AC
date_of_birth	date DEFAULT NULL
qualification	varchar(150) COLLATE utf8mb4_unicode_ci DEFAULT NULL
photo_file_name	varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Uploaded photo file name'
photo_path	varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Uploaded photo relative path or URL'
spouse_or_parent_name	varchar(150) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Name of spouse or parent'
relationship	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Relationship to spouse/parent'
kids_details	text COLLATE utf8mb4_unicode_ci COMMENT 'Children names and details'
id_proof_type	enum('AADHAAR','PAN','VOTER_ID','PASSPORT','DRIVING_LICENSE','RATION_CARD','NREGA_JOB_CARD','B COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Accepted Indian identification proof type'
id_proof_name	varchar(150) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Name on identification proof'
id_proof_number	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Identification proof number'
joining_date	date NOT NULL

permanent_address	text COLLATE utf8mb4_unicode_ci
permanent_city_district	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
permanent_state	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
permanent_pincode	varchar(10) COLLATE utf8mb4_unicode_ci DEFAULT NULL
temporary_address	text COLLATE utf8mb4_unicode_ci
temporary_city_district	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
temporary_state	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
temporary_pincode	varchar(10) COLLATE utf8mb4_unicode_ci DEFAULT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`employee\_id`)

UNIQUE KEY `employee\_code` (`employee\_code`)

UNIQUE KEY `email` (`email`)

KEY `idx\_employee\_code` (`employee\_code`)

KEY `idx\_department` (`department`)

KEY `idx\_email` (`email`)

KEY `idx\_active` (`is\_active`)

KEY `idx\_joining\_date` (`joining\_date`)

finance_transactions

Source: backend/migrations/production_schema.sql:901. Primary key: finance_transaction_id

Column	SQL declaration
finance_transaction_id	bigint NOT NULL AUTO_INCREMENT
transaction_date	date NOT NULL
transaction_type	enum('DEBIT','CREDIT') NOT NULL
category	varchar(100) NOT NULL
amount	decimal(12,2) NOT NULL
payment_mode	enum('CASH','ONLINE','BANK','UPI','OTHER') NOT NULL DEFAULT 'CASH'
reference_no	varchar(100) DEFAULT NULL
description	varchar(500) DEFAULT NULL
instructed_by	varchar(150) DEFAULT NULL
bill_copy_available	enum('YES','NO') DEFAULT NULL
item_list	text

received_by	varchar(150) DEFAULT NULL
received_date	date DEFAULT NULL
approval_status	enum('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
approved_by_user_id	int DEFAULT NULL
approved_at	datetime DEFAULT NULL
source_module	varchar(50) DEFAULT NULL
source_id	bigint DEFAULT NULL
created_by_user_id	int DEFAULT NULL
created_by_employee_id	int DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`finance\_transaction\_id`)

UNIQUE KEY `uk\_finance\_source` (`source\_module`, `source\_id`)

KEY `idx\_finance\_transaction\_date` (`transaction\_date`)

KEY `idx\_finance\_transaction\_type` (`transaction\_type`)

KEY `fk\_finance\_transaction\_user` (`created\_by\_user\_id`)

KEY `fk\_finance\_transaction\_employee` (`created\_by\_employee\_id`)

CONSTRAINT `fk\_finance\_transaction\_employee` FOREIGN KEY (`created\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`)

CONSTRAINT `fk\_finance\_transaction\_user` FOREIGN KEY (`created\_by\_user\_id`) REFERENCES `users` (`user\_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
created_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
created_by_user_id -> users.user_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
source_id -> cable_connection_sources.source_id	Inferred ID link; backend/migrations/production_schema.sql	Candidate only: same ID name as a unique primary key; confirm SQL join and live DDL.

internet_connection_materials

Source: backend/migrations/production_schema.sql:934. Primary key: internet_material_id

Column	SQL declaration
internet_material_id	bigint NOT NULL AUTO_INCREMENT
internet_customer_id	bigint NOT NULL
product_id	int DEFAULT NULL
item_name	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL
qty	decimal(10,2) NOT NULL DEFAULT '1.00'

unit	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PCS'
unit_rate	decimal(12,2) NOT NULL DEFAULT '0.00'
amount	decimal(12,2) NOT NULL DEFAULT '0.00'

Indexes and constraints

PRIMARY KEY (`internet\_material\_id`)

KEY `fk\_internet\_material\_customer` (`internet\_customer\_id`)

KEY `fk\_internet\_material\_product` (`product\_id`)

CONSTRAINT `fk\_internet\_material\_customer` FOREIGN KEY (`internet\_customer\_id`) REFERENCES `internet\_customers` (`internet\_customer\_id`) ON DELETE CASCADE

CONSTRAINT `fk\_internet\_material\_product` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON DELETE SET NULL

Child column -> parent column	Evidence	Delete/update rule or caveat
internet_customer_id -> internet_customers.internet_customer_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE
product_id -> products.product_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL

internet_connections

Source: backend/migrations/production_schema.sql:951. Primary key: internet_connection_id

Column	SQL declaration
internet_connection_id	bigint NOT NULL AUTO_INCREMENT
internet_customer_id	bigint NOT NULL
connection_date	date NOT NULL
connection_type	enum('NEW','RECONNECTION','LOCATION_CHANGE','DISCONNECT') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'NEW'
connection_status	enum('ACTIVE','DISCONNECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ACTIVE'
installed_by_employee_id	int DEFAULT NULL
old_address	varchar(500) COLLATE utf8mb4_unicode_ci DEFAULT NULL
new_address	varchar(500) COLLATE utf8mb4_unicode_ci DEFAULT NULL
new_door_no	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL
new_location_id	int DEFAULT NULL
new_area_id	int DEFAULT NULL
new_street_id	int DEFAULT NULL
connection_charge	decimal(12,2) NOT NULL DEFAULT '0.00'
connection_discount	decimal(12,2) NOT NULL DEFAULT '0.00'
labour_service_charge	decimal(12,2) NOT NULL DEFAULT '0.00'

remarks	varchar(500) COLLATE utf8mb4_unicode_ci DEFAULT NULL
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'APPROVED'

Indexes and constraints

PRIMARY KEY (`internet\_connection\_id`)

KEY `fk\_internet\_connection\_customer` (`internet\_customer\_id`)

CONSTRAINT `fk\_internet\_connection\_customer` FOREIGN KEY (`internet\_customer\_id`) REFERENCES `internet\_customers` (`internet\_customer\_id`) ON DELETE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
internet_customer_id -> internet_customers.internet_customer_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE

internet_customer_account_payments

Source: backend/migrations/production_schema.sql:975. Primary key: internet_payment_id

Column	SQL declaration
internet_payment_id	bigint NOT NULL AUTO_INCREMENT
internet_account_id	bigint NOT NULL
cash_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
online_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
received_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
paid_date	date NOT NULL
received_date	date NOT NULL
due_date	date DEFAULT NULL
balance_after_payment	decimal(12,2) NOT NULL DEFAULT '0.00'
payment_status	enum('PARTIAL','PAID') COLLATE utf8mb4_unicode_ci NOT NULL
received_by_user_id	int DEFAULT NULL
received_by_employee_id	int DEFAULT NULL
created_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`internet\_payment\_id`)

KEY `fk\_internet\_account\_payment` (`internet\_account\_id`)

CONSTRAINT `fk\_internet\_account\_payment` FOREIGN KEY (`internet\_account\_id`) REFERENCES `internet\_customer\_accounts` (`internet\_account\_id`) ON DELETE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
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internet_account_id ->
internet_customer_accounts.internet_account_id

Declared FK; backend/migrations/
production_schema.sql

ON DELETE
CASCADE

internet_customer_accounts

Source: backend/migrations/production_schema.sql:995. Primary key: internet_account_id

Column	SQL declaration
internet_account_id	bigint NOT NULL AUTO_INCREMENT
internet_customer_id	bigint NOT NULL
account_source	enum('LEGACY','CONNECTION','ROUTER','PACKAGE','SUBSCRIPTION') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'LEGACY'
router_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
router_discount	decimal(12,2) NOT NULL DEFAULT '0.00'
connection_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
labor_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
material_cost	decimal(12,2) NOT NULL DEFAULT '0.00'
material_discount	decimal(12,2) NOT NULL DEFAULT '0.00'
subscription_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
overall_discount	decimal(12,2) NOT NULL DEFAULT '0.00'
grand_total	decimal(12,2) NOT NULL DEFAULT '0.00'
customer_paid_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
office_received_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
office_balance_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
balance_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
due_date	date DEFAULT NULL
account_status	enum('PENDING','PARTIAL','PAID') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
created_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`internet\_account\_id`)

KEY `fk\_internet\_account\_customer` (`internet\_customer\_id`)

CONSTRAINT `fk\_internet\_account\_customer` FOREIGN KEY (`internet\_customer\_id`) REFERENCES `internet\_customers` (`internet\_customer\_id`) ON DELETE CASCADE

Child column -> parent column

Evidence

Delete/update rule or
caveat

internet_customer_id ->
internet_customers.internet_customer_id

Declared FK; backend/migrations/
production_schema.sql

ON DELETE CASCADE

internet_customer_complaints

Source: backend/migrations/production_schema.sql:1023. Primary key: internet_complaint_id

Column	SQL declaration
internet_complaint_id	bigint NOT NULL AUTO_INCREMENT
internet_customer_id	bigint NOT NULL
complaint_date	date NOT NULL
subject	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL
description	varchar(1000) COLLATE utf8mb4_unicode_ci DEFAULT NULL
complaint_status	enum('OPEN','IN_PROGRESS','RESOLVED','CLOSED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'OPEN'
created_by_user_id	int DEFAULT NULL
created_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`internet\_complaint\_id`)

KEY `idx\_internet\_complaint\_customer` (`internet\_customer\_id`)

CONSTRAINT `fk\_internet\_complaint\_customer` FOREIGN KEY (`internet\_customer\_id`) REFERENCES `internet\_customers` (`internet\_customer\_id`) ON DELETE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
internet_customer_id -> internet_customers.internet_customer_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE

internet_customer_packages

Source: backend/migrations/production_schema.sql:1038. Primary key: internet_customer_package_id

Column	SQL declaration
internet_customer_package_id	bigint NOT NULL AUTO_INCREMENT
internet_customer_id	bigint NOT NULL
package_id	int NOT NULL
package_price	decimal(12,2) NOT NULL DEFAULT '0.00'
start_date	date NOT NULL
end_date	date NOT NULL
is_active	tinyint(1) NOT NULL DEFAULT '1'
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'APPROVED'

updated_by_employee_id	int DEFAULT NULL
created_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`internet_customer_package_id`)
KEY `idx_internet_package_customer` (`internet_customer_id`)
KEY `fk_internet_package_master` (`package_id`)
CONSTRAINT `fk_internet_package_customer` FOREIGN KEY (`internet_customer_id`) REFERENCES `internet_customers` (`internet_customer_id`) ON DELETE CASCADE
CONSTRAINT `fk_internet_package_master` FOREIGN KEY (`package_id`) REFERENCES `cable_package_master` (`package_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
internet_customer_id -> internet_customers.internet_customer_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE
package_id -> cable_package_master.package_id	Declared FK; backend/migrations/ production_schema.sql	No explicit rule in source definition.
package_id -> internet_package_master.package_id	Declared FK; backend/controller/ internetCustomerController.js	No explicit rule in source definition.

internet_customer_routers

Source: backend/migrations/production_schema.sql:1058. Primary key: internet_router_id

Column	SQL declaration
internet_router_id	bigint NOT NULL AUTO_INCREMENT
internet_customer_id	bigint NOT NULL
router_type	enum('NEW','SERVICED','RETURNED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'NEW'
usage_category	enum('CUSTOMER_PAID','FREE_USE') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'CUSTOMER_PAID'
product_id	int NOT NULL
hsn_code	varchar(30) COLLATE utf8mb4_unicode_ci DEFAULT NULL
qty	decimal(10,2) NOT NULL DEFAULT '1.00'
unit	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PCS'
rate	decimal(12,2) NOT NULL DEFAULT '0.00'
discount	decimal(12,2) NOT NULL DEFAULT '0.00'
amount	decimal(12,2) NOT NULL DEFAULT '0.00'
remarks	varchar(500) COLLATE utf8mb4_unicode_ci DEFAULT NULL
update_reason	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL
reason_remarks	varchar(500) COLLATE utf8mb4_unicode_ci DEFAULT NULL

router_status	enum('ACTIVE','DISCONNECTED','FAULT','DAMAGED','RETURNED','REPLACED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ACTIVE'
returned_router_qty	decimal(10,2) NOT NULL DEFAULT '0.00'
returned_adapter_product_id	int DEFAULT NULL
returned_adapter_qty	decimal(10,2) NOT NULL DEFAULT '0.00'
refund_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
refund_payment_mode	enum('CASH','ONLINE','OTHER') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'CASH'
refund_status	enum('NOT_APPLICABLE','PENDING') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'NOT_APPLICABLE'
updated_by_employee_id	int DEFAULT NULL
stock_processed	tinyint(1) NOT NULL DEFAULT '0'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'APPROVED'

Indexes and constraints

PRIMARY KEY (`internet\_router\_id`)

KEY `idx\_internet\_router\_customer` (`internet\_customer\_id`)

KEY `fk\_internet\_router\_product` (`product\_id`)

CONSTRAINT `fk\_internet\_router\_customer` FOREIGN KEY (`internet\_customer\_id`) REFERENCES `internet\_customers` (`internet\_customer\_id`) ON DELETE CASCADE

CONSTRAINT `fk\_internet\_router\_product` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
internet_customer_id -> internet_customers.internet_customer_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE
product_id -> products.product_id	Declared FK; backend/migrations/ production_schema.sql	No explicit rule in source definition.

internet_customers

Source: backend/migrations/production_schema.sql:1093. Primary key: internet_customer_id

Column	SQL declaration
internet_customer_id	bigint NOT NULL AUTO_INCREMENT
customer_code	int NOT NULL
legacy_customer_no	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
network_type	enum('KRISHI','RAILWIRE','DMNET') COLLATE utf8mb4_unicode_ci NOT NULL
full_name	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL

net_id	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL
network_password	varchar(255) COLLATE utf8mb4_unicode_ci DEFAULT NULL
door_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
location_id	int NOT NULL
area_id	int NOT NULL
street_id	int NOT NULL
state	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'Tamil Nadu'
city	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
pincode	varchar(10) COLLATE utf8mb4_unicode_ci DEFAULT NULL
mobile_no	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL
alternate_mobile_no	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
aadhaar_no	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
source_name	enum('Customer Approach Office','Direct','Customer Approach Engineer') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'Direct'
installed_by_employee_id	int DEFAULT NULL
installed_date	date NOT NULL
status	enum('ACTIVE','INACTIVE') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ACTIVE'
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
created_by_user_id	int DEFAULT NULL
created_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`internet_customer_id`)
UNIQUE KEY `uk_internet_customer_code` (`customer_code`)
UNIQUE KEY `uk_internet_net_id` (`net_id`)
KEY `idx_internet_customer_name` (`full_name`)
KEY `idx_internet_customer_mobile` (`mobile_no`)
KEY `idx_internet_legacy_customer_no` (`legacy_customer_no`)

Child column -> parent column	Evidence	Delete/update rule or caveat
location_id -> cable_locations.location_id	Inferred ID link; backend/migrations/production_schema.sql	Candidate only: same ID name as a unique primary key; confirm SQL join and live DDL.
area_id -> cable_areas.area_id	Inferred ID link; backend/migrations/production_schema.sql	Candidate only: same ID name as a unique primary key; confirm SQL join and live DDL.
street_id -> cable_streets.street_id	Inferred ID link; backend/migrations/production_schema.sql	Candidate only: same ID name as a unique primary key; confirm SQL join and live DDL.

internet_package_master

Source: backend/controller/internetCustomerController.js:76. Primary key: package_id

Column	SQL declaration
package_id	INT AUTO_INCREMENT PRIMARY KEY
package_code	VARCHAR(50) NULL
package_name	VARCHAR(255) NOT NULL
provider_category	VARCHAR(50) NOT NULL
price	DECIMAL(12,2) NOT NULL DEFAULT 0
gst_percent	DECIMAL(5,2) NOT NULL DEFAULT 18
price_including_gst	DECIMAL(12,2) NOT NULL DEFAULT 0
description	TEXT NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

UNIQUE KEY uk_internet_package_provider_code (provider_category, package_code)

INDEX idx_internet_package_provider_active (provider_category, is_active)

internet_router_material_issues

Source: backend/controller/internetRouterStock.js:9. Primary key: internet_router_id

Column	SQL declaration
internet_router_id	BIGINT NOT NULL PRIMARY KEY
material_movement_id	BIGINT NOT NULL
qty	DECIMAL(10,2) NOT NULL

Indexes and constraints

INDEX idx_router_material_issue (material_movement_id)

Child column -> parent column	Evidence	Delete/update rule or caveat
material_movement_id -> technician_material_movements.material_movement_id	Inferred ID link; backend/ controller/internetRouterStock.js	Candidate only: same ID name as a unique primary key; confirm SQL join and live DDL.

internet_subscriptions

Source: backend/migrations/production_schema.sql:1128. Primary key: internet_subscription_id

Column	SQL declaration
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internet_subscription_id	bigint NOT NULL AUTO_INCREMENT
internet_customer_id	bigint NOT NULL
internet_customer_package_id	bigint NOT NULL
subscription_month	int NOT NULL
subscription_year	int NOT NULL
billing_basis	enum('MONTH','DAYS','YEAR') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'MONTH'
period_value	decimal(10,2) NOT NULL DEFAULT '1.00'
period_count	decimal(10,4) NOT NULL DEFAULT '1.0000'
additional_months	int NOT NULL DEFAULT '0'
additional_days	int NOT NULL DEFAULT '0'
additional_years	int NOT NULL DEFAULT '0'
free_period_value	decimal(10,2) NOT NULL DEFAULT '0.00'
free_period_unit	enum('MONTH','DAYS','YEAR') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'MONTH'
start_date	date NOT NULL
end_date	date NOT NULL
collect_date	date DEFAULT NULL
collected_by_employee_id	int DEFAULT NULL
renewed_by	enum('LOGGED_IN_USER','EMPLOYEE','CUSTOMER','ADMIN') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'LOGGED_IN_USER'
payment_mode	enum('DASHBOARD','CASH','ONLINE','BANK_PAYMENT') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'DASHBOARD'
payment_reference	varchar(150) COLLATE utf8mb4_unicode_ci DEFAULT NULL
payment_mapped_employee_id	int DEFAULT NULL
cash_admin_locked	tinyint(1) NOT NULL DEFAULT '0'
renewed_by_employee_id	int DEFAULT NULL
amount	decimal(12,2) NOT NULL DEFAULT '0.00'
paid_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
balance_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
payment_status	enum('PENDING','PARTIAL','PAID') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'APPROVED'
created_at	timestamp NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`internet\_subscription\_id`)

KEY `idx\_internet\_subscription\_period` (`internet\_customer\_id`,`subscription\_month`,`subscription\_year`)

KEY `fk\_internet\_subscription\_package` (`internet\_customer\_package\_id`)

CONSTRAINT `fk\_internet\_subscription\_customer` FOREIGN KEY (`internet\_customer\_id`) REFERENCES `internet\_customers` (`internet\_customer\_id`) ON DELETE CASCADE

CONSTRAINT `fk\_internet\_subscription\_package` FOREIGN KEY (`internet\_customer\_package\_id`) REFERENCES `internet\_customer\_packages` (`internet\_customer\_package\_id`) ON DELETE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
internet_customer_id -> internet_customers.internet_customer_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE
internet_customer_package_id -> internet_customer_packages.internet_customer_package_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE

material_master

Source: backend/migrations/production_schema.sql:1166. Primary key: material_id

Column	SQL declaration
material_id	int NOT NULL AUTO_INCREMENT
product_id	int DEFAULT NULL COMMENT 'Optional linked inventory product'
material_code	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
material_name	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL
description	text COLLATE utf8mb4_unicode_ci
unit	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PCS'
standard_rate	decimal(12,2) NOT NULL DEFAULT '0.00'
gst_percent	decimal(5,2) NOT NULL DEFAULT '0.00'
is_active	tinyint(1) NOT NULL DEFAULT '1'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`material\_id`)

UNIQUE KEY `material\_code` (`material\_code`)

KEY `product\_id` (`product\_id`)

KEY `idx\_material\_code` (`material\_code`)

KEY `idx\_material\_name` (`material\_name`)

KEY `idx\_active` (`is\_active`)

CONSTRAINT `material\_master\_ibfk\_1` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `material\_master\_chk\_1` CHECK ((`standard\_rate` >= 0))

CONSTRAINT `material\_master\_chk\_2` CHECK ((`gst\_percent` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

notifications

Source: backend/migrations/production_schema.sql:1190. Primary key: notification_id

Column	SQL declaration
notification_id	int NOT NULL AUTO_INCREMENT
source_key	varchar(180) COLLATE utf8mb4_unicode_ci NOT NULL
notification_type	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
title	varchar(180) COLLATE utf8mb4_unicode_ci NOT NULL
message	varchar(500) COLLATE utf8mb4_unicode_ci NOT NULL
severity	enum('INFO','WARNING','CRITICAL') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'INFO'
reference_type	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL
reference_id	int DEFAULT NULL
reference_no	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
navigation_url	varchar(500) COLLATE utf8mb4_unicode_ci DEFAULT NULL
target_user_id	int DEFAULT NULL
is_read	tinyint(1) NOT NULL DEFAULT '0'
is_active	tinyint(1) NOT NULL DEFAULT '1'
resolved_at	timestamp NULL DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`notification_id`)
UNIQUE KEY `source_key` (`source_key`)
KEY `idx_active_read` (`is_active`,`is_read`)
KEY `idx_type_active` (`notification_type`,`is_active`)
KEY `idx_severity` (`severity`)
KEY `idx_notification_target` (`target_user_id`,`is_active`)

products

Source: backend/migrations/production_schema.sql:1216. Primary key: product_id

Column	SQL declaration
product_id	int NOT NULL AUTO_INCREMENT

product_name	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL
product_code	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL COMMENT 'SKU/Product code'
brand_id	int DEFAULT NULL COMMENT 'Brand foreign key'
category_id	int DEFAULT NULL COMMENT 'Category foreign key'
barcode	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
description	text COLLATE utf8mb4_unicode_ci
product_type	enum('MATERIAL','SERVICE','LABOR') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'MATERIAL'
purchase_price	decimal(12,2) NOT NULL DEFAULT '0.00'
selling_price	decimal(12,2) NOT NULL DEFAULT '0.00'
gst_percent	decimal(5,2) NOT NULL DEFAULT '0.00'
hsn_code	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'GST HSN code'
unit	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PCS' COMMENT 'Unit of measurement (PCS, BOX, etc.)'
reorder_level	decimal(10,2) NOT NULL DEFAULT '0.00'
status	enum('ACTIVE','INACTIVE','DISCONTINUED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ACTIVE'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`product\_id`)

UNIQUE KEY `product\_code` (`product\_code`)

UNIQUE KEY `uk\_product\_code` (`product\_code`)

UNIQUE KEY `barcode` (`barcode`)

KEY `idx\_brand\_id` (`brand\_id`)

KEY `idx\_category\_id` (`category\_id`)

KEY `idx\_product\_code` (`product\_code`)

KEY `idx\_status` (`status`)

KEY `idx\_reorder` (`reorder\_level`, `status`)

CONSTRAINT `products\_ibfk\_1` FOREIGN KEY (`brand\_id`) REFERENCES `brands` (`brand\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `products\_ibfk\_2` FOREIGN KEY (`category\_id`) REFERENCES `categories` (`category\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `products\_chk\_1` CHECK ((`purchase\_price` >= 0))

CONSTRAINT `products\_chk\_2` CHECK ((`selling\_price` >= 0))

CONSTRAINT `products\_chk\_3` CHECK (((`gst\_percent` >= 0) and (`gst\_percent` <= 100)))

CONSTRAINT `products\_chk\_4` CHECK ((`reorder\_level` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
brand_id -> brands.brand_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
category_id -> categories.category_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

purchase_items

Source: backend/migrations/production_schema.sql:1252. Primary key: purchase_item_id

Column	SQL declaration
purchase_item_id	int NOT NULL AUTO_INCREMENT
purchase_id	int NOT NULL
product_id	int NOT NULL
qty	decimal(10,2) NOT NULL
purchase_price	decimal(12,2) NOT NULL
discount_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_percent	decimal(5,2) DEFAULT '0.00'
tax_percent	decimal(5,2) NOT NULL DEFAULT '0.00'
tax_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
amount	decimal(12,2) NOT NULL COMMENT 'Line total'
received_qty	decimal(10,2) DEFAULT '0.00' COMMENT 'Actual qty received'
remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`purchase_item_id`)
KEY `idx_purchase_id` (`purchase_id`)
KEY `idx_product_id` (`product_id`)
CONSTRAINT `purchase_items_ibfk_1` FOREIGN KEY (`purchase_id`) REFERENCES `purchase_master` (`purchase_id`) ON DELETE CASCADE ON UPDATE CASCADE
CONSTRAINT `purchase_items_ibfk_2` FOREIGN KEY (`product_id`) REFERENCES `products` (`product_id`) ON UPDATE CASCADE
CONSTRAINT `purchase_items_chk_1` CHECK ((`qty` > 0))
CONSTRAINT `purchase_items_chk_2` CHECK ((`purchase_price` >= 0))
CONSTRAINT `purchase_items_chk_3` CHECK ((`discount_amount` >= 0))
CONSTRAINT `purchase_items_chk_4` CHECK ((`tax_percent` >= 0))
CONSTRAINT `purchase_items_chk_5` CHECK ((`tax_amount` >= 0))
CONSTRAINT `purchase_items_chk_6` CHECK ((`amount` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
purchase_id -> purchase_master.purchase_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
product_id -> products.product_id	Declared FK; backend/migrations/ production_schema.sql	ON UPDATE CASCADE

purchase_master

Source: backend/migrations/production_schema.sql:1280. Primary key: purchase_id

Column	SQL declaration
purchase_id	int NOT NULL AUTO_INCREMENT
purchase_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL COMMENT 'Purchase order number'
supplier_id	int NOT NULL
invoice_no	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Supplier invoice number'
invoice_date	date DEFAULT NULL
purchase_date	date NOT NULL DEFAULT (curdate())
total_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_percent	decimal(5,2) DEFAULT '0.00'
tax_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
net_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
paid_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
balance_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
purchase_status	enum('DRAFT','RECEIVED','COMPLETED','CANCELLED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'DRAFT'
payment_status	enum('PENDING','PARTIAL','PAID') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
approved_by_employee_id	int DEFAULT NULL
approved_at	timestamp NULL DEFAULT NULL
remarks	text COLLATE utf8mb4_unicode_ci
received_date	date DEFAULT NULL
created_by_employee_id	int DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`purchase\_id`)

UNIQUE KEY `purchase\_no` (`purchase\_no`)

UNIQUE KEY `uk_purchase_no` (`purchase_no`)
KEY `created_by_employee_id` (`created_by_employee_id`)
KEY `idx_supplier_id` (`supplier_id`)
KEY `idx_purchase_date` (`purchase_date`)
KEY `idx_purchase_status` (`purchase_status`)
KEY `idx_payment_status` (`payment_status`)
KEY `idx_supplier_status` (`supplier_id`, `purchase_status`)
CONSTRAINT `purchase_master_ibfk_1` FOREIGN KEY (`supplier_id`) REFERENCES `suppliers` (`supplier_id`) ON UPDATE CASCADE
CONSTRAINT `purchase_master_ibfk_2` FOREIGN KEY (`created_by_employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE SET NULL
CONSTRAINT `purchase_master_chk_1` CHECK ((`total_amount` >= 0))
CONSTRAINT `purchase_master_chk_2` CHECK ((`discount_amount` >= 0))
CONSTRAINT `purchase_master_chk_3` CHECK ((`tax_amount` >= 0))
CONSTRAINT `purchase_master_chk_4` CHECK ((`net_amount` >= 0))
CONSTRAINT `purchase_master_chk_5` CHECK ((`paid_amount` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
supplier_id -> suppliers.supplier_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE
created_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

quotation_items

Source: backend/migrations/production_schema.sql:1323. Primary key: quotation_item_id

Column	SQL declaration
quotation_item_id	int NOT NULL AUTO_INCREMENT
quotation_id	int NOT NULL
product_id	int DEFAULT NULL COMMENT 'NULL for custom items'
item_name	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL
description	text COLLATE utf8mb4_unicode_ci
qty	decimal(10,2) NOT NULL
selling_price	decimal(12,2) NOT NULL
discount_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_percent	decimal(5,2) DEFAULT '0.00'
tax_percent	decimal(5,2) NOT NULL DEFAULT '0.00'
tax_amount	decimal(12,2) NOT NULL DEFAULT '0.00'

amount	decimal(12,2) NOT NULL
notes	text COLLATE utf8mb4_unicode_ci
line_no	smallint DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`quotation_item_id`)
KEY `idx_quotation_id` (`quotation_id`)
KEY `idx_product_id` (`product_id`)
CONSTRAINT `quotation_items_ibfk_1` FOREIGN KEY (`quotation_id`) REFERENCES `quotation_master` (`quotation_id`) ON DELETE CASCADE ON UPDATE CASCADE
CONSTRAINT `quotation_items_ibfk_2` FOREIGN KEY (`product_id`) REFERENCES `products` (`product_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `quotation_items_chk_1` CHECK ((`qty` > 0))
CONSTRAINT `quotation_items_chk_2` CHECK ((`selling_price` >= 0))
CONSTRAINT `quotation_items_chk_3` CHECK ((`discount_amount` >= 0))
CONSTRAINT `quotation_items_chk_4` CHECK ((`tax_percent` >= 0))
CONSTRAINT `quotation_items_chk_5` CHECK ((`tax_amount` >= 0))
CONSTRAINT `quotation_items_chk_6` CHECK ((`amount` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
quotation_id -> quotation_master.quotation_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

quotation_master

Source: backend/migrations/production_schema.sql:1353. Primary key: quotation_id

Column	SQL declaration
quotation_id	int NOT NULL AUTO_INCREMENT
quotation_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
quotation_version	int NOT NULL DEFAULT '1'
quotation_date	date NOT NULL DEFAULT (curdate())
valid_until	date DEFAULT NULL COMMENT 'Quote expiry date'
customer_id	int NOT NULL
prepared_by_employee_id	int DEFAULT NULL
approved_by_employee_id	int DEFAULT NULL
requirement_details	text COLLATE utf8mb4_unicode_ci COMMENT 'Customer requirements/site details'

total_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_percent	decimal(5,2) DEFAULT '0.00'
cgst_percent	decimal(5,2) NOT NULL DEFAULT '0.00'
sgst_percent	decimal(5,2) NOT NULL DEFAULT '0.00'
tax_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
net_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
quotation_status	enum('DRAFT','SENT','APPROVED','ACCEPTED','CANCELLED','REJECTED','EXPIRED','CONVERTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'DRAFT'
sent_date	date DEFAULT NULL
approved_date	date DEFAULT NULL
rejected_reason	text COLLATE utf8mb4_unicode_ci
remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`quotation\_id`)

UNIQUE KEY `quotation\_no` (`quotation\_no`)

UNIQUE KEY `uk\_quotation\_no` (`quotation\_no`)

KEY `prepared\_by\_employee\_id` (`prepared\_by\_employee\_id`)

KEY `approved\_by\_employee\_id` (`approved\_by\_employee\_id`)

KEY `idx\_customer\_id` (`customer\_id`)

KEY `idx\_quotation\_date` (`quotation\_date`)

KEY `idx\_status` (`quotation\_status`)

KEY `idx\_valid\_until` (`valid\_until`)

KEY `idx\_customer\_status` (`customer\_id`, `quotation\_status`)

CONSTRAINT `quotation\_master\_ibfk\_1` FOREIGN KEY (`customer\_id`) REFERENCES `customers` (`customer\_id`) ON UPDATE CASCADE

CONSTRAINT `quotation\_master\_ibfk\_2` FOREIGN KEY (`prepared\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

CONSTRAINT `quotation\_master\_ibfk\_3` FOREIGN KEY (`approved\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

CONSTRAINT `quotation\_master\_chk\_1` CHECK ((`total\_amount` >= 0))

CONSTRAINT `quotation\_master\_chk\_2` CHECK ((`discount\_amount` >= 0))

CONSTRAINT `quotation\_master\_chk\_3` CHECK ((`tax\_amount` >= 0))

CONSTRAINT `quotation\_master\_chk\_4` CHECK ((`net\_amount` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
customer_id -> customers.customer_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE
prepared_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
approved_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

quotation_templates

Source: backend/migrations/create_quotation_templates.sql:1. Primary key: template_id

Column	SQL declaration
template_id	INT AUTO_INCREMENT PRIMARY KEY
template_name	VARCHAR(150) NOT NULL
items_json	JSON NOT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

UNIQUE KEY uk_quotation_template_name (template_name)

role_permissions

Source: backend/migrations/production_schema.sql:1397. Primary key: role, permission_key

Column	SQL declaration
role	enum('MANAGER','EMPLOYEE','SALES','SERVICE') NOT NULL
permission_key	varchar(80) NOT NULL
can_view	tinyint(1) NOT NULL DEFAULT '0'
can_create	tinyint(1) NOT NULL DEFAULT '0'
can_update	tinyint(1) NOT NULL DEFAULT '0'
can_delete	tinyint(1) NOT NULL DEFAULT '0'
updated_by	int DEFAULT NULL
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`role`,`permission\_key`)

KEY `fk\_role\_permissions\_updated\_by` (`updated\_by`)

CONSTRAINT `fk\_role\_permissions\_updated\_by` FOREIGN KEY (`updated\_by`) REFERENCES `users` (`user\_id`) ON DELETE SET NULL

Child column -> parent column	Evidence	Delete/update rule or caveat
updated_by -> users.user_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

sales_items

Source: backend/migrations/production_schema.sql:1412. Primary key: sales_item_id

Column	SQL declaration
sales_item_id	int NOT NULL AUTO_INCREMENT
sales_id	int NOT NULL
product_id	int DEFAULT NULL COMMENT 'NULL for custom items'
item_name	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL
description	text COLLATE utf8mb4_unicode_ci
qty	decimal(10,2) NOT NULL
selling_price	decimal(12,2) NOT NULL
discount_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_percent	decimal(5,2) DEFAULT '0.00'
tax_percent	decimal(5,2) NOT NULL DEFAULT '0.00'
tax_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
amount	decimal(12,2) NOT NULL
line_no	smallint DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`sales_item_id`)
KEY `idx_sales_id` (`sales_id`)
KEY `idx_product_id` (`product_id`)
CONSTRAINT `sales_items_ibfk_1` FOREIGN KEY (`sales_id`) REFERENCES `sales_master` (`sales_id`) ON DELETE CASCADE ON UPDATE CASCADE
CONSTRAINT `sales_items_ibfk_2` FOREIGN KEY (`product_id`) REFERENCES `products` (`product_id`) ON DELETE SET NULL ON UPDATE CASCADE
CONSTRAINT `sales_items_chk_1` CHECK ((`qty` > 0))
CONSTRAINT `sales_items_chk_2` CHECK ((`selling_price` >= 0))
CONSTRAINT `sales_items_chk_3` CHECK ((`discount_amount` >= 0))
CONSTRAINT `sales_items_chk_4` CHECK ((`tax_percent` >= 0))
CONSTRAINT `sales_items_chk_5` CHECK ((`tax_amount` >= 0))
CONSTRAINT `sales_items_chk_6` CHECK ((`amount` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
sales_id -> sales_master.sales_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
product_id -> products.product_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

sales_master

Source: backend/migrations/production_schema.sql:1441. Primary key: sales_id

Column	SQL declaration
sales_id	int NOT NULL AUTO_INCREMENT
invoice_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
invoice_date	date NOT NULL DEFAULT (curdate())
customer_id	int NOT NULL
quotation_id	int DEFAULT NULL COMMENT 'Source quotation (if any)'
work_order_id	int DEFAULT NULL COMMENT 'Associated work order (if any)'
total_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_percent	decimal(5,2) DEFAULT '0.00'
tax_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
net_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
paid_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
balance_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
payment_mode	enum('CASH','CARD','UPI','BANK','CHEQUE','CREDIT') COLLATE utf8mb4_unicode_ci DEFAULT 'CREDIT'
payment_status	enum('PENDING','PARTIAL','PAID','OVERDUE') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
sales_status	enum('DRAFT','COMPLETED','CANCELLED','RETURNED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'DRAFT'
due_date	date DEFAULT NULL COMMENT 'Payment due date'
remarks	text COLLATE utf8mb4_unicode_ci
created_by_employee_id	int DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`sales_id`)
UNIQUE KEY `invoice_no` (`invoice_no`)
UNIQUE KEY `uk_invoice_no` (`invoice_no`)

KEY `quotation_id` (`quotation_id`)
KEY `work_order_id` (`work_order_id`)
KEY `created_by_employee_id` (`created_by_employee_id`)
KEY `idx_customer_id` (`customer_id`)
KEY `idx_invoice_date` (`invoice_date`)
KEY `idx_sales_status` (`sales_status`)
KEY `idx_payment_status` (`payment_status`)
KEY `idx_due_date` (`due_date`)
KEY `idx_customer_status` (`customer_id`, `sales_status`)
KEY `idx_invoice_date_status` (`invoice_date`, `payment_status`)
CONSTRAINT `sales_master_ibfk_1` FOREIGN KEY (`customer_id`) REFERENCES `customers` (`customer_id`) ON UPDATE CASCADE
CONSTRAINT `sales_master_ibfk_2` FOREIGN KEY (`quotation_id`) REFERENCES `quotation_master` (`quotation_id`) ON DELETE SET NULL
CONSTRAINT `sales_master_ibfk_3` FOREIGN KEY (`work_order_id`) REFERENCES `work_orders` (`work_order_id`) ON DELETE SET NULL
CONSTRAINT `sales_master_ibfk_4` FOREIGN KEY (`created_by_employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE SET NULL
CONSTRAINT `sales_master_chk_1` CHECK ((`total_amount` >= 0))
CONSTRAINT `sales_master_chk_2` CHECK ((`discount_amount` >= 0))
CONSTRAINT `sales_master_chk_3` CHECK ((`tax_amount` >= 0))
CONSTRAINT `sales_master_chk_4` CHECK ((`net_amount` >= 0))
CONSTRAINT `sales_master_chk_5` CHECK ((`paid_amount` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
customer_id -> customers.customer_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE
quotation_id -> quotation_master.quotation_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
work_order_id -> work_orders.work_order_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
created_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

service_tickets

Source: backend/migrations/production_schema.sql:1488. Primary key: service_ticket_id

Column	SQL declaration
service_ticket_id	int NOT NULL AUTO_INCREMENT
ticket_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL

customer_id	int NOT NULL
product_id	int DEFAULT NULL COMMENT 'Product under service'
sales_id	int DEFAULT NULL COMMENT 'Related sales invoice'
assigned_to_employee_id	int DEFAULT NULL
complaint_details	text COLLATE utf8mb4_unicode_ci NOT NULL
service_status	enum('OPEN','ASSIGNED','IN_PROGRESS','RESOLVED','CLOSED','CANCELLED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'OPEN'
priority	enum('LOW','MEDIUM','HIGH','URGENT') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'MEDIUM'
opened_date	date NOT NULL DEFAULT (curdate())
closed_date	date DEFAULT NULL
resolution_notes	text COLLATE utf8mb4_unicode_ci
resolution_time_hours	int DEFAULT NULL COMMENT 'Time to resolve'
created_by_employee_id	int DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`service\_ticket\_id`)

UNIQUE KEY `ticket\_no` (`ticket\_no`)

UNIQUE KEY `uk\_ticket\_no` (`ticket\_no`)

KEY `product\_id` (`product\_id`)

KEY `sales\_id` (`sales\_id`)

KEY `created\_by\_employee\_id` (`created\_by\_employee\_id`)

KEY `idx\_customer\_id` (`customer\_id`)

KEY `idx\_service\_status` (`service\_status`)

KEY `idx\_priority` (`priority`)

KEY `idx\_opened\_date` (`opened\_date`)

KEY `idx\_assigned\_to` (`assigned\_to\_employee\_id`)

KEY `idx\_status\_priority` (`service\_status`,`priority`)

CONSTRAINT `service\_tickets\_ibfk\_1` FOREIGN KEY (`customer\_id`) REFERENCES `customers` (`customer\_id`) ON UPDATE CASCADE

CONSTRAINT `service\_tickets\_ibfk\_2` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON DELETE SET NULL

CONSTRAINT `service\_tickets\_ibfk\_3` FOREIGN KEY (`sales\_id`) REFERENCES `sales\_master` (`sales\_id`) ON DELETE SET NULL

CONSTRAINT `service\_tickets\_ibfk\_4` FOREIGN KEY (`assigned\_to\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

CONSTRAINT `service\_tickets\_ibfk\_5` FOREIGN KEY (`created\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

Child column -> parent column	Evidence	Delete/update rule or caveat
customer_id -> customers.customer_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
sales_id -> sales_master.sales_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
assigned_to_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
created_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

stock_ledger

Source: backend/migrations/production_schema.sql:1525. Primary key: stock_ledger_id

Column	SQL declaration
stock_ledger_id	int NOT NULL AUTO_INCREMENT
product_id	int NOT NULL
transaction_type	enum('PURCHASE','SALE','RETURN','ADJUSTMENT','INSTALLATION','SCRAP') COLLATE utf8mb4_unicode_ci NOT NULL
transaction_id	int DEFAULT NULL
reference_no	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
qty_in	decimal(10,2) NOT NULL DEFAULT '0.00'
qty_out	decimal(10,2) NOT NULL DEFAULT '0.00'
balance_qty	decimal(10,2) NOT NULL
unit_cost	decimal(12,2) DEFAULT NULL
remarks	text COLLATE utf8mb4_unicode_ci
recorded_by_employee_id	int DEFAULT NULL
transaction_date	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`stock_ledger_id`)
KEY `recorded_by_employee_id` (`recorded_by_employee_id`)
KEY `idx_product_id` (`product_id`)
KEY `idx_transaction_type` (`transaction_type`)
KEY `idx_transaction_date` (`transaction_date`)
KEY `idx_product_date` (`product_id`, `transaction_date`)
KEY `idx_reference` (`reference_no`)

CONSTRAINT `stock\_ledger\_ibfk\_1` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON UPDATE CASCADE

CONSTRAINT `stock\_ledger\_ibfk\_2` FOREIGN KEY (`recorded\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

Child column -> parent column	Evidence	Delete/update rule or caveat
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE
recorded_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

stock_master

Source: backend/migrations/production_schema.sql:1551. Primary key: stock_id

Column	SQL declaration
stock_id	int NOT NULL AUTO_INCREMENT
product_id	int NOT NULL
available_qty	decimal(10,2) NOT NULL DEFAULT '0.00'
reserved_qty	decimal(10,2) NOT NULL DEFAULT '0.00'
minimum_stock	decimal(10,2) NOT NULL DEFAULT '0.00'
maximum_stock	decimal(10,2) NOT NULL DEFAULT '0.00'
reorder_qty	decimal(10,2) NOT NULL DEFAULT '0.00'
last_purchase_price	decimal(12,2) DEFAULT NULL
last_sale_price	decimal(12,2) DEFAULT NULL
last_stock_check_date	date DEFAULT NULL
remarks	text COLLATE utf8mb4_unicode_ci
last_updated	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`stock\_id`)

UNIQUE KEY `product\_id` (`product\_id`)

KEY `idx\_available\_qty` (`available\_qty`)

KEY `idx\_below\_minimum` (`available\_qty`, `minimum\_stock`)

KEY `idx\_last\_updated` (`last\_updated`)

CONSTRAINT `stock\_master\_ibfk\_1` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON DELETE CASCADE ON UPDATE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE

supplier_payments

Source: backend/migrations/production_schema.sql:1573. Primary key: payment_id

Column	SQL declaration
payment_id	int NOT NULL AUTO_INCREMENT
supplier_id	int NOT NULL
purchase_id	int DEFAULT NULL COMMENT 'Against which PO'
amount	decimal(12,2) NOT NULL
payment_date	date NOT NULL DEFAULT (curdate())
payment_mode	enum('CASH','CARD','UPI','BANK','CHEQUE','ONLINE') COLLATE utf8mb4_unicode_ci NOT NULL
reference_no	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Cheque/Transaction number'
narration	text COLLATE utf8mb4_unicode_ci
paid_by_employee_id	int DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`payment_id`)
KEY `paid_by_employee_id` (`paid_by_employee_id`)
KEY `idx_supplier_id` (`supplier_id`)
KEY `idx_purchase_id` (`purchase_id`)
KEY `idx_payment_date` (`payment_date`)
KEY `idx_reference_no` (`reference_no`)
CONSTRAINT `supplier_payments_ibfk_1` FOREIGN KEY (`supplier_id`) REFERENCES `suppliers` (`supplier_id`) ON UPDATE CASCADE
CONSTRAINT `supplier_payments_ibfk_2` FOREIGN KEY (`purchase_id`) REFERENCES `purchase_master` (`purchase_id`) ON DELETE SET NULL
CONSTRAINT `supplier_payments_ibfk_3` FOREIGN KEY (`paid_by_employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE SET NULL
CONSTRAINT `supplier_payments_chk_1` CHECK ((`amount` > 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
supplier_id -> suppliers.supplier_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE
purchase_id -> purchase_master.purchase_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
paid_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

suppliers

Source: backend/migrations/production_schema.sql:1597. Primary key: supplier_id

Column	SQL declaration
supplier_id	int NOT NULL AUTO_INCREMENT
supplier_name	varchar(150) COLLATE utf8mb4_unicode_ci NOT NULL
contact_person	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
phone	varchar(20) COLLATE utf8mb4_unicode_ci NOT NULL
alternate_phone	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
email	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
gst_no	varchar(15) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'GST registration number'
pan_no	varchar(10) COLLATE utf8mb4_unicode_ci DEFAULT NULL
address	text COLLATE utf8mb4_unicode_ci NOT NULL
city	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
state	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
pincode	varchar(10) COLLATE utf8mb4_unicode_ci DEFAULT NULL
opening_balance	decimal(12,2) NOT NULL DEFAULT '0.00' COMMENT 'Supplier credit balance'
is_active	tinyint(1) NOT NULL DEFAULT '1'
payment_terms	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'e.g., Net 30, 2/10 Net 30'
bank_account_no	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
bank_name	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
ifsc_code	varchar(11) COLLATE utf8mb4_unicode_ci DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`supplier\_id`)

UNIQUE KEY `supplier\_name` (`supplier\_name`)

KEY `idx\_supplier\_name` (`supplier\_name`)

KEY `idx\_gst\_no` (`gst\_no`)

KEY `idx\_city` (`city`)

KEY `idx\_active` (`is\_active`)

KEY `idx\_email` (`email`)

technician_material_movements

Source: backend/migrations/production_schema.sql:1628. Primary key: material_movement_id

Column	SQL declaration
material_movement_id	bigint NOT NULL AUTO_INCREMENT
movement_no	varchar(30) NOT NULL

movement_type	enum('ISSUE','SALE','FAULT','RETURN') NOT NULL
employee_id	int NOT NULL
product_id	int NOT NULL
qty	decimal(10,2) NOT NULL
unit_price	decimal(12,2) NOT NULL DEFAULT '0.00'
total_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
commission_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
paid_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
balance_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
payment_status	enum('PENDING','PARTIAL','PAID') NOT NULL DEFAULT 'PENDING'
customer_type	enum('CATV','NET','CCTV','ANONYMOUS') DEFAULT NULL
cable_customer_id	bigint DEFAULT NULL
service_customer_id	int DEFAULT NULL
anonymous_name	varchar(150) DEFAULT NULL
anonymous_mobile	varchar(20) DEFAULT NULL
reason	varchar(500) DEFAULT NULL
remarks	text
movement_date	datetime NOT NULL DEFAULT CURRENT_TIMESTAMP
created_by_user_id	int DEFAULT NULL
created_by_employee_id	int DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`material\_movement\_id`)

UNIQUE KEY `uk\_material\_movement\_no` (`movement\_no`)

KEY `idx\_material\_movement\_employee` (`employee\_id`)

KEY `idx\_material\_movement\_product` (`product\_id`)

KEY `idx\_material\_movement\_date` (`movement\_date`)

KEY `fk\_material\_movement\_catv\_customer` (`cable\_customer\_id`)

KEY `fk\_material\_movement\_service\_customer` (`service\_customer\_id`)

CONSTRAINT `fk\_material\_movement\_catv\_customer` FOREIGN KEY (`cable\_customer\_id`) REFERENCES `cable\_tv\_customers` (`cable\_customer\_id`)

CONSTRAINT `fk\_material\_movement\_employee` FOREIGN KEY (`employee\_id`) REFERENCES `employees` (`employee\_id`)

CONSTRAINT `fk\_material\_movement\_product` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`)

CONSTRAINT `fk\_material\_movement\_service\_customer` FOREIGN KEY (`service\_customer\_id`) REFERENCES `customers` (`customer\_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
cable_customer_id -> cable_tv_customers.cable_customer_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
service_customer_id -> customers.customer_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.

technician_material_sale_adjustments

Source: backend/controller/materialSalesController.js:122. Primary key: material_sale_adjustment_id

Column	SQL declaration
material_sale_adjustment_id	BIGINT AUTO_INCREMENT PRIMARY KEY
material_movement_id	BIGINT NOT NULL
adjustment_type	ENUM('RETURN','FAULT') NOT NULL
qty	DECIMAL(10,2) NOT NULL
remarks	TEXT NULL
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
requested_by_user_id	INT NULL
requested_by_employee_id	INT NULL
reviewed_by_user_id	INT NULL
reviewed_at	DATETIME NULL
review_remarks	TEXT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

INDEX idx_material_adjustment_sale (material_movement_id)

INDEX idx_material_adjustment_status (approval_status)

CONSTRAINT fk_material_adjustment_sale FOREIGN KEY (material_movement_id) REFERENCES technician_material_movements(material_movement_id)

Child column -> parent column	Evidence	Delete/update rule or caveat
material_movement_id -> technician_material_movements.material_movement_id	Declared FK; backend/controller/materialSalesController.js	No explicit rule in source definition.

technician_material_sale_payments

Source: backend/migrations/production_schema.sql:1666. Primary key: material_sale_payment_id

Column	SQL declaration
material_sale_payment_id	bigint NOT NULL AUTO_INCREMENT
material_movement_id	bigint NOT NULL
cash_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
online_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
received_amount	decimal(12,2) NOT NULL
balance_after_payment	decimal(12,2) NOT NULL
payment_status	enum('PARTIAL','PAID') NOT NULL
received_by_user_id	int DEFAULT NULL
received_by_employee_id	int DEFAULT NULL
received_date	date NOT NULL
due_date	date DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`material\_sale\_payment\_id`)

KEY `idx\_material\_sale\_payment\_movement` (`material\_movement\_id`)

CONSTRAINT `fk\_material\_sale\_payment\_movement` FOREIGN KEY (`material\_movement\_id`) REFERENCES `technician\_material\_movements` (`material\_movement\_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
material_movement_id -> technician_material_movements.material_movement_id	Declared FK; backend/migrations/ production_schema.sql	No explicit rule in source definition.

technician_material_stock

Source: backend/migrations/production_schema.sql:1685. Primary key: technician_material_stock_id

Column	SQL declaration
technician_material_stock_id	bigint NOT NULL AUTO_INCREMENT
employee_id	int NOT NULL
product_id	int NOT NULL
available_qty	decimal(10,2) NOT NULL DEFAULT '0.00'
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`technician\_material\_stock\_id`)

UNIQUE KEY `uk\_technician\_product` (`employee\_id`,`product\_id`)

KEY `fk\_technician\_stock\_product` (`product\_id`)

CONSTRAINT `fk\_technician\_stock\_employee` FOREIGN KEY (`employee\_id`) REFERENCES `employees` (`employee\_id`)

CONSTRAINT `fk\_technician\_stock\_product` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`)

Child column -> parent column	Evidence	Delete/update rule or caveat
employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	No explicit rule in source definition.

users

Source: backend/migrations/production_schema.sql:1699. Primary key: user_id

Column	SQL declaration
user_id	int NOT NULL AUTO_INCREMENT
employee_id	int DEFAULT NULL
employee_code	varchar(50) COLLATE utf8mb4_unicode_ci DEFAULT NULL
username	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
password	varchar(255) COLLATE utf8mb4_unicode_ci NOT NULL COMMENT 'Bcrypt hashed password'
email	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
contact_number	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
first_name	varchar(100) COLLATE utf8mb4_unicode_ci NOT NULL
last_name	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
role	enum('ADMIN','MANAGER','EMPLOYEE','SALES','SERVICE') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'EMPLOYEE'
is_active	tinyint(1) NOT NULL DEFAULT '1' COMMENT '1=Active, 0=Inactive'
date_registered	date NOT NULL DEFAULT (curdate())
last_login	datetime DEFAULT NULL
login_attempt_count	int DEFAULT '0' COMMENT 'For lockout mechanism'
locked_until	datetime DEFAULT NULL COMMENT 'Account lock timestamp'
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`user\_id`)

UNIQUE KEY `username` (`username`)

UNIQUE KEY `email` (`email`)

UNIQUE KEY `idx\_users\_employee\_id` (`employee\_id`)

KEY `idx\_email` (`email`)

KEY `idx\_username` (`username`)

KEY `idx\_role` (`role`)

KEY `idx\_active\_date` (`is\_active`,`date\_registered`)

CONSTRAINT `fk\_users\_employee` FOREIGN KEY (`employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE RESTRICT ON UPDATE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
employee_id -> employees.employee_id	Declared FK; backend/migrations/ production_schema.sql	ON DELETE RESTRICT ON UPDATE CASCADE

warranty_master

Source: backend/migrations/production_schema.sql:1729. Primary key: warranty_id

Column	SQL declaration
warranty_id	int NOT NULL AUTO_INCREMENT
warranty_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
customer_id	int NOT NULL
sales_id	int DEFAULT NULL
product_id	int NOT NULL
serial_no	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Product serial number'
warranty_start_date	date NOT NULL DEFAULT (curdate())
warranty_end_date	date NOT NULL
warranty_type	enum('MANUFACTURER','EXTENDED','VOID') COLLATE utf8mb4_unicode_ci DEFAULT 'MANUFACTURER'
warranty_status	enum('ACTIVE','EXPIRED','CLAIMED','VOID') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ACTIVE'
coverage_type	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Parts, Labor, Both'
warranty_cost	decimal(12,2) DEFAULT '0.00'
claims_count	int DEFAULT '0'
remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`warranty\_id`)

UNIQUE KEY `warranty\_no` (`warranty\_no`)

UNIQUE KEY `uk\_warranty\_no` (`warranty\_no`)

UNIQUE KEY `uk\_serial\_no` (`serial\_no`)

KEY `sales\_id` (`sales\_id`)

KEY `idx\_customer\_id` (`customer\_id`)

KEY `idx\_warranty\_status` (`warranty\_status`)

KEY `idx\_warranty\_end\_date` (`warranty\_end\_date`)

KEY `idx\_product\_id` (`product\_id`)

CONSTRAINT `warranty\_master\_ibfk\_1` FOREIGN KEY (`customer\_id`) REFERENCES `customers` (`customer\_id`) ON UPDATE CASCADE

CONSTRAINT `warranty\_master\_ibfk\_2` FOREIGN KEY (`sales\_id`) REFERENCES `sales\_master` (`sales\_id`) ON DELETE SET NULL

CONSTRAINT `warranty\_master\_ibfk\_3` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON UPDATE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
customer_id -> customers.customer_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE
sales_id -> sales_master.sales_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE

work_order_employees

Source: backend/migrations/production_schema.sql:1761. Primary key: work_order_employee_id

Column	SQL declaration
work_order_employee_id	int NOT NULL AUTO_INCREMENT
work_order_id	int NOT NULL
employee_id	int NOT NULL
assigned_date	date NOT NULL DEFAULT (curdate())
role_in_work	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL COMMENT 'Technician, Helper, Supervisor, etc.'
status	enum('ASSIGNED','STARTED','COMPLETED','REMOVED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'ASSIGNED'
remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`work\_order\_employee\_id`)

UNIQUE KEY `uk\_work\_order\_employee` (`work\_order\_id`,`employee\_id`)

KEY `idx\_work\_order\_id` (`work\_order\_id`)

KEY `idx\_employee\_id` (`employee\_id`)

KEY `idx\_status` (`status`)

CONSTRAINT `work\_order\_employees\_ibfk\_1` FOREIGN KEY (`work\_order\_id`) REFERENCES `work\_orders` (`work\_order\_id`) ON DELETE CASCADE ON UPDATE CASCADE

CONSTRAINT `work\_order\_employees\_ibfk\_2` FOREIGN KEY (`employee\_id`) REFERENCES `employees` (`employee\_id`) ON UPDATE CASCADE

Child column -> parent column	Evidence	Delete/update rule or caveat
work_order_id -> work_orders.work_order_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE

employee_id -> employees.employee_id Declared FK; backend/migrations/
production_schema.sql ON UPDATE CASCADE

work_order_items

Source: backend/migrations/production_schema.sql:1780. Primary key: work_order_item_id

Column	SQL declaration
work_order_item_id	int NOT NULL AUTO_INCREMENT
work_order_id	int NOT NULL
quotation_item_id	int DEFAULT NULL COMMENT 'Source quotation item if created from quotation'
product_id	int DEFAULT NULL COMMENT 'NULL for custom work items'
item_name	varchar(200) COLLATE utf8mb4_unicode_ci NOT NULL
description	text COLLATE utf8mb4_unicode_ci
qty	decimal(10,2) NOT NULL
selling_price	decimal(12,2) NOT NULL
discount_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
discount_percent	decimal(5,2) DEFAULT '0.00'
tax_percent	decimal(5,2) NOT NULL DEFAULT '0.00'
tax_amount	decimal(12,2) NOT NULL DEFAULT '0.00'
amount	decimal(12,2) NOT NULL
line_no	smallint DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`work\_order\_item\_id`)

KEY `idx\_work\_order\_id` (`work\_order\_id`)

KEY `idx\_product\_id` (`product\_id`)

KEY `idx\_quotation\_item\_id` (`quotation\_item\_id`)

CONSTRAINT `work\_order\_items\_ibfk\_1` FOREIGN KEY (`work\_order\_id`) REFERENCES `work\_orders` (`work\_order\_id`) ON DELETE CASCADE ON UPDATE CASCADE

CONSTRAINT `work\_order\_items\_ibfk\_2` FOREIGN KEY (`quotation\_item\_id`) REFERENCES `quotation\_items` (`quotation\_item\_id`) ON DELETE SET NULL

CONSTRAINT `work\_order\_items\_ibfk\_3` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `work\_order\_items\_chk\_1` CHECK ((`qty` > 0))

CONSTRAINT `work\_order\_items\_chk\_2` CHECK ((`selling\_price` >= 0))

CONSTRAINT `work\_order\_items\_chk\_3` CHECK ((`discount\_amount` >= 0))

CONSTRAINT `work\_order\_items\_chk\_4` CHECK ((`tax\_percent` >= 0))

CONSTRAINT `work\_order\_items\_chk\_5` CHECK ((`tax\_amount` >= 0))

CONSTRAINT `work\_order\_items\_chk\_6` CHECK ((`amount` >= 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
work_order_id -> work_orders.work_order_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
quotation_item_id -> quotation_items.quotation_item_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE

work_order_material_issues

Source: backend/migrations/production_schema.sql:1812. Primary key: issue_id

Column	SQL declaration
issue_id	int NOT NULL AUTO_INCREMENT
issue_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
work_order_id	int NOT NULL
material_id	int DEFAULT NULL
product_id	int DEFAULT NULL COMMENT 'Inventory product issued'
issued_qty	decimal(10,2) NOT NULL
issued_date	date NOT NULL DEFAULT (curdate())
issued_to_employee_id	int DEFAULT NULL
issued_by_employee_id	int DEFAULT NULL
remarks	text COLLATE utf8mb4_unicode_ci
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
approved_by_employee_id	int DEFAULT NULL
approved_at	timestamp NULL DEFAULT NULL
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`issue\_id`)

UNIQUE KEY `issue\_no` (`issue\_no`)

KEY `issued\_to\_employee\_id` (`issued\_to\_employee\_id`)

KEY `issued\_by\_employee\_id` (`issued\_by\_employee\_id`)

KEY `idx\_work\_order\_id` (`work\_order\_id`)

KEY `idx\_material\_id` (`material\_id`)

KEY `idx\_product\_id` (`product\_id`)

KEY `idx\_issued\_date` (`issued\_date`)

CONSTRAINT `work\_order\_material\_issues\_ibfk\_1` FOREIGN KEY (`work\_order\_id`) REFERENCES `work\_orders` (`work\_order\_id`) ON DELETE CASCADE ON UPDATE CASCADE

CONSTRAINT `work\_order\_material\_issues\_ibfk\_2` FOREIGN KEY (`material\_id`) REFERENCES `material\_master` (`material\_id`) ON DELETE SET NULL

CONSTRAINT `work\_order\_material\_issues\_ibfk\_3` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `work\_order\_material\_issues\_ibfk\_4` FOREIGN KEY (`issued\_to\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

CONSTRAINT `work\_order\_material\_issues\_ibfk\_5` FOREIGN KEY (`issued\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

CONSTRAINT `work\_order\_material\_issues\_chk\_1` CHECK ((`issued\_qty` > 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
work_order_id -> work_orders.work_order_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
material_id -> material_master.material_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
issued_to_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
issued_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

work_order_material_returns

Source: backend/migrations/production_schema.sql:1844. Primary key: return_id

Column	SQL declaration
return_id	int NOT NULL AUTO_INCREMENT
return_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
issue_id	int DEFAULT NULL
work_order_id	int NOT NULL
material_id	int DEFAULT NULL
product_id	int DEFAULT NULL
returned_qty	decimal(10,2) NOT NULL
return_date	date NOT NULL DEFAULT (curdate())
returned_by_employee_id	int DEFAULT NULL
received_by_employee_id	int DEFAULT NULL
condition_status	enum('GOOD','DAMAGED','SCRAP') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'GOOD'
remarks	text COLLATE utf8mb4_unicode_ci

created_at timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`return\_id`)

UNIQUE KEY `return\_no` (`return\_no`)

KEY `material\_id` (`material\_id`)

KEY `product\_id` (`product\_id`)

KEY `returned\_by\_employee\_id` (`returned\_by\_employee\_id`)

KEY `received\_by\_employee\_id` (`received\_by\_employee\_id`)

KEY `idx\_issue\_id` (`issue\_id`)

KEY `idx\_work\_order\_id` (`work\_order\_id`)

KEY `idx\_return\_date` (`return\_date`)

CONSTRAINT `work\_order\_material\_returns\_ibfk\_1` FOREIGN KEY (`issue\_id`) REFERENCES `work\_order\_material\_issues` (`issue\_id`) ON DELETE SET NULL

CONSTRAINT `work\_order\_material\_returns\_ibfk\_2` FOREIGN KEY (`work\_order\_id`) REFERENCES `work\_orders` (`work\_order\_id`) ON DELETE CASCADE ON UPDATE CASCADE

CONSTRAINT `work\_order\_material\_returns\_ibfk\_3` FOREIGN KEY (`material\_id`) REFERENCES `material\_master` (`material\_id`) ON DELETE SET NULL

CONSTRAINT `work\_order\_material\_returns\_ibfk\_4` FOREIGN KEY (`product\_id`) REFERENCES `products` (`product\_id`) ON DELETE SET NULL ON UPDATE CASCADE

CONSTRAINT `work\_order\_material\_returns\_ibfk\_5` FOREIGN KEY (`returned\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

CONSTRAINT `work\_order\_material\_returns\_ibfk\_6` FOREIGN KEY (`received\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

CONSTRAINT `work\_order\_material\_returns\_chk\_1` CHECK ((`returned\_qty` > 0))

Child column -> parent column	Evidence	Delete/update rule or caveat
issue_id -> work_order_material_issues.issue_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
work_order_id -> work_orders.work_order_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE CASCADE ON UPDATE CASCADE
material_id -> material_master.material_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
product_id -> products.product_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL ON UPDATE CASCADE
returned_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
received_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

work_orders

Source: backend/migrations/production_schema.sql:1877. Primary key: work_order_id

Column	SQL declaration
work_order_id	int NOT NULL AUTO_INCREMENT
work_order_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
quotation_id	int DEFAULT NULL COMMENT 'Source quotation'
customer_id	int NOT NULL
work_type	enum('INSTALLATION','SERVICE','REPAIR','MAINTENANCE','OTHER') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'INSTALLATION'
work_status	enum('PENDING','IN_PROGRESS','ON_HOLD','COMPLETED','CANCELLED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
approval_status	enum('PENDING','APPROVED','REJECTED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
priority	enum('LOW','MEDIUM','HIGH','URGENT') COLLATE utf8mb4_unicode_ci DEFAULT 'MEDIUM'
start_date	date DEFAULT NULL
completion_date	date DEFAULT NULL
site_address	text COLLATE utf8mb4_unicode_ci NOT NULL
site_contact_person	varchar(100) COLLATE utf8mb4_unicode_ci DEFAULT NULL
site_contact_phone	varchar(20) COLLATE utf8mb4_unicode_ci DEFAULT NULL
work_notes	text COLLATE utf8mb4_unicode_ci COMMENT 'Work description and special instructions'
assigned_to_employee_id	int DEFAULT NULL
supervisor_id	int DEFAULT NULL COMMENT 'Field supervisor'
created_by_employee_id	int DEFAULT NULL
completion_remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`work_order_id`)
UNIQUE KEY `work_order_no` (`work_order_no`)
UNIQUE KEY `uk_work_order_no` (`work_order_no`)
KEY `assigned_to_employee_id` (`assigned_to_employee_id`)
KEY `supervisor_id` (`supervisor_id`)
KEY `created_by_employee_id` (`created_by_employee_id`)
KEY `idx_quotation_id` (`quotation_id`)
KEY `idx_customer_id` (`customer_id`)
KEY `idx_work_status` (`work_status`)
KEY `idx_start_date` (`start_date`)
KEY `idx_work_type_status` (`work_type`,`work_status`)

CONSTRAINT `work\_orders\_ibfk\_1` FOREIGN KEY (`quotation\_id`) REFERENCES `quotation\_master` (`quotation\_id`) ON DELETE SET NULL

CONSTRAINT `work\_orders\_ibfk\_2` FOREIGN KEY (`customer\_id`) REFERENCES `customers` (`customer\_id`) ON UPDATE CASCADE

CONSTRAINT `work\_orders\_ibfk\_3` FOREIGN KEY (`assigned\_to\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

CONSTRAINT `work\_orders\_ibfk\_4` FOREIGN KEY (`supervisor\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

CONSTRAINT `work\_orders\_ibfk\_5` FOREIGN KEY (`created\_by\_employee\_id`) REFERENCES `employees` (`employee\_id`) ON DELETE SET NULL

Child column -> parent column	Evidence	Delete/update rule or caveat
quotation_id -> quotation_master.quotation_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
customer_id -> customers.customer_id	Declared FK; backend/migrations/production_schema.sql	ON UPDATE CASCADE
assigned_to_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
supervisor_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
created_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

workflow_approvals

Source: backend/migrations/production_schema.sql:1917. Primary key: workflow_id

Column	SQL declaration
workflow_id	int NOT NULL AUTO_INCREMENT
module_name	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
reference_id	int NOT NULL
reference_no	varchar(50) COLLATE utf8mb4_unicode_ci NOT NULL
workflow_status	enum('PENDING','APPROVED','REJECTED','CANCELLED') COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'PENDING'
requested_by_employee_id	int DEFAULT NULL
approved_by_employee_id	int DEFAULT NULL
requested_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
reviewed_at	timestamp NULL DEFAULT NULL
remarks	text COLLATE utf8mb4_unicode_ci
created_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	timestamp NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Indexes and constraints

PRIMARY KEY (`workflow_id`)
UNIQUE KEY `uk_workflow_reference` (`module_name`,`reference_id`)
KEY `idx_module_status` (`module_name`,`workflow_status`)
KEY `idx_reference_no` (`reference_no`)
KEY `requested_by_employee_id` (`requested_by_employee_id`)
KEY `approved_by_employee_id` (`approved_by_employee_id`)
CONSTRAINT `workflow_approvals_ibfk_1` FOREIGN KEY (`requested_by_employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE SET NULL
CONSTRAINT `workflow_approvals_ibfk_2` FOREIGN KEY (`approved_by_employee_id`) REFERENCES `employees` (`employee_id`) ON DELETE SET NULL

Child column -> parent column	Evidence	Delete/update rule or caveat
requested_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL
approved_by_employee_id -> employees.employee_id	Declared FK; backend/migrations/production_schema.sql	ON DELETE SET NULL

6. Schema evolution, verification and maintenance

Alternative CREATE definitions and runtime additions

The following definitions differ from the selected table definition. Only different/new column declarations are shown. Their existence does not mean they replace the baseline: an existing table is unchanged by CREATE IF NOT EXISTS. Index/constraint differences require reading the referenced definition. Runtime ALTER routines and conditional/dynamic declarations need separate review.

cable_network_master | backend/migrations/create_cable_tv_module.sql:21

Column	Alternative / additional declaration
network_id	INT AUTO_INCREMENT PRIMARY KEY
network_code	VARCHAR(20) NOT NULL
network_name	VARCHAR(100) NOT NULL
customer_no_start	INT NULL
customer_no_end	INT NULL
remarks	VARCHAR(255) NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_locations | backend/migrations/create_cable_tv_module.sql:41

Column	Alternative / additional declaration
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location_id	INT AUTO_INCREMENT PRIMARY KEY
location_name	VARCHAR(150) NOT NULL
post_short_code	VARCHAR(20) NULL
city	VARCHAR(100) NOT NULL
pincode	VARCHAR(10) NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_areas | backend/migrations/create_cable_tv_module.sql:63

Column	Alternative / additional declaration
area_id	INT AUTO_INCREMENT PRIMARY KEY
network_id	INT NULL
location_id	INT NOT NULL
area_name	VARCHAR(150) NOT NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_streets | backend/migrations/create_cable_tv_module.sql:90

Column	Alternative / additional declaration
street_id	INT AUTO_INCREMENT PRIMARY KEY
area_id	INT NOT NULL
street_name	VARCHAR(150) NOT NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_connection_sources | backend/migrations/create_cable_tv_module.sql:112

Column	Alternative / additional declaration
source_id	INT AUTO_INCREMENT PRIMARY KEY
source_name	VARCHAR(100) NOT NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_mso_master | backend/migrations/create_cable_tv_module.sql:128

Column	Alternative / additional declaration
mso_id	INT AUTO_INCREMENT PRIMARY KEY
mso_name	VARCHAR(100) NOT NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_package_master | backend/migrations/create_cable_tv_module.sql:144

Column	Alternative / additional declaration
package_id	INT AUTO_INCREMENT PRIMARY KEY
package_name	VARCHAR(150) NOT NULL
package_type	ENUM('MSO_PACKAGE','ADDON','ALACARTE','BROADCAST') NOT NULL DEFAULT 'MSO_PACKAGE'
price	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (price >= 0)
description	TEXT NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_approval_groups | backend/migrations/create_cable_tv_module.sql:164

Column	Alternative / additional declaration
approval_group_id	BIGINT AUTO_INCREMENT PRIMARY KEY
approval_group_no	VARCHAR(50) NOT NULL
group_type	ENUM('NEW_CUSTOMER_ONBOARDING','CUSTOMER_UPDATE','STB_UPDATE','CONNECTION_UPDATE','MAT NOT NULL
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
requested_by_user_id	INT NULL
approved_by_user_id	INT NULL
requested_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
approved_at	TIMESTAMP NULL
rejected_reason	TEXT NULL
remarks	TEXT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_tv_customers | backend/migrations/create_cable_tv_module.sql:195

Column	Alternative / additional declaration
cable_customer_id	BIGINT AUTO_INCREMENT PRIMARY KEY

approval_group_id	BIGINT NULL
erp_customer_id	INT NULL
network_id	INT NOT NULL
network_type	VARCHAR(20) NULL
customer_type	ENUM('REGULAR','BUSINESS') NOT NULL DEFAULT 'REGULAR'
legacy_customer_no	VARCHAR(50) NULL
customer_code	INT NOT NULL
full_name	VARCHAR(150) NOT NULL
door_no	VARCHAR(50) NOT NULL
location_id	INT NOT NULL
area_id	INT NOT NULL
street_id	INT NOT NULL
city	VARCHAR(100) NOT NULL
pincode	VARCHAR(10) NULL
mobile_no	VARCHAR(20) NOT NULL
aadhaar_no	VARCHAR(12) NULL
alternate_mobile_no	VARCHAR(20) NULL
source_id	INT NULL
installed_by_employee_id	INT NULL
labour_service_charge	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (labour_service_charge >= 0)
status	ENUM('ACTIVE','INACTIVE','DISCONNECTED','SHIFTED','TRANSFERRED','RETRIEVED','FAULT','UPGRADE') NOT NULL DEFAULT 'ACTIVE'
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	INT NULL
approved_by_user_id	INT NULL
approved_at	TIMESTAMP NULL
rejected_reason	TEXT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_stb_master | backend/migrations/create_cable_tv_module.sql:274

Column	Alternative / additional declaration
stb_master_id	BIGINT AUTO_INCREMENT PRIMARY KEY
stb_number	VARCHAR(100) NOT NULL
box_type	ENUM('HD','SD') NOT NULL DEFAULT 'HD'

stock_type	ENUM('NEW','SERVICED','RETURNED','FAULT','DAMAGED','BURNT','NOT_SERVICEABLE') NOT NULL DEFAULT 'NEW'
mso_id	INT NULL
stb_amount	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (stb_amount >= 0)
full_set_amount	DECIMAL(12,2) NOT NULL DEFAULT 800 CHECK (full_set_amount >= 0)
assigned_employee_id	INT NULL
status	ENUM('AVAILABLE','NOT_AVAILABLE') NOT NULL DEFAULT 'AVAILABLE'
updated_date	DATE NOT NULL DEFAULT (CURDATE())
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_customer_stbs | backend/migrations/create_cable_tv_module.sql:306

Column	Alternative / additional declaration
customer_stb_id	BIGINT AUTO_INCREMENT PRIMARY KEY
approval_group_id	BIGINT NULL
cable_customer_id	BIGINT NOT NULL
stb_master_id	BIGINT NULL
stb_type	ENUM('NEW','SERVICED','RETURNED','FAULT','DAMAGED','UPGRADE','REPLACED','EXCHANGE','CUSTOMI NOT NULL DEFAULT 'NEW'
issue_mode	ENUM('FULL_SET','BOX_ONLY') NOT NULL DEFAULT 'BOX_ONLY'
installed_mso_id	INT NULL
exchange_original_mso_id	INT NULL
stb_no	VARCHAR(100) NOT NULL
stb_image_path	VARCHAR(255) NULL
stb_amount	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (stb_amount >= 0)
stb_discount	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (stb_discount >= 0)
labour_service_charge	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (labour_service_charge >= 0)
refund_amount	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (refund_amount >= 0)
refund_payment_mode	ENUM('CASH','ONLINE','BANK','UPI','OTHER') NOT NULL DEFAULT 'CASH'
installed_by_employee_id	INT NULL
entered_by_employee_id	INT NULL
installed_date	DATE NOT NULL DEFAULT (CURDATE())
updated_date	DATE NULL
update_reason	VARCHAR(50) NULL
reason_remarks	VARCHAR(500) NULL

status	ENUM('ACTIVE','RETRIEVED','FAULT','DISCONNECTED','UPGRADE','RETURNED','FAULTY','REPLACED') NO DEFAULT 'ACTIVE'
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	INT NULL
approved_by_user_id	INT NULL
approved_at	TIMESTAMP NULL
rejected_reason	TEXT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_customer_stb_accessories | backend/migrations/create_cable_tv_module.sql:375

Column	Alternative / additional declaration
stb_accessory_id	BIGINT AUTO_INCREMENT PRIMARY KEY
approval_group_id	BIGINT NULL
cable_customer_id	BIGINT NOT NULL
customer_stb_id	BIGINT NOT NULL
product_id	INT NOT NULL
movement_type	ENUM('ISSUE','RETURN') NOT NULL DEFAULT 'ISSUE'
accessory_name	VARCHAR(200) NOT NULL
qty	DECIMAL(10,2) NOT NULL DEFAULT 1
unit	VARCHAR(20) NOT NULL DEFAULT 'PCS'
issued_by_employee_id	INT NULL
issued_date	DATE NULL
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	INT NULL
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_connections | backend/migrations/create_cable_tv_module.sql:418

Column	Alternative / additional declaration
connection_id	BIGINT AUTO_INCREMENT PRIMARY KEY
approval_group_id	BIGINT NULL
cable_customer_id	BIGINT NOT NULL
connection_date	DATE NOT NULL DEFAULT (CURDATE())
disconnection_date	DATE NULL
connection_type	ENUM('NEW','RECONNECTION','SHIFTED','TRANSFERRED') NOT NULL DEFAULT 'NEW'

old_door_no	VARCHAR(50) NULL
new_door_no	VARCHAR(50) NULL
old_location_id	INT NULL
old_area_id	INT NULL
old_street_id	INT NULL
new_location_id	INT NULL
new_area_id	INT NULL
new_street_id	INT NULL
old_address	VARCHAR(500) NULL
new_address	VARCHAR(500) NULL
connected_by_employee_id	INT NULL
entered_by_employee_id	INT NULL
connection_charge	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (connection_charge >= 0)
connection_discount	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (connection_discount >= 0)
labour_service_charge	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (labour_service_charge >= 0)
status	ENUM('ACTIVE','DISCONNECTED','SHIFTED','TRANSFERRED') NOT NULL DEFAULT 'ACTIVE'
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
remarks	TEXT NULL
created_by_user_id	INT NULL
approved_by_user_id	INT NULL
approved_at	TIMESTAMP NULL
rejected_reason	TEXT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_connection_materials | backend/migrations/create_cable_tv_module.sql:479

Column	Alternative / additional declaration
connection_material_id	BIGINT AUTO_INCREMENT PRIMARY KEY
approval_group_id	BIGINT NULL
connection_id	BIGINT NOT NULL
product_id	INT NULL
material_id	INT NULL
item_name	VARCHAR(200) NOT NULL
qty	DECIMAL(10,2) NOT NULL CHECK (qty > 0)
unit	VARCHAR(20) NOT NULL DEFAULT 'PCS'

unit_rate	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (unit_rate >= 0)
amount	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (amount >= 0)
issued_by_employee_id	INT NULL
updated_by_employee_id	INT NULL
updated_date	DATE NULL
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	INT NULL
approved_by_user_id	INT NULL
approved_at	TIMESTAMP NULL
rejected_reason	TEXT NULL
remarks	TEXT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

cable_customer_packages | backend/migrations/create_cable_tv_module.sql:535

Column	Alternative / additional declaration
customer_package_id	BIGINT AUTO_INCREMENT PRIMARY KEY
approval_group_id	BIGINT NULL
cable_customer_id	BIGINT NOT NULL
package_id	INT NOT NULL
package_type	ENUM('ADDON','ALACARTE','BROADCASTER') NOT NULL DEFAULT 'ADDON'
package_price	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (package_price >= 0)
start_date	DATE NOT NULL DEFAULT (CURDATE())
end_date	DATE NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
updated_by_employee_id	INT NULL
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	INT NULL
approved_by_user_id	INT NULL
approved_at	TIMESTAMP NULL
rejected_reason	TEXT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_subscriptions | backend/migrations/create_cable_tv_module.sql:587

Column	Alternative / additional declaration
subscription_id	BIGINT AUTO_INCREMENT PRIMARY KEY

approval_group_id	BIGINT NULL
cable_customer_id	BIGINT NOT NULL
customer_package_id	BIGINT NOT NULL
subscription_month	TINYINT NOT NULL CHECK (subscription_month BETWEEN 1 AND 12)
subscription_year	SMALLINT NOT NULL CHECK (subscription_year >= 2000)
days_in_month	INT NOT NULL CHECK (days_in_month BETWEEN 28 AND 31)
billing_basis	ENUM('DAY','MONTH','YEAR') NOT NULL DEFAULT 'MONTH'
number_of_days_or_months	DECIMAL(8,2) NOT NULL DEFAULT 1 CHECK (number_of_days_or_months > 0)
amount	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (amount >= 0)
paid_amount	DECIMAL(12,2) NOT NULL DEFAULT 0 CHECK (paid_amount >= 0)
balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
collect_date	DATE NULL
start_date	DATE NULL
expiry_date	DATE NULL
collected_by_employee_id	INT NULL
payment_mapped_employee_id	INT NULL
payment_mode	ENUM('CASH','ONLINE','OFFICE','UPI','CARD','BANK','CHEQUE') NULL
payment_status	ENUM('PENDING','PARTIAL','PAID','CANCELLED') NOT NULL DEFAULT 'PENDING'
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
remarks	TEXT NULL
created_by_user_id	INT NULL
approved_by_user_id	INT NULL
approved_at	TIMESTAMP NULL
rejected_reason	TEXT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_stb_issue_master | backend/migrations/create_cable_tv_module.sql:658

Column	Alternative / additional declaration
stb_issue_id	BIGINT AUTO_INCREMENT PRIMARY KEY
stb_master_id	BIGINT NOT NULL
cable_customer_id	BIGINT NOT NULL
customer_stb_id	BIGINT NULL
issued_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
issued_by_employee_id	INT NULL

issue_status	ENUM('ISSUED','RETURNED','CANCELLED') NOT NULL DEFAULT 'ISSUED'
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

cable_customer_accounts | backend/migrations/create_cable_tv_module.sql:690

Column	Alternative / additional declaration
account_id	BIGINT AUTO_INCREMENT PRIMARY KEY
approval_group_id	BIGINT NOT NULL
cable_customer_id	BIGINT NOT NULL
stb_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
connection_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
labor_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
material_cost	DECIMAL(12,2) NOT NULL DEFAULT 0
material_discount	DECIMAL(12,2) NOT NULL DEFAULT 0
subscription_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
sub_total	DECIMAL(12,2) NOT NULL DEFAULT 0
discount	DECIMAL(12,2) NOT NULL DEFAULT 0
overall_discount	DECIMAL(12,2) NOT NULL DEFAULT 0
grand_total	DECIMAL(12,2) NOT NULL DEFAULT 0
customer_paid_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
office_received_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
office_balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
due_date	DATE NULL
account_status	ENUM('PENDING','PARTIAL','PAID','RECEIVED') NOT NULL DEFAULT 'PENDING'
received_by_user_id	INT NULL
received_by_employee_id	INT NULL
received_at	TIMESTAMP NULL
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	INT NULL
approved_by_user_id	INT NULL
approved_at	TIMESTAMP NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_customer_account_payments | backend/migrations/create_cable_tv_module.sql:732

Column	Alternative / additional declaration
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payment_id	BIGINT AUTO_INCREMENT PRIMARY KEY
account_id	BIGINT NOT NULL
cash_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
online_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
received_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
paid_date	DATE NOT NULL
received_date	DATE NOT NULL
due_date	DATE NULL
balance_after_payment	DECIMAL(12,2) NOT NULL DEFAULT 0
payment_status	ENUM('PARTIAL','PAID') NOT NULL
received_by_user_id	INT NOT NULL
received_by_employee_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

cable_subscription_payments | backend/migrations/create_cable_tv_module.sql:752

Column	Alternative / additional declaration
subscription_payment_id	BIGINT AUTO_INCREMENT PRIMARY KEY
subscription_id	BIGINT NOT NULL
cable_customer_id	BIGINT NOT NULL
received_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
collected_date	DATE NOT NULL
payment_mode	VARCHAR(30) NULL
payment_reference	VARCHAR(150) NULL
received_by_employee_id	INT NULL
comments	VARCHAR(500) NULL
balance_after_payment	DECIMAL(12,2) NOT NULL DEFAULT 0
payment_status	ENUM('PARTIAL','PAID') NOT NULL
created_by_user_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

finance_transactions | backend/migrations/create_cable_tv_module.sql:780

Column	Alternative / additional declaration
finance_transaction_id	BIGINT AUTO_INCREMENT PRIMARY KEY
transaction_date	DATE NOT NULL
transaction_type	ENUM('DEBIT','CREDIT') NOT NULL
category	VARCHAR(100) NOT NULL

amount	DECIMAL(12,2) NOT NULL
payment_mode	ENUM('CASH','ONLINE','BANK','UPI','OTHER') NOT NULL DEFAULT 'CASH'
reference_no	VARCHAR(100) NULL
description	VARCHAR(500) NULL
source_module	VARCHAR(50) NULL
source_id	BIGINT NULL
created_by_user_id	INT NULL
created_by_employee_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

cable_tv_complaints | backend/migrations/create_cable_tv_module.sql:843

Column	Alternative / additional declaration
complaint_id	BIGINT AUTO_INCREMENT PRIMARY KEY
complaint_no	VARCHAR(30) NOT NULL
complainant_type	ENUM('CATV','NET','CCTV','ANONYMOUS') NOT NULL DEFAULT 'CATV'
cable_customer_id	BIGINT NULL
service_customer_id	INT NULL
anonymous_name	VARCHAR(150) NULL
anonymous_mobile	VARCHAR(20) NULL
reported_mobile	VARCHAR(20) NULL
anonymous_address	VARCHAR(500) NULL
nature_of_complaint	VARCHAR(250) NOT NULL
complaint_description	TEXT NULL
status	ENUM('OPEN','IN_PROGRESS','HOLD','PENDING','COMPLETED') NOT NULL DEFAULT 'OPEN'
assigned_employee_id	INT NULL
registered_by_user_id	INT NULL
registered_by_employee_id	INT NULL
registered_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
completed_at	DATETIME NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_tv_complaint_attempts | backend/migrations/create_cable_tv_module.sql:873

Column	Alternative / additional declaration
complaint_attempt_id	BIGINT AUTO_INCREMENT PRIMARY KEY
complaint_id	BIGINT NOT NULL

attempt_no	INT NOT NULL
status	ENUM('OPEN','IN_PROGRESS','HOLD','PENDING','COMPLETED') NOT NULL
assigned_employee_id	INT NULL
start_time	DATETIME NULL
end_time	DATETIME NULL
reason	TEXT NULL
remedy	TEXT NULL
notes	TEXT NULL
entered_by_user_id	INT NULL
entered_by_employee_id	INT NULL
entered_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP

technician_material_stock | backend/migrations/create_cable_tv_module.sql:892

Column	Alternative / additional declaration
technician_material_stock_id	BIGINT AUTO_INCREMENT PRIMARY KEY
employee_id	INT NOT NULL
product_id	INT NOT NULL
available_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

technician_material_movements | backend/migrations/create_cable_tv_module.sql:903

Column	Alternative / additional declaration
material_movement_id	BIGINT AUTO_INCREMENT PRIMARY KEY
movement_no	VARCHAR(30) NOT NULL
movement_type	ENUM('ISSUE','SALE','FAULT','RETURN') NOT NULL
employee_id	INT NOT NULL
product_id	INT NOT NULL
qty	DECIMAL(10,2) NOT NULL
unit_price	DECIMAL(12,2) NOT NULL DEFAULT 0
total_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
commission_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
paid_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
payment_status	ENUM('PENDING','PARTIAL','PAID') NOT NULL DEFAULT 'PENDING'
customer_type	ENUM('CATV','NET','CCTV','ANONYMOUS') NULL
cable_customer_id	BIGINT NULL

service_customer_id	INT NULL
anonymous_name	VARCHAR(150) NULL
anonymous_mobile	VARCHAR(20) NULL
reason	VARCHAR(500) NULL
remarks	TEXT NULL
movement_date	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
created_by_user_id	INT NULL
created_by_employee_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

technician_material_sale_payments | backend/migrations/create_cable_tv_module.sql:937

Column	Alternative / additional declaration
material_sale_payment_id	BIGINT AUTO_INCREMENT PRIMARY KEY
material_movement_id	BIGINT NOT NULL
cash_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
online_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
received_amount	DECIMAL(12,2) NOT NULL
balance_after_payment	DECIMAL(12,2) NOT NULL
payment_status	ENUM('PARTIAL','PAID') NOT NULL
received_by_user_id	INT NULL
received_by_employee_id	INT NULL
received_date	DATE NOT NULL
due_date	DATE NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

employee_attendance | backend/migrations/create_employee_attendance.sql:1

Column	Alternative / additional declaration
attendance_id	INT AUTO_INCREMENT PRIMARY KEY
employee_id	INT NOT NULL
attendance_date	DATE NOT NULL
status	ENUM('PRESENT','ABSENT','HALF_DAY','LEAVE','HOLIDAY') NOT NULL DEFAULT 'PRESENT'
check_in	TIME NULL
check_out	TIME NULL
remarks	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

notifications | backend/migrations/create_notifications.sql:1

Column	Alternative / additional declaration
notification_id	INT AUTO_INCREMENT PRIMARY KEY
source_key	VARCHAR(180) NOT NULL UNIQUE
notification_type	VARCHAR(50) NOT NULL
title	VARCHAR(180) NOT NULL
message	VARCHAR(500) NOT NULL
severity	ENUM('INFO','WARNING','CRITICAL') NOT NULL DEFAULT 'INFO'
reference_type	VARCHAR(50)
reference_id	INT
reference_no	VARCHAR(100)
navigation_url	VARCHAR(500)
is_read	TINYINT(1) NOT NULL DEFAULT 0
is_active	TINYINT(1) NOT NULL DEFAULT 1
resolved_at	TIMESTAMP NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

role_permissions | backend/migrations/create_role_permissions.sql:1

Column	Alternative / additional declaration
role	ENUM('MANAGER','EMPLOYEE','SALES','SERVICE') NOT NULL
permission_key	VARCHAR(80) NOT NULL
can_view	TINYINT(1) NOT NULL DEFAULT 0
can_create	TINYINT(1) NOT NULL DEFAULT 0
can_update	TINYINT(1) NOT NULL DEFAULT 0
can_delete	TINYINT(1) NOT NULL DEFAULT 0
updated_by	INT NULL
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_tv_complaints | backend/controller/cableTvComplaintController.js:12

Column	Alternative / additional declaration
complaint_id	BIGINT AUTO_INCREMENT PRIMARY KEY
complaint_no	VARCHAR(30) NOT NULL
complainant_type	ENUM('CATV','NET','CCTV','ANONYMOUS') NOT NULL DEFAULT 'CATV'
cable_customer_id	BIGINT NULL
internet_customer_id	BIGINT NULL

service_customer_id	INT NULL
anonymous_name	VARCHAR(150) NULL
anonymous_mobile	VARCHAR(20) NULL
reported_mobile	VARCHAR(20) NULL
anonymous_address	VARCHAR(500) NULL
complaint_subject	VARCHAR(250) NULL
nature_of_complaint	VARCHAR(250) NOT NULL
complaint_description	TEXT NULL
status	ENUM('OPEN','IN_PROGRESS','HOLD','PENDING','COMPLETED') NOT NULL DEFAULT 'OPEN'
assigned_employee_id	INT NULL
registered_by_user_id	INT NULL
registered_by_employee_id	INT NULL
registered_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
completed_at	DATETIME NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_tv_complaint_attempts | backend/controller/cableTvComplaintController.js:82

Column	Alternative / additional declaration
complaint_attempt_id	BIGINT AUTO_INCREMENT PRIMARY KEY
complaint_id	BIGINT NOT NULL
attempt_no	INT NOT NULL
status	ENUM('OPEN','IN_PROGRESS','HOLD','PENDING','COMPLETED') NOT NULL
assigned_employee_id	INT NULL
start_time	DATETIME NULL
end_time	DATETIME NULL
reason	TEXT NULL
remedy	TEXT NULL
notes	TEXT NULL
entered_by_user_id	INT NULL
entered_by_employee_id	INT NULL
entered_at	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP

cable_stb_master | backend/controller/cableTvController.js:126

Column	Alternative / additional declaration
stb_master_id	BIGINT AUTO_INCREMENT PRIMARY KEY

stb_number	VARCHAR(100) NOT NULL
box_type	ENUM('HD','SD') NOT NULL DEFAULT 'HD'
stock_type	ENUM('NEW','SERVICED','RETURNED','FAULT') NOT NULL DEFAULT 'NEW'
mso_id	INT NULL
stb_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
status	ENUM('AVAILABLE','NOT_AVAILABLE') NOT NULL DEFAULT 'AVAILABLE'
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_stb_issue_master | backend/controller/cableTvController.js:144

Column	Alternative / additional declaration
stb_issue_id	BIGINT AUTO_INCREMENT PRIMARY KEY
stb_master_id	BIGINT NOT NULL
cable_customer_id	BIGINT NOT NULL
customer_stb_id	BIGINT NULL
issued_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
issued_by_employee_id	INT NULL
issue_status	ENUM('ISSUED','RETURNED','CANCELLED') NOT NULL DEFAULT 'ISSUED'
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

cable_customer_accounts | backend/controller/cableTvController.js:162

Column	Alternative / additional declaration
account_id	BIGINT AUTO_INCREMENT PRIMARY KEY
approval_group_id	BIGINT NOT NULL
cable_customer_id	BIGINT NOT NULL
stb_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
connection_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
labor_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
material_cost	DECIMAL(12,2) NOT NULL DEFAULT 0
material_discount	DECIMAL(12,2) NOT NULL DEFAULT 0
subscription_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
sub_total	DECIMAL(12,2) NOT NULL DEFAULT 0
discount	DECIMAL(12,2) NOT NULL DEFAULT 0
overall_discount	DECIMAL(12,2) NOT NULL DEFAULT 0
grand_total	DECIMAL(12,2) NOT NULL DEFAULT 0

customer_paid_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
office_received_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
office_balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
due_date	DATE NULL
account_status	ENUM('PENDING','PARTIAL','PAID','RECEIVED','NA') NOT NULL DEFAULT 'PENDING'
received_by_user_id	INT NULL
received_by_employee_id	INT NULL
received_at	TIMESTAMP NULL
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	INT NULL
approved_by_user_id	INT NULL
approved_at	TIMESTAMP NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_customer_stb_accessories | backend/controller/cableTvController.js:214

Column	Alternative / additional declaration
stb_accessory_id	BIGINT AUTO_INCREMENT PRIMARY KEY
approval_group_id	BIGINT NULL
cable_customer_id	BIGINT NOT NULL
customer_stb_id	BIGINT NOT NULL
product_id	INT NOT NULL
movement_type	ENUM('ISSUE','RETURN') NOT NULL DEFAULT 'ISSUE'
accessory_name	VARCHAR(200) NOT NULL
qty	DECIMAL(10,2) NOT NULL DEFAULT 1
unit	VARCHAR(20) NOT NULL DEFAULT 'PCS'
issued_by_employee_id	INT NULL
issued_date	DATE NULL
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	INT NULL
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

cable_customer_account_payments | backend/controller/cableTvController.js:448

Column

Alternative / additional declaration

payment_id	BIGINT AUTO_INCREMENT PRIMARY KEY
account_id	BIGINT NOT NULL
cash_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
online_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
received_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
paid_date	DATE NOT NULL
received_date	DATE NOT NULL
due_date	DATE NULL
balance_after_payment	DECIMAL(12,2) NOT NULL DEFAULT 0
payment_status	ENUM('PARTIAL','PAID') NOT NULL
received_by_user_id	INT NOT NULL
received_by_employee_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

cable_subscription_payments | backend/controller/cableTvController.js:469

Column	Alternative / additional declaration
subscription_payment_id	BIGINT AUTO_INCREMENT PRIMARY KEY
subscription_id	BIGINT NOT NULL
cable_customer_id	BIGINT NOT NULL
received_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
collected_date	DATE NOT NULL
payment_mode	VARCHAR(30) NULL
payment_reference	VARCHAR(150) NULL
received_by_employee_id	INT NULL
comments	VARCHAR(500) NULL
balance_after_payment	DECIMAL(12,2) NOT NULL DEFAULT 0
payment_status	ENUM('PARTIAL','PAID') NOT NULL
created_by_user_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

internet_customers | backend/controller/internetCustomerController.js:56

Column	Alternative / additional declaration
internet_customer_id	BIGINT AUTO_INCREMENT PRIMARY KEY
customer_code	INT NOT NULL
legacy_customer_no	VARCHAR(100) NULL
network_type	ENUM('KRISHI','RAILWIRE','DMNET') NOT NULL

full_name	VARCHAR(200) NOT NULL
net_id	VARCHAR(150) NOT NULL
network_password	VARCHAR(255) NULL
door_no	VARCHAR(50) NOT NULL
location_id	INT NOT NULL
area_id	INT NOT NULL
street_id	INT NOT NULL
state	VARCHAR(100) NOT NULL DEFAULT 'Tamil Nadu'
city	VARCHAR(100) NOT NULL
pincode	VARCHAR(10) NULL
mobile_no	VARCHAR(20) NOT NULL
alternate_mobile_no	VARCHAR(20) NULL
aadhaar_no	VARCHAR(20) NULL
source_name	ENUM('Customer Approach Office','Direct','Customer Approach Engineer') NOT NULL DEFAULT 'Direct'
installed_by_employee_id	INT NULL
installed_date	DATE NOT NULL
status	ENUM('ACTIVE','INACTIVE') NOT NULL DEFAULT 'ACTIVE'
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_by_user_id	INT NULL
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

internet_customer_packages | backend/controller/internetCustomerController.js:86

Column	Alternative / additional declaration
internet_customer_package_id	BIGINT AUTO_INCREMENT PRIMARY KEY
internet_customer_id	BIGINT NOT NULL
package_id	INT NOT NULL
package_price	DECIMAL(12,2) NOT NULL DEFAULT 0
start_date	DATE NOT NULL
end_date	DATE NOT NULL
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

internet_customer_routers | backend/controller/internetCustomerController.js:94

Column	Alternative / additional declaration
internet_router_id	BIGINT AUTO_INCREMENT PRIMARY KEY

internet_customer_id	BIGINT NOT NULL
router_type	ENUM('NEW','SERVICED','RETURNED') NOT NULL DEFAULT 'NEW'
product_id	INT NOT NULL
hsn_code	VARCHAR(30) NULL
qty	DECIMAL(10,2) NOT NULL DEFAULT 1
unit	VARCHAR(20) NOT NULL DEFAULT 'PCS'
rate	DECIMAL(12,2) NOT NULL DEFAULT 0
amount	DECIMAL(12,2) NOT NULL DEFAULT 0

internet_connections | backend/controller/internetCustomerController.js:103

Column	Alternative / additional declaration
internet_connection_id	BIGINT AUTO_INCREMENT PRIMARY KEY
internet_customer_id	BIGINT NOT NULL
connection_date	DATE NOT NULL
connection_type	ENUM('NEW','RECONNECTION','LOCATION_CHANGE') NOT NULL DEFAULT 'NEW'
connection_charge	DECIMAL(12,2) NOT NULL DEFAULT 0
connection_discount	DECIMAL(12,2) NOT NULL DEFAULT 0
labour_service_charge	DECIMAL(12,2) NOT NULL DEFAULT 0
remarks	VARCHAR(500) NULL

internet_connection_materials | backend/controller/internetCustomerController.js:110

Column	Alternative / additional declaration
internet_material_id	BIGINT AUTO_INCREMENT PRIMARY KEY
internet_customer_id	BIGINT NOT NULL
product_id	INT NULL
item_name	VARCHAR(200) NOT NULL
qty	DECIMAL(10,2) NOT NULL DEFAULT 1
unit	VARCHAR(20) NOT NULL DEFAULT 'PCS'
unit_rate	DECIMAL(12,2) NOT NULL DEFAULT 0
amount	DECIMAL(12,2) NOT NULL DEFAULT 0

internet_subscriptions | backend/controller/internetCustomerController.js:117

Column	Alternative / additional declaration
internet_subscription_id	BIGINT AUTO_INCREMENT PRIMARY KEY
internet_customer_id	BIGINT NOT NULL
internet_customer_package_id	BIGINT NOT NULL

subscription_month	INT NOT NULL
subscription_year	INT NOT NULL
start_date	DATE NOT NULL
end_date	DATE NOT NULL
amount	DECIMAL(12,2) NOT NULL DEFAULT 0
paid_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
payment_status	ENUM('PENDING','PARTIAL','PAID') NOT NULL DEFAULT 'PENDING'
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

internet_customer_accounts | backend/controller/internetCustomerController.js:127

Column	Alternative / additional declaration
internet_account_id	BIGINT AUTO_INCREMENT PRIMARY KEY
internet_customer_id	BIGINT NOT NULL
router_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
router_discount	DECIMAL(12,2) NOT NULL DEFAULT 0
connection_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
labor_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
material_cost	DECIMAL(12,2) NOT NULL DEFAULT 0
material_discount	DECIMAL(12,2) NOT NULL DEFAULT 0
subscription_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
overall_discount	DECIMAL(12,2) NOT NULL DEFAULT 0
grand_total	DECIMAL(12,2) NOT NULL DEFAULT 0
customer_paid_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
office_received_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
office_balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
due_date	DATE NULL
account_status	ENUM('PENDING','PARTIAL','PAID') NOT NULL DEFAULT 'PENDING'
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

workflow_approvals | backend/controller/internetCustomerController.js:225

Column	Alternative / additional declaration
workflow_id	INT AUTO_INCREMENT PRIMARY KEY

module_name	VARCHAR(50) NOT NULL
reference_id	INT NOT NULL
reference_no	VARCHAR(50) NOT NULL
workflow_status	ENUM('PENDING','APPROVED','REJECTED','CANCELLED') NOT NULL DEFAULT 'PENDING'
requested_by_employee_id	INT NULL
approved_by_employee_id	INT NULL
requested_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
reviewed_at	TIMESTAMP NULL
remarks	TEXT NULL
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

internet_customer_account_payments | backend/controller/internetCustomerController.js:233

Column	Alternative / additional declaration
internet_payment_id	BIGINT AUTO_INCREMENT PRIMARY KEY
internet_account_id	BIGINT NOT NULL
cash_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
online_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
received_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
paid_date	DATE NOT NULL
received_date	DATE NOT NULL
due_date	DATE NULL
balance_after_payment	DECIMAL(12,2) NOT NULL DEFAULT 0
payment_status	ENUM('PARTIAL','PAID') NOT NULL
received_by_user_id	INT NULL
received_by_employee_id	INT NULL
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

internet_customer_complaints | backend/controller/internetCustomerController.js:242

Column	Alternative / additional declaration
internet_complaint_id	BIGINT AUTO_INCREMENT PRIMARY KEY
internet_customer_id	BIGINT NOT NULL
complaint_date	DATE NOT NULL
subject	VARCHAR(200) NOT NULL
description	VARCHAR(1000) NULL
complaint_status	ENUM('OPEN','IN_PROGRESS','RESOLVED','CLOSED') NOT NULL DEFAULT 'OPEN'

created_by_user_id	INT NULL
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

technician_material_stock | backend/controller/materialSalesController.js:28

Column	Alternative / additional declaration
technician_material_stock_id	BIGINT AUTO_INCREMENT PRIMARY KEY
employee_id	INT NOT NULL
product_id	INT NOT NULL
available_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

technician_material_movements | backend/controller/materialSalesController.js:40

Column	Alternative / additional declaration
material_movement_id	BIGINT AUTO_INCREMENT PRIMARY KEY
movement_no	VARCHAR(30) NOT NULL
movement_type	ENUM('ISSUE','SALE','FAULT','RETURN') NOT NULL
employee_id	INT NOT NULL
product_id	INT NOT NULL
qty	DECIMAL(10,2) NOT NULL
unit_price	DECIMAL(12,2) NOT NULL DEFAULT 0
total_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
commission_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
paid_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
payment_status	ENUM('PENDING','PARTIAL','PAID') NOT NULL DEFAULT 'PENDING'
sale_status	ENUM('ISSUED','SOLD') NOT NULL DEFAULT 'SOLD'
customer_type	ENUM('CATV','NET','CCTV','ANONYMOUS') NULL
cable_customer_id	BIGINT NULL
service_customer_id	INT NULL
anonymous_name	VARCHAR(150) NULL
anonymous_mobile	VARCHAR(20) NULL
reason	VARCHAR(500) NULL
remarks	TEXT NULL
movement_date	DATETIME NOT NULL DEFAULT CURRENT_TIMESTAMP
created_by_user_id	INT NULL
created_by_employee_id	INT NULL

created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
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technician_material_sale_payments | backend/controller/materialSalesController.js:103

Column	Alternative / additional declaration
material_sale_payment_id	BIGINT AUTO_INCREMENT PRIMARY KEY
material_movement_id	BIGINT NOT NULL
cash_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
online_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
received_amount	DECIMAL(12,2) NOT NULL
balance_after_payment	DECIMAL(12,2) NOT NULL
payment_status	ENUM('PARTIAL','PAID') NOT NULL
received_by_user_id	INT NULL
received_by_employee_id	INT NULL
received_date	DATE NOT NULL
due_date	DATE NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

notifications | backend/controller/notificationController.js:14

Column	Alternative / additional declaration
notification_id	INT AUTO_INCREMENT PRIMARY KEY
source_key	VARCHAR(180) NOT NULL UNIQUE
notification_type	VARCHAR(50) NOT NULL
title	VARCHAR(180) NOT NULL
message	VARCHAR(500) NOT NULL
severity	ENUM('INFO','WARNING','CRITICAL') NOT NULL DEFAULT 'INFO'
reference_type	VARCHAR(50)
reference_id	INT
reference_no	VARCHAR(100)
navigation_url	VARCHAR(500)
target_user_id	INT NULL
is_read	TINYINT(1) NOT NULL DEFAULT 0
is_active	TINYINT(1) NOT NULL DEFAULT 1
resolved_at	TIMESTAMP NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

stock_master | backend/controller/purchaseController.js:104

Column	Alternative / additional declaration
stock_id	INT AUTO_INCREMENT PRIMARY KEY
product_id	INT NOT NULL UNIQUE
available_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
reserved_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
minimum_stock	DECIMAL(10,2) NOT NULL DEFAULT 0
maximum_stock	DECIMAL(10,2) NOT NULL DEFAULT 0
reorder_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
last_purchase_price	DECIMAL(12,2)
last_sale_price	DECIMAL(12,2)
last_stock_check_date	DATE
remarks	TEXT
last_updated	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

stock_ledger | backend/controller/purchaseController.js:125

Column	Alternative / additional declaration
stock_ledger_id	INT AUTO_INCREMENT PRIMARY KEY
product_id	INT NOT NULL
transaction_type	ENUM('PURCHASE','SALE','RETURN','ADJUSTMENT','INSTALLATION','SCRAP') NOT NULL
transaction_id	INT
reference_no	VARCHAR(100)
qty_in	DECIMAL(10,2) NOT NULL DEFAULT 0
qty_out	DECIMAL(10,2) NOT NULL DEFAULT 0
balance_qty	DECIMAL(10,2) NOT NULL
unit_cost	DECIMAL(12,2)
remarks	TEXT
recorded_by_employee_id	INT
transaction_date	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

workflow_approvals | backend/controller/quotationController.js:46

Column	Alternative / additional declaration
workflow_id	INT AUTO_INCREMENT PRIMARY KEY
module_name	VARCHAR(50) NOT NULL
reference_id	INT NOT NULL
reference_no	VARCHAR(50) NOT NULL

workflow_status	ENUM('PENDING','APPROVED','REJECTED','CANCELLED') NOT NULL DEFAULT 'PENDING'
requested_by_employee_id	INT
approved_by_employee_id	INT
requested_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
reviewed_at	TIMESTAMP NULL
remarks	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

stock_master | backend/controller/stockController.js:14

Column	Alternative / additional declaration
stock_id	INT AUTO_INCREMENT PRIMARY KEY
product_id	INT NOT NULL UNIQUE
available_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
reserved_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
minimum_stock	DECIMAL(10,2) NOT NULL DEFAULT 0
maximum_stock	DECIMAL(10,2) NOT NULL DEFAULT 0
reorder_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
last_purchase_price	DECIMAL(12,2)
last_sale_price	DECIMAL(12,2)
last_stock_check_date	DATE
remarks	TEXT
last_updated	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

stock_ledger | backend/controller/stockController.js:35

Column	Alternative / additional declaration
stock_ledger_id	INT AUTO_INCREMENT PRIMARY KEY
product_id	INT NOT NULL
transaction_type	ENUM('PURCHASE','SALE','RETURN','ADJUSTMENT','INSTALLATION','SCRAP') NOT NULL
transaction_id	INT
reference_no	VARCHAR(100)
qty_in	DECIMAL(10,2) NOT NULL DEFAULT 0
qty_out	DECIMAL(10,2) NOT NULL DEFAULT 0
balance_qty	DECIMAL(10,2) NOT NULL
unit_cost	DECIMAL(12,2)
remarks	TEXT

recorded_by_employee_id	INT
transaction_date	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

finance_transactions | backend/controller/transactionController.js:5

Column	Alternative / additional declaration
finance_transaction_id	BIGINT AUTO_INCREMENT PRIMARY KEY
transaction_date	DATE NOT NULL
transaction_type	ENUM('DEBIT','CREDIT') NOT NULL
category	VARCHAR(100) NOT NULL
amount	DECIMAL(12,2) NOT NULL
payment_mode	ENUM('CASH','ONLINE','BANK','UPI','OTHER') NOT NULL DEFAULT 'CASH'
reference_no	VARCHAR(100) NULL
description	VARCHAR(500) NULL
instructed_by	VARCHAR(150) NULL
bill_copy_available	ENUM('YES','NO') NULL
item_list	TEXT NULL
received_by	VARCHAR(150) NULL
received_date	DATE NULL
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
approved_by_user_id	INT NULL
approved_at	DATETIME NULL
source_module	VARCHAR(50) NULL
source_id	BIGINT NULL
created_by_user_id	INT NULL
created_by_employee_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

workflow_approvals | backend/controller/workflowController.js:59

Column	Alternative / additional declaration
workflow_id	INT AUTO_INCREMENT PRIMARY KEY
module_name	VARCHAR(50) NOT NULL
reference_id	INT NOT NULL
reference_no	VARCHAR(50) NOT NULL
workflow_status	ENUM('PENDING','APPROVED','REJECTED','CANCELLED') NOT NULL DEFAULT 'PENDING'
requested_by_employee_id	INT

approved_by_employee_id	INT
requested_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
reviewed_at	TIMESTAMP NULL
remarks	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

workflow_approvals | backend/controller/workOrderController.js:47

Column	Alternative / additional declaration
workflow_id	INT AUTO_INCREMENT PRIMARY KEY
module_name	VARCHAR(50) NOT NULL
reference_id	INT NOT NULL
reference_no	VARCHAR(50) NOT NULL
workflow_status	ENUM('PENDING','APPROVED','REJECTED','CANCELLED') NOT NULL DEFAULT 'PENDING'
requested_by_employee_id	INT
approved_by_employee_id	INT
requested_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
reviewed_at	TIMESTAMP NULL
remarks	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

stock_master | backend/controller/workOrderController.js:64

Column	Alternative / additional declaration
stock_id	INT AUTO_INCREMENT PRIMARY KEY
product_id	INT NOT NULL UNIQUE
available_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
reserved_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
minimum_stock	DECIMAL(10,2) NOT NULL DEFAULT 0
maximum_stock	DECIMAL(10,2) NOT NULL DEFAULT 0
reorder_qty	DECIMAL(10,2) NOT NULL DEFAULT 0
last_purchase_price	DECIMAL(12,2)
last_sale_price	DECIMAL(12,2)
last_stock_check_date	DATE
remarks	TEXT
last_updated	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

stock_ledger | backend/controller/workOrderController.js:85

Column	Alternative / additional declaration
stock_ledger_id	INT AUTO_INCREMENT PRIMARY KEY
product_id	INT NOT NULL
transaction_type	ENUM('PURCHASE','SALE','RETURN','ADJUSTMENT','INSTALLATION','SCRAP') NOT NULL
transaction_id	INT
reference_no	VARCHAR(100)
qty_in	DECIMAL(10,2) NOT NULL DEFAULT 0
qty_out	DECIMAL(10,2) NOT NULL DEFAULT 0
balance_qty	DECIMAL(10,2) NOT NULL
unit_cost	DECIMAL(12,2)
remarks	TEXT
recorded_by_employee_id	INT
transaction_date	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

work_order_items | backend/controller/workOrderController.js:110

Column	Alternative / additional declaration
work_order_item_id	INT AUTO_INCREMENT PRIMARY KEY
work_order_id	INT NOT NULL
quotation_item_id	INT
product_id	INT
item_name	VARCHAR(200) NOT NULL
description	TEXT
qty	DECIMAL(10,2) NOT NULL
selling_price	DECIMAL(12,2) NOT NULL DEFAULT 0
discount_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
discount_percent	DECIMAL(5,2) DEFAULT 0
tax_percent	DECIMAL(5,2) NOT NULL DEFAULT 0
tax_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
amount	DECIMAL(12,2) NOT NULL DEFAULT 0
line_no	SMALLINT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

material_master | backend/controller/workOrderController.js:136

Column	Alternative / additional declaration
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material_id	INT AUTO_INCREMENT PRIMARY KEY
product_id	INT
material_code	VARCHAR(50) NOT NULL UNIQUE
material_name	VARCHAR(200) NOT NULL
description	TEXT
unit	VARCHAR(20) NOT NULL DEFAULT 'PCS'
standard_rate	DECIMAL(12,2) NOT NULL DEFAULT 0
gst_percent	DECIMAL(5,2) NOT NULL DEFAULT 0
is_active	TINYINT(1) NOT NULL DEFAULT 1
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

work_order_material_issues | backend/controller/workOrderController.js:156

Column	Alternative / additional declaration
issue_id	INT AUTO_INCREMENT PRIMARY KEY
issue_no	VARCHAR(50) NOT NULL UNIQUE
work_order_id	INT NOT NULL
material_id	INT
product_id	INT
issued_qty	DECIMAL(10,2) NOT NULL
issued_date	DATE NOT NULL DEFAULT (CURDATE())
issued_to_employee_id	INT
issued_by_employee_id	INT
remarks	TEXT
approval_status	ENUM('PENDING','APPROVED','REJECTED') NOT NULL DEFAULT 'PENDING'
approved_by_employee_id	INT
approved_at	TIMESTAMP NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

work_order_material_returns | backend/controller/workOrderController.js:187

Column	Alternative / additional declaration
return_id	INT AUTO_INCREMENT PRIMARY KEY
return_no	VARCHAR(50) NOT NULL UNIQUE
issue_id	INT
work_order_id	INT NOT NULL
material_id	INT

product_id	INT
returned_qty	DECIMAL(10,2) NOT NULL
return_date	DATE NOT NULL DEFAULT (CURDATE())
returned_by_employee_id	INT
received_by_employee_id	INT
condition_status	ENUM('GOOD','DAMAGED','SCRAP') NOT NULL DEFAULT 'GOOD'
remarks	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

audit_log | backend/utils/auditLogSchema.js:3

Column	Alternative / additional declaration
audit_id	INT AUTO_INCREMENT PRIMARY KEY
user_id	INT NULL
module	VARCHAR(100)
action	VARCHAR(50)
table_name	VARCHAR(100)
record_id	INT NULL
old_values	JSON NULL
new_values	JSON NULL
ip_address	VARCHAR(45) NULL
browser_info	VARCHAR(255) NULL
change_reason	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

employee_attendance | backend/utils/businessModuleSchema.js:3

Column	Alternative / additional declaration
attendance_id	INT AUTO_INCREMENT PRIMARY KEY
employee_id	INT NOT NULL
attendance_date	DATE NOT NULL
status	ENUM('PRESENT','ABSENT','HALF_DAY','LEAVE','HOLIDAY') NOT NULL DEFAULT 'PRESENT'
check_in	TIME NULL
check_out	TIME NULL
remarks	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

sales_master | backend/utils/businessModuleSchema.js:23

Column	Alternative / additional declaration
sales_id	INT AUTO_INCREMENT PRIMARY KEY
invoice_no	VARCHAR(50) NOT NULL UNIQUE
invoice_date	DATE NOT NULL DEFAULT (CURDATE())
customer_id	INT NOT NULL
quotation_id	INT NULL
work_order_id	INT NULL
total_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
discount_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
discount_percent	DECIMAL(5,2) DEFAULT 0
tax_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
net_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
paid_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
balance_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
payment_mode	ENUM('CASH','CARD','UPI','BANK','CHEQUE','CREDIT') DEFAULT 'CREDIT'
paid_date	DATE NULL
payment_reference	VARCHAR(150) NULL
payment_status	ENUM('PENDING','PARTIAL','PAID','OVERDUE') NOT NULL DEFAULT 'PENDING'
sales_status	ENUM('DRAFT','COMPLETED','CANCELLED','RETURNED') NOT NULL DEFAULT 'DRAFT'
due_date	DATE NULL
remarks	TEXT
created_by_employee_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

customer_payments | backend/utils/businessModuleSchema.js:76

Column	Alternative / additional declaration
payment_id	INT AUTO_INCREMENT PRIMARY KEY
customer_id	INT NOT NULL
sales_id	INT NULL
amount	DECIMAL(12,2) NOT NULL
payment_date	DATE NOT NULL DEFAULT (CURDATE())
payment_mode	ENUM('CASH','CARD','UPI','BANK','CHEQUE','ONLINE') NOT NULL
reference_no	VARCHAR(100) NULL
payment_against	ENUM('INVOICE','ADVANCE','ADJUSTMENT') DEFAULT 'INVOICE'

narration	TEXT
received_by_employee_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

supplier_payments | backend/utils/businessModuleSchema.js:99

Column	Alternative / additional declaration
payment_id	INT AUTO_INCREMENT PRIMARY KEY
supplier_id	INT NOT NULL
purchase_id	INT NULL
amount	DECIMAL(12,2) NOT NULL
payment_date	DATE NOT NULL DEFAULT (CURDATE())
payment_mode	ENUM('CASH','CARD','UPI','BANK','CHEQUE','ONLINE') NOT NULL
reference_no	VARCHAR(100) NULL
narration	TEXT
paid_by_employee_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP

service_tickets | backend/utils/businessModuleSchema.js:125

Column	Alternative / additional declaration
service_ticket_id	INT AUTO_INCREMENT PRIMARY KEY
ticket_no	VARCHAR(50) NOT NULL UNIQUE
customer_id	INT NOT NULL
product_id	INT NULL
sales_id	INT NULL
assigned_to_employee_id	INT NULL
complaint_details	TEXT NOT NULL
service_status	ENUM('OPEN','ASSIGNED','IN_PROGRESS','RESOLVED','CLOSED','CANCELLED') NOT NULL DEFAULT 'OPEN'
priority	ENUM('LOW','MEDIUM','HIGH','URGENT') NOT NULL DEFAULT 'MEDIUM'
opened_date	DATE NOT NULL DEFAULT (CURDATE())
closed_date	DATE NULL
resolution_notes	TEXT
resolution_time_hours	INT NULL
created_by_employee_id	INT NULL
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

warranty_master | backend/utills/businessModuleSchema.js:160

Column	Alternative / additional declaration
warranty_id	INT AUTO_INCREMENT PRIMARY KEY
warranty_no	VARCHAR(50) NOT NULL UNIQUE
customer_id	INT NOT NULL
sales_id	INT NULL
product_id	INT NOT NULL
serial_no	VARCHAR(100) NULL
warranty_start_date	DATE NOT NULL DEFAULT (CURDATE())
warranty_end_date	DATE NOT NULL
warranty_type	ENUM('MANUFACTURER','EXTENDED','VOID') DEFAULT 'MANUFACTURER'
warranty_status	ENUM('ACTIVE','EXPIRED','CLAIMED','VOID') NOT NULL DEFAULT 'ACTIVE'
coverage_type	VARCHAR(100) NULL
warranty_cost	DECIMAL(12,2) DEFAULT 0
claims_count	INT DEFAULT 0
remarks	TEXT
created_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Migration file inventory

Migration source	Purpose by filename
backend/migrations/add_catv_package_updated_date.sql	add catv package updated date
backend/migrations/add_consumable_spare_parts_product_type.sql	add consumable spare parts product type
backend/migrations/add_customer_reference_fields.sql	add customer reference fields
backend/migrations/add_customer_salutation.sql	add customer salutation
backend/migrations/add_sales_payment_fields.sql	add sales payment fields
backend/migrations/add_senior_manager_role.sql	add senior manager role
backend/migrations/create_cable_tv_module.sql	create cable tv module
backend/migrations/create_employee_attendance.sql	create employee attendance
backend/migrations/create_notifications.sql	create notifications
backend/migrations/create_quotation_templates.sql	create quotation templates
backend/migrations/create_role_permissions.sql	create role permissions
backend/migrations/diagnose_cable_customer_9496_approval.sql	diagnose cable customer 9496 approval
backend/migrations/production_schema.sql	production schema
backend/migrations/update_employee_department_enum.sql	update employee department enum

Static ALTER statement inventory

These are source excerpts, not executable migration instructions. JavaScript template placeholders and dynamic ensureColumn routines cannot be resolved by this inventory. Some excerpts need surrounding code to interpret conditional execution. Before deployment use SHOW CREATE TABLE and INFORMATION_SCHEMA against the intended database, compare the current definition, and review the whole migration/helper.

Source	ALTER excerpt
backend/migrations/ add_catv_package_updated_date.sql	ALTER TABLE cable_customer_packages ADD COLUMN updated_date DATE NULL
backend/migrations/ add_consumable_spare_parts_product_type.sql	ALTER TABLE products MODIFY COLUMN product_type ENUM('MATERIAL', 'CONSUMABLE') NOT NULL DEFAULT 'MATERIAL'
backend/migrations/ add_customer_reference_fields.sql	ALTER TABLE customers ADD COLUMN marketing_employee_id INT NULL AFTER customer_id, ADD COLUMN marketing_employee VARCHAR(255) NULL AFTER marketing_employee_id, ADD INDEX idx_marketing_employee_id (marketing_employee_id) ON DELETE SET NULL ON UPDATE CASCADE
backend/migrations/ add_customer_salutation.sql	ALTER TABLE customers ADD COLUMN salutation ENUM('Mr/Mrs/Ms','Mr.','Mrs.','Ms.','M/S') NULL AFTER customer_id, ADD INDEX idx_salutation (salutation)
backend/migrations/ add_sales_payment_fields.sql	ALTER TABLE commits implicitly. -- 1. Check which payment columns already exist.
backend/migrations/ add_sales_payment_fields.sql	ALTER TABLE sales_master ADD COLUMN paid_date DATE NULL AFTER payment_mode
backend/migrations/ add_sales_payment_fields.sql	ALTER TABLE sales_master ADD COLUMN payment_reference VARCHAR(150) NULL AFTER payment_mode
backend/migrations/ add_senior_manager_role.sql	ALTER TABLE users MODIFY COLUMN role ENUM('ADMIN','MANAGER','EMPLOYEE','SALES','PURCHASE','STORE','INSTALLATION') CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci NOT NULL DEFAULT 'EMPLOYEE'
backend/migrations/ add_senior_manager_role.sql	ALTER TABLE role_permissions MODIFY COLUMN role ENUM('MANAGER','EMPLOYEE','SALES','PURCHASE','STORE','INSTALLATION') NOT NULL
backend/migrations/ update_employee_department_enum.sql	ALTER TABLE employees MODIFY COLUMN department ENUM('ADMIN','ENGINEER','TECHNICAL','STAFF','SALES','PURCHASE','STORE','INSTALLATION','SERVICE') NULL DEFAULT 'SERVICE'
backend/controller/ cableTvComplaintController.js	ALTER TABLE cable_tv_complaints ADD COLUMN complainant_type ENUM('CATV','NET','INTERNET') DEFAULT 'CATV' AFTER complaint_no
backend/controller/ cableTvComplaintController.js	ALTER TABLE cable_tv_complaints ADD COLUMN service_customer_id INT NULL AFTER complaint_no
backend/controller/ cableTvComplaintController.js	ALTER TABLE cable_tv_complaints ADD INDEX idx_cable_tv_complaint_service_customer_id (service_customer_id)
backend/controller/ cableTvComplaintController.js	ALTER TABLE cable_tv_complaints ADD COLUMN internet_customer_id BIGINT NULL AFTER service_customer_id
backend/controller/ cableTvComplaintController.js	ALTER TABLE cable_tv_complaints ADD INDEX idx_cable_tv_complaint_internet_customer_id (internet_customer_id)
backend/controller/ cableTvComplaintController.js	ALTER TABLE cable_tv_complaints ADD COLUMN reported_mobile VARCHAR(20) NULL AFTER internet_customer_id
backend/controller/ cableTvComplaintController.js	ALTER TABLE cable_tv_complaints ADD COLUMN complaint_subject VARCHAR(250) NULL AFTER reported_mobile
backend/controller/cableTvController.js	ALTER TABLE cable_package_master ADD COLUMN service_category ENUM('CATV','INTERNET') AFTER package_type
backend/controller/cableTvController.js	ALTER TABLE cable_package_master ADD COLUMN \${column} \${definition}

backend/controller/cableTvController.js	ALTER TABLE cable_stb_master DROP INDEX uk_cable_stb_master_number
backend/controller/cableTvController.js	ALTER TABLE cable_stb_master ADD INDEX idx_cable_stb_master_number (stb_number
backend/controller/cableTvController.js	ALTER TABLE cable_customer_stbs ADD COLUMN stb_master_id BIGINT NULL AFTER ca
backend/controller/cableTvController.js	ALTER TABLE cable_customer_stbs ADD INDEX idx_cable_stbs_master (stb_master_id
backend/controller/cableTvController.js	ALTER TABLE cable_customer_stbs ADD CONSTRAINT fk_cable_stbs_master FOREIGN K cable_stb_master(stb_master_id
backend/controller/cableTvController.js	ALTER TABLE cable_customer_stbs DROP INDEX uk_cable_stb_no
backend/controller/cableTvController.js	ALTER TABLE cable_customer_packages ADD COLUMN removal_status ENUM('NONE','PI DEFAULT 'NONE' AFTER approval_status"), ['removal_requested_at', 'ALTER TABLE cable removal_requested_at DATETIME NULL AFTER removal_status']]
backend/controller/cableTvController.js	ALTER TABLE cable_tv_customers MODIFY status ENUM('ACTIVE','INACTIVE','DISCONNECTED','SHIFTED','TRANSFERRED','RETRIEVED', DEFAULT 'ACTIVE
backend/controller/cableTvController.js	ALTER TABLE cable_customer_stbs MODIFY stb_type ENUM('NEW','SERVICED','RETURNED','FAULT','DAMAGED','UPGRADE','REPLACED','EX NULL DEFAULT 'NEW
backend/controller/cableTvController.js	ALTER TABLE cable_customer_stbs MODIFY status ENUM('ACTIVE','RETRIEVED','FAULT','DISCONNECTED','UPGRADE','RETURNED','FAUL 'ACTIVE
backend/controller/cableTvController.js	ALTER TABLE cable_stb_master MODIFY status ENUM('AVAILABLE','IN_SERVICE','NOT_ NULL DEFAULT 'AVAILABLE
backend/controller/cableTvController.js	ALTER TABLE cable_stb_master MODIFY stock_type ENUM('NEW','SERVICED','RETURNED','FAULT','DAMAGED','BURNT','NOT_SERVICEABL
backend/controller/cableTvController.js	ALTER TABLE cable_connections MODIFY connection_type ENUM('NEW','RECONNECTIO NULL DEFAULT 'NEW
backend/controller/cableTvController.js	ALTER TABLE cable_subscriptions MODIFY billing_basis ENUM('DAY','MONTH','YEAR') NC
backend/controller/cableTvController.js	ALTER TABLE cable_subscriptions MODIFY payment_mode ENUM('CASH','ONLINE','OFFLI
backend/controller/cableTvController.js	ALTER TABLE cable_connections ADD COLUMN connection_discount DECIMAL(12,2) NOT connection_charge
backend/controller/cableTvController.js	ALTER TABLE cable_connections ADD COLUMN old_network_id INT NULL AFTER connec TABLE cable_connections ADD COLUMN new_network_id INT NULL AFTER old_network_id cable_connections ADD COLUMN old_door_no VARCHAR(50) NULL AFTER connection_ty cable_connections ADD COLUMN new_door_no VARCHAR(50) NULL AFTER old_door_no cable_connections ADD COLUMN old_location_id INT NULL AFTER connection_type'], ['old cable_connections ADD COLUMN old_area_id INT NULL AFTER old_location_id'], ['old_stre ADD COLUMN old_street_id INT NULL AFTER old_area_id'], ['new_location_id', 'ALTER TAI new_location_id INT NULL AFTER old_street_id'], ['new_area_id', 'ALTER TABLE cable_con NULL AFTER new_location_id'], ['new_street_id', 'ALTER TABLE cable_connections ADD C new_area_id'], ['old_address', 'ALTER TABLE cable_connections ADD COLUMN old_addres new_street_id'], ['new_address', 'ALTER TABLE cable_connections ADD COLUMN new_ad old_address']]
backend/controller/cableTvController.js	ALTER TABLE cable_customer_accounts ADD COLUMN overall_discount DECIMAL(12,2) N ['material_discount', 'ALTER TABLE cable_customer_accounts ADD COLUMN material_disc 0 AFTER material_cost'], ['customer_paid_amount', 'ALTER TABLE cable_customer_accoun DECIMAL(12,2) NOT NULL DEFAULT 0 AFTER grand_total'], ['balance_amount', 'ALTER T/ COLUMN balance_amount DECIMAL(12,2) NOT NULL DEFAULT 0 AFTER customer_paid_ cable_customer_accounts ADD COLUMN due_date DATE NULL AFTER balance_amount'], cable_customer_accounts ADD COLUMN office_received_amount DECIMAL(12,2) NOT NU customer_paid_amount'], ['office_balance_amount', 'ALTER TABLE cable_customer_accoun DECIMAL(12,2) NOT NULL DEFAULT 0 AFTER office_received_amount'], ['received_by_err cable_customer_accounts ADD COLUMN received_by_employee_id INT NULL AFTER rece

backend/controller/cableTvController.js	ALTER TABLE cable_tv_customers ADD COLUMN network_type VARCHAR(20) NULL AFTER
backend/controller/cableTvController.js	ALTER TABLE cable_tv_customers ADD COLUMN customer_type ENUM('REGULAR','BUSINESS') DEFAULT 'REGULAR' AFTER network_type
backend/controller/cableTvController.js	ALTER TABLE cable_tv_customers MODIFY customer_type ENUM('REGULAR','BUSINESS') DEFAULT 'REGULAR'
backend/controller/cableTvController.js	ALTER TABLE cable_tv_customers ADD COLUMN network_customer_no VARCHAR(30) NULL AFTER status, ["topup_base_amount", "ALTER TABLE cable_tv_customers ADD COLUMN topup_base_amount DECIMAL(12,2) NOT NULL DEFAULT 0 AFTER customer_type"], ["topup_gst_percent", "ALTER TABLE cable_tv_customers ADD COLUMN topup_gst_percent DECIMAL(12,2) NOT NULL DEFAULT 0 AFTER topup_base_amount"], ["recharge_amount", "ALTER TABLE cable_tv_customers ADD COLUMN recharge_amount DECIMAL(12,2) NOT NULL DEFAULT 0 AFTER topup_gst_percent"]]
backend/controller/cableTvController.js	ALTER TABLE cable_tv_customers MODIFY status ENUM('ACTIVE','INACTIVE','DISCONNECTED','SHIFTED','TRANSFERRED','RETRIEVED','REMOVED') NOT NULL DEFAULT 'ACTIVE'
backend/controller/cableTvController.js	ALTER TABLE cable_tv_customers MODIFY customer_code INT NOT NULL
backend/controller/cableTvController.js	ALTER TABLE cable_connections ADD COLUMN entered_by_employee_id INT NULL AFTER cable_customer_stbs, ['cable_customer_stbs', 'entered_by_employee_id', 'ALTER TABLE cable_customer_stbs ADD COLUMN installed_by_employee_id INT NULL AFTER installed_by_employee_id']]
backend/controller/cableTvController.js	ALTER TABLE cable_customer_accounts MODIFY account_status ENUM('PENDING','PARTIAL') DEFAULT 'PENDING'
backend/controller/cableTvController.js	ALTER TABLE cable_customer_account_payments ADD COLUMN received_by_employee_id INT NULL AFTER account_id
backend/controller/cableTvController.js	ALTER TABLE cable_customer_stbs ADD COLUMN updated_date DATE NULL AFTER installed_date, TABLE cable_customer_stbs ADD COLUMN update_reason VARCHAR(50) NULL AFTER updated_date, TABLE cable_customer_stbs ADD COLUMN reason_remarks VARCHAR(500) NULL AFTER update_reason, TABLE cable_customer_stbs ADD COLUMN issue_mode ENUM('FULL_SET','BOX_ONLY') AFTER stb_type, ['refund_amount', "ALTER TABLE cable_customer_stbs ADD COLUMN refund_amount DECIMAL(12,2) NOT NULL DEFAULT 0 AFTER labour_service_charge"], ['refund_payment_mode', "ALTER TABLE cable_customer_stbs ADD COLUMN refund_payment_mode ENUM('CASH','ONLINE','BANK','UPI','OTHER') NOT NULL DEFAULT 'CASH'"]]
backend/controller/cableTvController.js	ALTER TABLE cable_stb_master ADD COLUMN updated_date DATE NULL AFTER status
backend/controller/cableTvController.js	ALTER TABLE cable_customer_stb_accessories ADD COLUMN movement_type ENUM('ISSUE','RETURN','REPAIR') AFTER product_id
backend/controller/cableTvController.js	ALTER TABLE cable_stb_master ADD COLUMN full_set_amount DECIMAL(12,2) NOT NULL
backend/controller/cableTvController.js	ALTER TABLE cable_stb_master ADD COLUMN assigned_employee_id INT NULL AFTER full_set_amount
backend/controller/cableTvController.js	ALTER TABLE cable_connection_materials ADD COLUMN updated_by_employee_id INT NULL AFTER updated_date, ['updated_date', 'ALTER TABLE cable_connection_materials ADD COLUMN updated_date DATE NULL AFTER updated_by_employee_id']]
backend/controller/cableTvController.js	ALTER TABLE cable_customer_packages ADD COLUMN package_type ENUM('ADDON','BUNDLE') DEFAULT 'ADDON' AFTER package_id, ['updated_date', 'ALTER TABLE cable_customer_packages ADD COLUMN updated_date DATE NULL'], ['updated_by_employee_id', 'ALTER TABLE cable_customer_packages ADD COLUMN updated_by_employee_id INT NULL AFTER is_active']]
backend/controller/cableTvController.js	ALTER TABLE cable_subscriptions ADD COLUMN payment_reference VARCHAR(150) NULL AFTER payment_mapped_employee_id, ['payment_mapped_employee_id', 'ALTER TABLE cable_subscriptions ADD COLUMN payment_mapped_employee_id INT NULL AFTER collected_by_employee_id'], ['received_count', 'ALTER TABLE cable_subscriptions ADD COLUMN received_count DECIMAL(8,2) NOT NULL DEFAULT 1 AFTER number_of_days_or_months']]
backend/controller/internetCustomerController.js	ALTER TABLE internet_customers ADD COLUMN legacy_customer_no VARCHAR(100) NULL AFTER idx_internet_legacy_customer_no (legacy_customer_no)
backend/controller/internetCustomerController.js	ALTER TABLE internet_customer_accounts ADD COLUMN \${column} \${definition}
backend/controller/internetCustomerController.js	ALTER TABLE internet_subscriptions MODIFY billing_basis ENUM('MONTH','DAYS','YEAR')

backend/controller/ internetCustomerController.js	ALTER TABLE internet_subscriptions MODIFY period_count DECIMAL(10,4) NOT NULL DE
backend/controller/ internetCustomerController.js	ALTER TABLE internet_subscriptions MODIFY renewed_by ENUM('LOGGED_IN_USER','EI NULL DEFAULT 'LOGGED_IN_USER
backend/controller/ internetCustomerController.js	ALTER TABLE internet_subscriptions MODIFY payment_mode ENUM('DASHBOARD','CASH','ONLINE','BANK_PAYMENT','ACCOUNT') NOT NULL DEFA
backend/controller/ internetCustomerController.js	ALTER TABLE internet_connections MODIFY connection_type ENUM('NEW','RECONNECTION','LOCATION_CHANGE','DISCONNECT') NOT NULL DEFA
backend/controller/materialSalesController.js	ALTER TABLE technician_material_movements ADD COLUMN commission_amount DECIM total_amount
backend/controller/materialSalesController.js	ALTER TABLE technician_material_movements ADD COLUMN paid_amount DECIMAL(12,2 total_amount
backend/controller/materialSalesController.js	ALTER TABLE technician_material_movements ADD COLUMN balance_amount DECIMAL(12,2 paid_amount
backend/controller/materialSalesController.js	ALTER TABLE technician_material_movements ADD COLUMN payment_status ENUM('PE DEFAULT 'PENDING' AFTER balance_amount
backend/controller/materialSalesController.js	ALTER TABLE technician_material_movements ADD COLUMN sale_status ENUM('ISSUED AFTER payment_status
backend/controller/notificationController.js	ALTER TABLE notifications ADD COLUMN target_user_id INT NULL AFTER navigation_url
backend/controller/notificationController.js	ALTER TABLE notifications ADD INDEX idx_notification_target (target_user_id, is_active
backend/controller/productController.js	ALTER TABLE products ADD COLUMN product_type ENUM('MATERIAL','CONSUMABLE','S NULL DEFAULT 'MATERIAL' AFTER description
backend/controller/productController.js	ALTER TABLE products MODIFY COLUMN product_type ENUM('MATERIAL','CONSUMABI NOT NULL DEFAULT 'MATERIAL
backend/controller/purchaseController.js	ALTER TABLE purchase_master ADD COLUMN approval_status ENUM('PENDING','APPR 'PENDING' AFTER payment_status'], ['approved_by_employee_id', 'ALTER TABLE purchas approved_by_employee_id INT NULL AFTER approval_status'], ['approved_at', 'ALTER TAB approved_at TIMESTAMP NULL AFTER approved_by_employee_id']]
backend/controller/quotationController.js	ALTER TABLE quotation_master MODIFY quotation_status ENUM('DRAFT','SENT','APPROVED','ACCEPTED','CANCELLED','REJECTED','EXPIRED','C 'DRAFT
backend/controller/supplierController.js	ALTER TABLE suppliers ADD COLUMN website_address VARCHAR(255) NULL AFTER en
backend/controller/transactionController.js	ALTER TABLE finance_transactions ADD COLUMN instructed_by VARCHAR(150) NULL AF "ALTER TABLE finance_transactions ADD COLUMN bill_copy_available ENUM('YES','NO') I 'ALTER TABLE finance_transactions ADD COLUMN item_list TEXT NULL AFTER bill_copy_ finance_transactions ADD COLUMN received_by VARCHAR(150) NULL AFTER item_list'], [finance_transactions ADD COLUMN received_date DATE NULL AFTER received_by'], ['appi finance_transactions ADD COLUMN approval_status ENUM('PENDING','APPROVED','REJE AFTER received_date'], ['approved_by_user_id', 'ALTER TABLE finance_transactions ADD (AFTER approval_status'], ['approved_at', 'ALTER TABLE finance_transactions ADD COLUM approved_by_user_id']]
backend/controller/userController.js	ALTER TABLE users ADD COLUMN employee_id INT NULL AFTER user_id
backend/controller/userController.js	ALTER TABLE users ADD COLUMN employee_code VARCHAR(50) NULL AFTER employe
backend/controller/userController.js	ALTER TABLE users ADD CONSTRAINT fk_users_employee FOREIGN KEY (employee_id, ON UPDATE CASCADE ON DELETE RESTRICT
backend/controller/workflowController.js	ALTER TABLE cable_stb_master MODIFY stock_type ENUM('NEW','SERVICED','RETURN NULL DEFAULT 'NEW

backend/controller/workflowController.js	ALTER TABLE cable_stb_master MODIFY status ENUM('AVAILABLE','IN_SERVICE','NOT_AVAILABLE') NULL DEFAULT 'AVAILABLE'
backend/controller/workOrderController.js	ALTER TABLE products ADD COLUMN product_type ENUM('MATERIAL','SERVICE','LABOR') NULL AFTER description
backend/utills/businessModuleSchema.js	ALTER TABLE sales_master ADD COLUMN paid_date DATE NULL AFTER payment_mode
backend/utills/businessModuleSchema.js	ALTER TABLE sales_master ADD COLUMN payment_reference VARCHAR(150) NULL AFTER payment_date
backend/utills/customerSchema.js	ALTER TABLE customers ADD COLUMN salutation \${CUSTOMER_SALUTATION_DDL} NULL AFTER name
backend/utills/customerSchema.js	ALTER TABLE customers MODIFY COLUMN salutation \${CUSTOMER_SALUTATION_DDL} NULL AFTER name
backend/utills/customerSchema.js	ALTER TABLE customers ADD INDEX idx_salutation (salutation)
backend/utills/customerSchema.js	ALTER TABLE customers ADD COLUMN marketing_employee_id INT NULL AFTER customer_id
backend/utills/customerSchema.js	ALTER TABLE customers ADD COLUMN referral_details VARCHAR(255) NULL AFTER marketing_employee_id
backend/utills/customerSchema.js	ALTER TABLE customers ADD INDEX idx_marketing_employee_id (marketing_employee_id)
backend/utills/customerSchema.js	ALTER TABLE customers ADD CONSTRAINT fk_customers_marketing_employee FOREIGN KEY (marketing_employee_id) REFERENCES employees(employee_id) ON DELETE SET NULL ON UPDATE CASCADE
backend/utills/customerSchema.js	ALTER TABLE customers ADD COLUMN approval_status ENUM('PENDING','APPROVED','REJECTED') NULL AFTER referral_details
backend/utills/customerSchema.js	ALTER TABLE customers ADD COLUMN created_by_user_id INT NULL AFTER approval_status
backend/utills/customerSchema.js	ALTER TABLE customers ADD COLUMN approved_by_user_id INT NULL AFTER created_by_user_id
backend/utills/customerSchema.js	ALTER TABLE customers ADD COLUMN approved_at TIMESTAMP NULL AFTER approved_by_user_id
backend/utills/customerSchema.js	ALTER TABLE customers ADD INDEX idx_customers_approval_status (approval_status)
backend/utills/employeeSchema.js	ALTER TABLE employees MODIFY COLUMN department \${EMPLOYEE_DEPARTMENT_DDL} NULL AFTER name

Focused verification references

The backend package has no aggregate test command: npm test is a placeholder that exits with failure. Existing tests use standalone Node scripts and/or node:test. Inspect a test first; run only tests whose database behavior and prerequisites are understood. Commands for mocked node:test files include: node --test backend/tests/internetConnectionActions.test.js and node --test backend/tests/internetSubscriptionEmail.test.js. Other scripts may use plain node and may require local data. Frontend tests use Karma/Jasmine via npm test; npm run build verifies Angular compilation.

Test source	Covered subject by filename
backend/tests/cableAccountStockDeferral.test.js	cable Account Stock Deferral
backend/tests/cablePackageApproval.test.js	cable Package Approval
backend/tests/cablePackageMasterEdit.test.js	cable Package Master Edit
backend/tests/cableStbSwap.test.js	cable Stb Swap
backend/tests/internetConnectionActions.test.js	internet Connection Actions
backend/tests/internetConnectionDetails.test.js	internet Connection Details
backend/tests/internetPackageMaster.test.js	internet Package Master
backend/tests/internetSubscriptionEmail.test.js	internet Subscription Email
backend/tests/internetSubscriptionInvoice.test.js	internet Subscription Invoice
backend/tests/quotationTemplates.test.js	quotation Templates

backend/scripts/assignNetCollector.test.js	assign Net Collector
backend/scripts/catvPackageRows.test.js	catv Package Rows
backend/scripts/catvSubscriptionPackageTotal.test.js	catv Subscription Package Total
backend/scripts/enrollmentAccountSync.test.js	enrollment Account Sync
backend/scripts/internetCustomerInformation.test.js	internet Customer Information
backend/scripts/internetCustomerNumber.test.js	internet Customer Number
backend/scripts/internetEnrollment.test.js	internet Enrollment
backend/scripts/internetEnrollmentFormTotals.test.js	internet Enrollment Form Totals
backend/scripts/internetRouterStock.test.js	internet Router Stock
backend/scripts/internetSubscriptionRenewal.test.js	internet Subscription Renewal
backend/scripts/materialIssueCorrection.test.js	material Issue Correction
backend/scripts/salaryReadAccess.test.js	salary Read Access
backend/scripts/salesBalance.test.js	sales Balance
backend/scripts/seniorManagerRole.test.js	senior Manager Role

Safe development workflow

Record the failing operation, role, endpoint and error. Inspect existing worktree changes. Reproduce with read-only requests and isolate UI, API, permission or data failure. Make the smallest change to the responsible controller, mapping or form. Verify the affected path and an adjacent working path. Use focused tests and the relevant build, then review the diff. Preserve admin access, approvals, payment allocations, stock ledger behavior and existing roles; do not broaden grants to work around a failure.

Deploying a module change

1. Compare intended database DDL to the source snapshot and required migrations. 2. Back up the intended database and inspect exact IDs for any approved data correction. 3. Apply reviewed schema changes in dependency order in a controlled environment. 4. Run focused module checks. 5. Build frontend and confirm runtime API/base-href configuration. 6. Deploy/restart backend and frontend through the existing hosting procedure. 7. Smoke-test login, permitted/denied actions, affected workflow and reports. Production deployment has not been performed as part of this documentation task.

Troubleshooting

Symptom	Inspect
Login/API unavailable	PORT, app-config.ts, backend listener, network request URL and DB pool error; avoid logging secrets.
401/403	Token presence/expiry, fresh login after role changes, permission_key and action columns, specific route middleware.
Unknown column or enum error	SHOW CREATE TABLE vs migration/runtime ensureSchema; senior manager and product/department enums.
Incorrect linked record	Customer family ID, header vs child PK, nested customer ownership and discriminator for workflow/audit.
Unexpected stock or totals	Approval-time stock effects, material/return ledger, account source, subscription period and payment allocations.
Email invoice failure	Registered recipient, billing period, one PDF attachment under 4 MB, supported SMTP variables; route and email tests.

Source coverage and limitations

This guide explains all mounted backend routers and all detected active lazy frontend routes. Endpoint middleware is copied from source declarations, but `global/router.use` middleware must also be consulted. Static SQL table and request-field inventories are navigation aids, not complete runtime execution traces or formal OpenAPI schemas. Dynamic SQL, destructured request payloads and conditional ALTER logic require reading the concrete handler. Unmounted `contactMgmtRouter.js` is not an active API merely because the file exists. Earlier specification/PPT files can describe an older design. Live data, hosting state and real user grants are outside this offline review.

Complete database table / primary-key catalog

Includes tables without any detected relationship; CREATE definitions are source snapshots, not verified live DDL.

audit_log	brands	cable_approval_groups
cable_areas	cable_connection_materials	cable_connection_sources
cable_connections	cable_customer_account_payments	cable_customer_accounts
cable_customer_packages	cable_customer_stb_accessories	cable_customer_stbs
cable_locations	cable_mso_master	cable_network_master
cable_package_master	cable_stb_issue_master	cable_stb_master
cable_streets	cable_subscription_payments	cable_subscriptions
cable_tv_complaint_attempts	cable_tv_complaints	cable_tv_customers
categories	customer_payments	customers
employee_attendance	employee_salary	employee_salary_items
employees	finance_transactions	internet_connection_materials
internet_connections	internet_customer_account_payments	internet_customer_accounts

Complete database table / primary-key catalog

Includes tables without any detected relationship; CREATE definitions are source snapshots, not verified live DDL.

internet_customer_complaints	internet_customer_packages	internet_customer_routers
internet_customers	internet_package_master	internet_router_material_issues
internet_subscriptions	material_master	notifications
products	purchase_items	purchase_master
quotation_items	quotation_master	quotation_templates
role_permissions	sales_items	sales_master
service_tickets	stock_ledger	stock_master
supplier_payments	suppliers	technician_material_movements
technician_material_sale_adjustments	technician_material_sale_payments	technician_material_stock
users	warranty_master	work_order_employees
work_order_items	work_order_material_issues	work_order_material_returns
work_orders	workflow_approvals	

Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 1-7 of 69

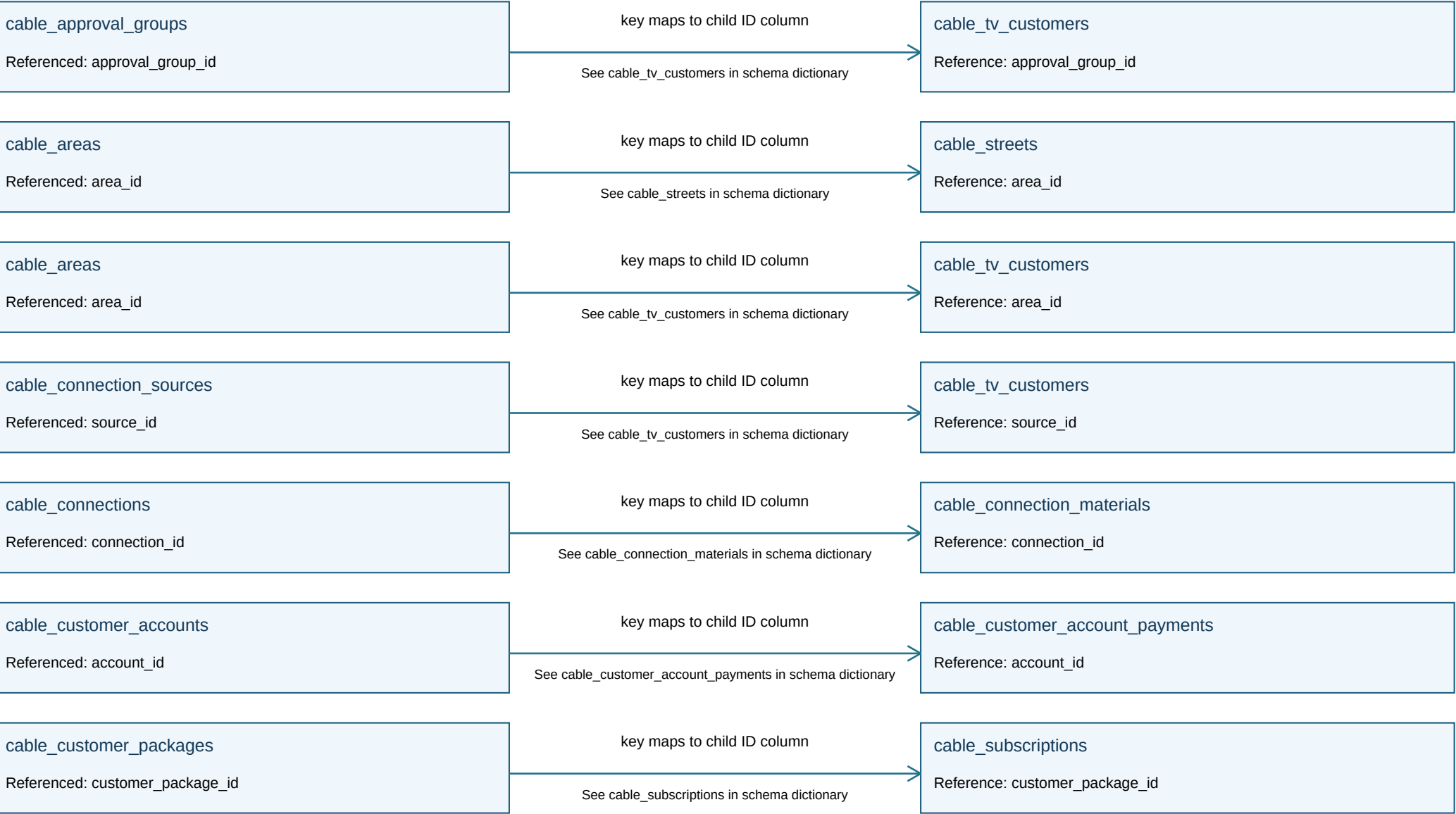
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 8-14 of 69

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 15-21 of 69

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 22-28 of 69

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 29-35 of 69

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 36-42 of 69

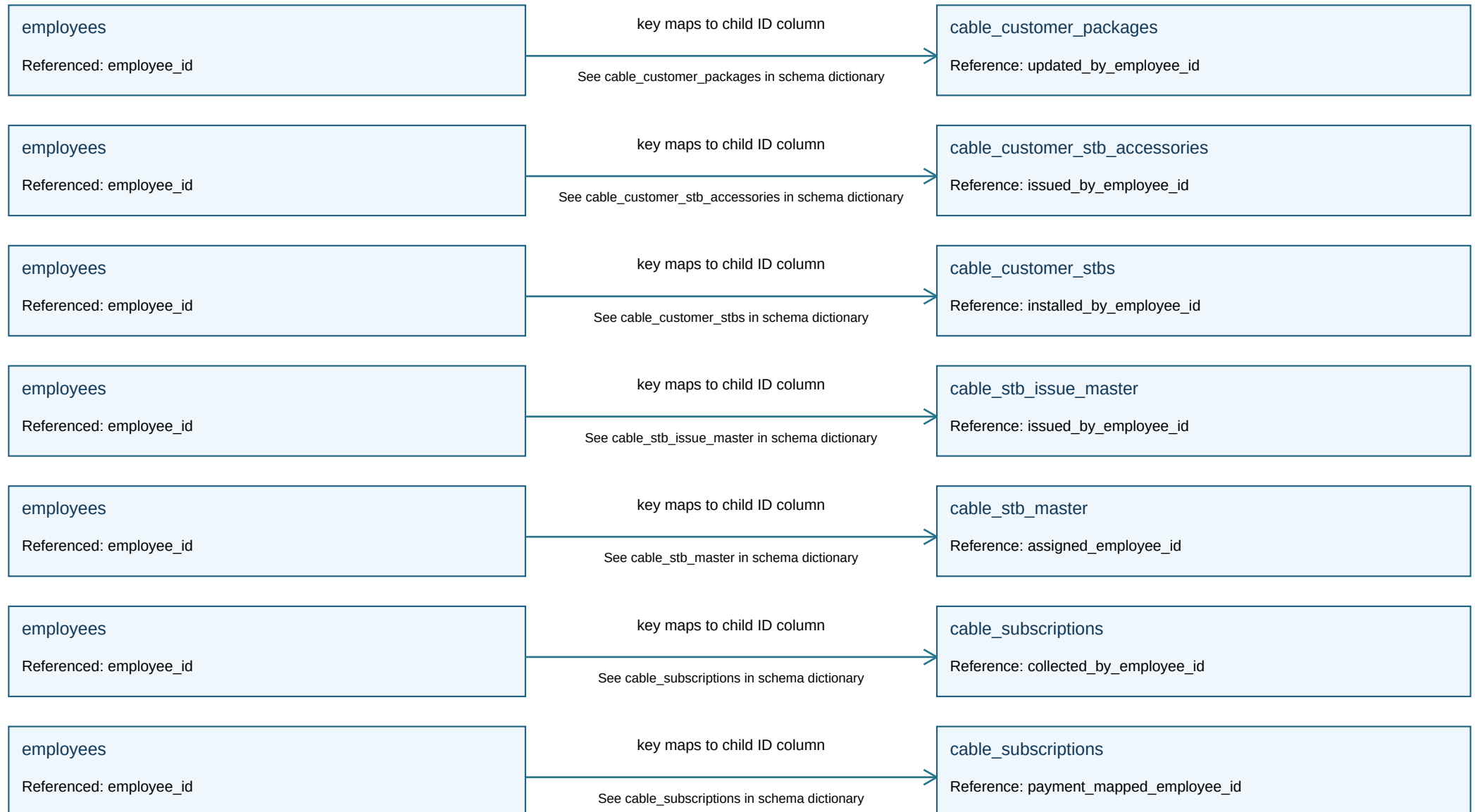
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 43-49 of 69

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 50-56 of 69

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 57-63 of 69

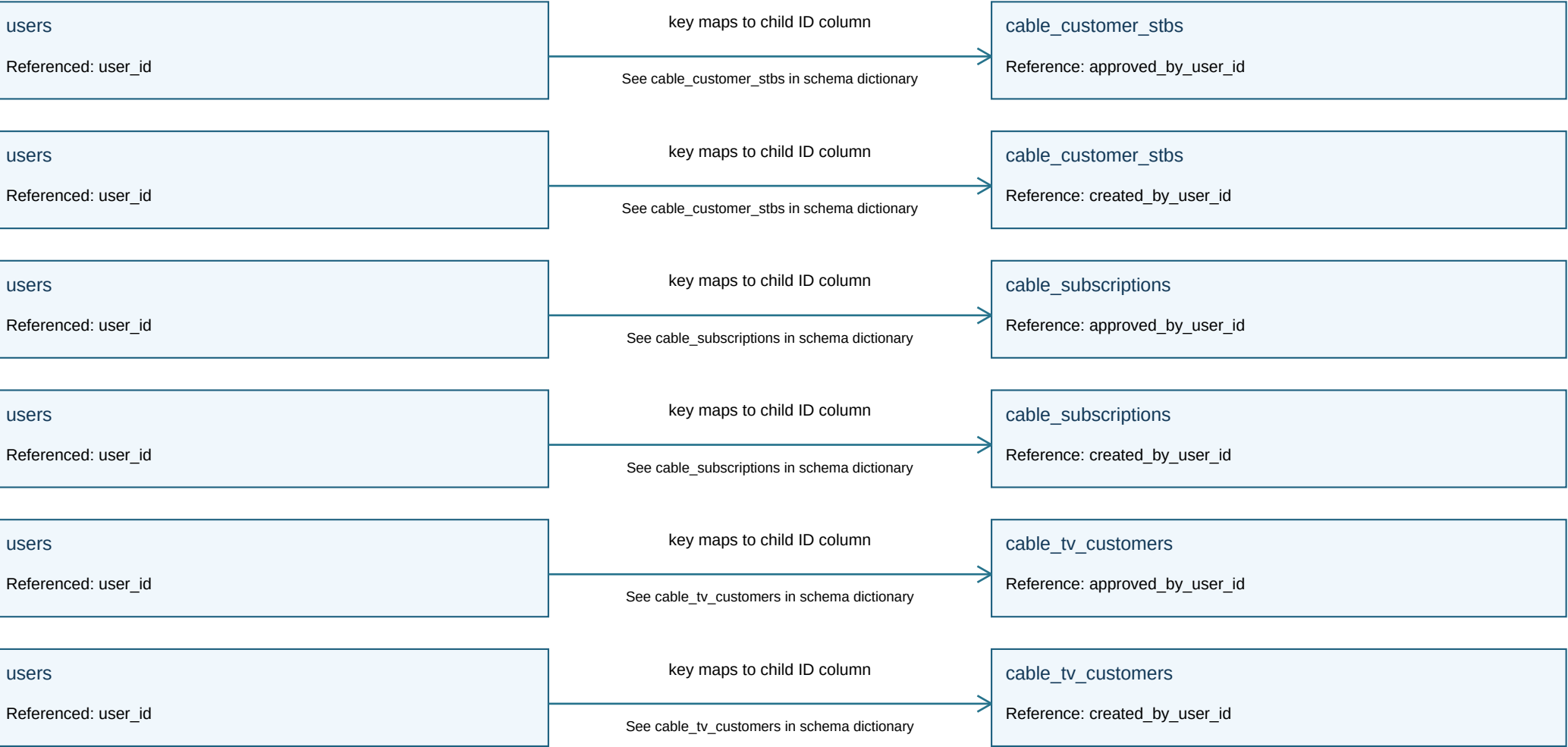
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 64-69 of 69

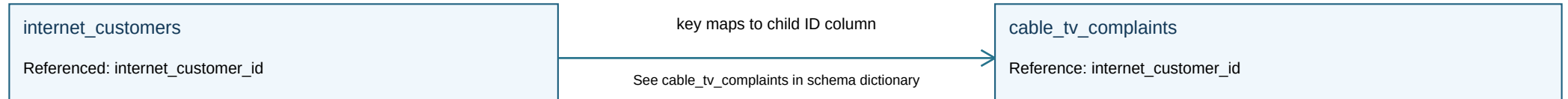
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Cable TV customer and billing relationships

Inferred ID link | parent referenced key -> child reference | rows 1-1 of 1

Dashed arrows are inferred ID-name candidates; validate joins and live DDL before relying on them.



Commercial, installation and service relationships

Declared FK | parent referenced key -> child reference | rows 1-7 of 45

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Commercial, installation and service relationships

Declared FK | parent referenced key -> child reference | rows 8-14 of 45

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Commercial, installation and service relationships

Declared FK | parent referenced key -> child reference | rows 15-21 of 45

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Commercial, installation and service relationships

Declared FK | parent referenced key -> child reference | rows 22-28 of 45

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Commercial, installation and service relationships

Declared FK | parent referenced key -> child reference | rows 29-35 of 45

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Commercial, installation and service relationships

Declared FK | parent referenced key -> child reference | rows 36-42 of 45

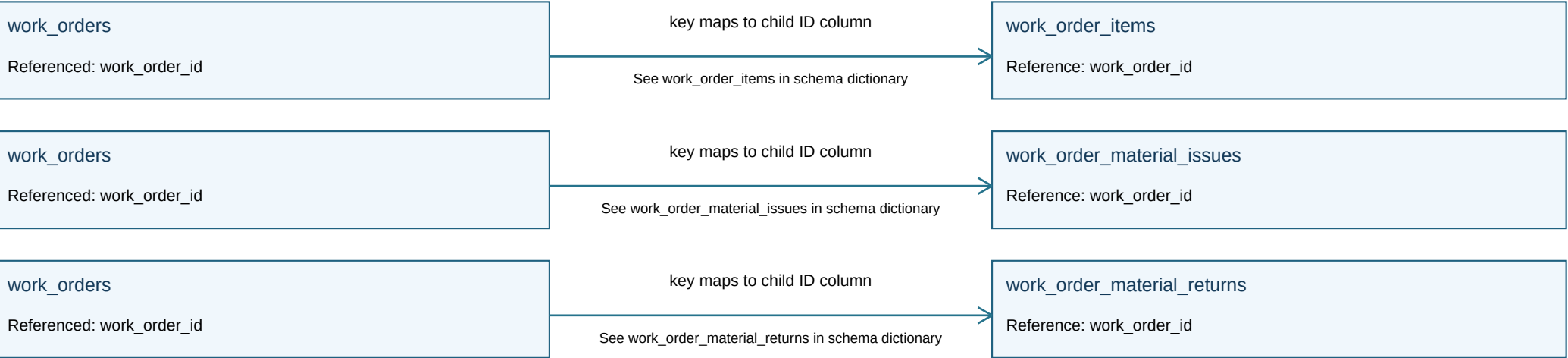
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Commercial, installation and service relationships

Declared FK | parent referenced key -> child reference | rows 43-45 of 45

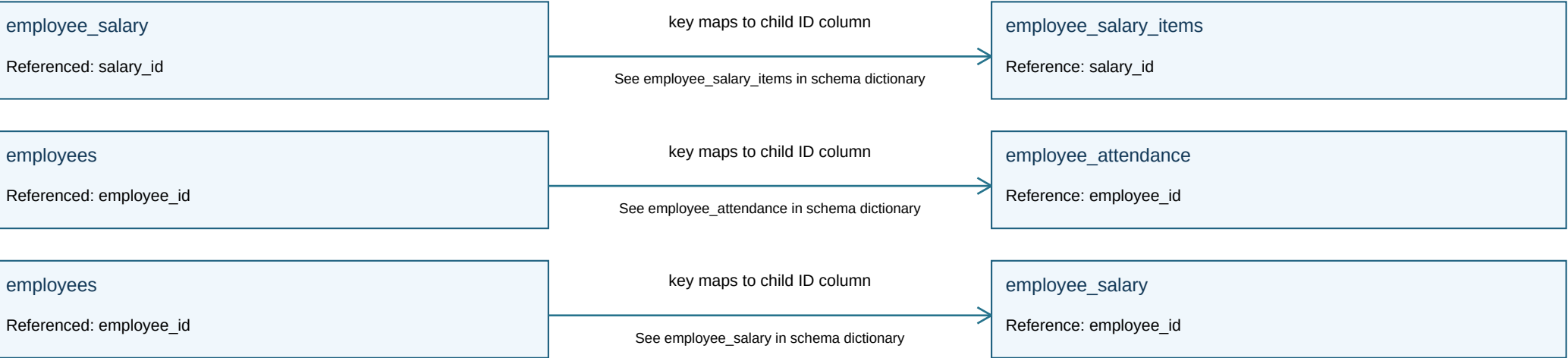
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Employees, attendance and salary relationships

Declared FK | parent referenced key -> child reference | rows 1-3 of 3

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Internet customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 1-7 of 13

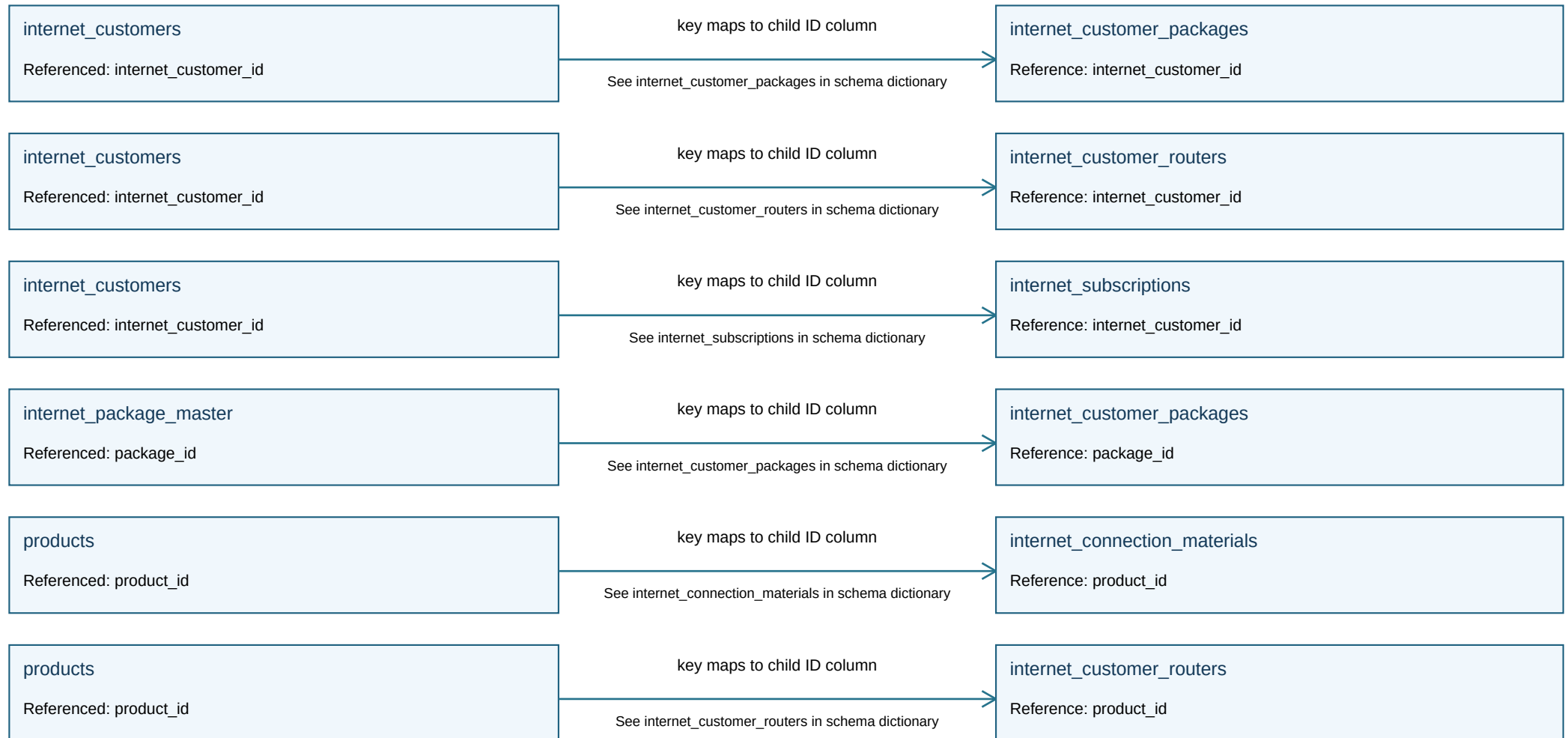
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Internet customer and billing relationships

Declared FK | parent referenced key -> child reference | rows 8-13 of 13

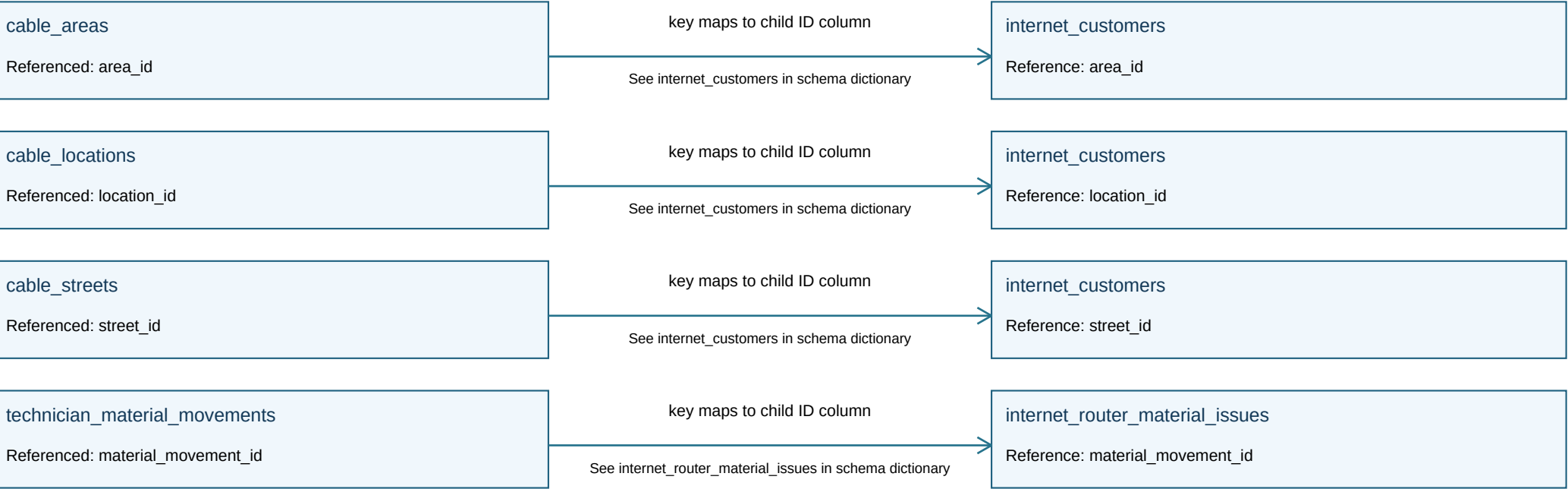
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Internet customer and billing relationships

Inferred ID link | parent referenced key -> child reference | rows 1-4 of 4

Dashed arrows are inferred ID-name candidates; validate joins and live DDL before relying on them.



Masters, purchasing, inventory and administration relationships

Declared FK | parent referenced key -> child reference | rows 1-7 of 20

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Masters, purchasing, inventory and administration relationships

Declared FK | parent referenced key -> child reference | rows 8-14 of 20

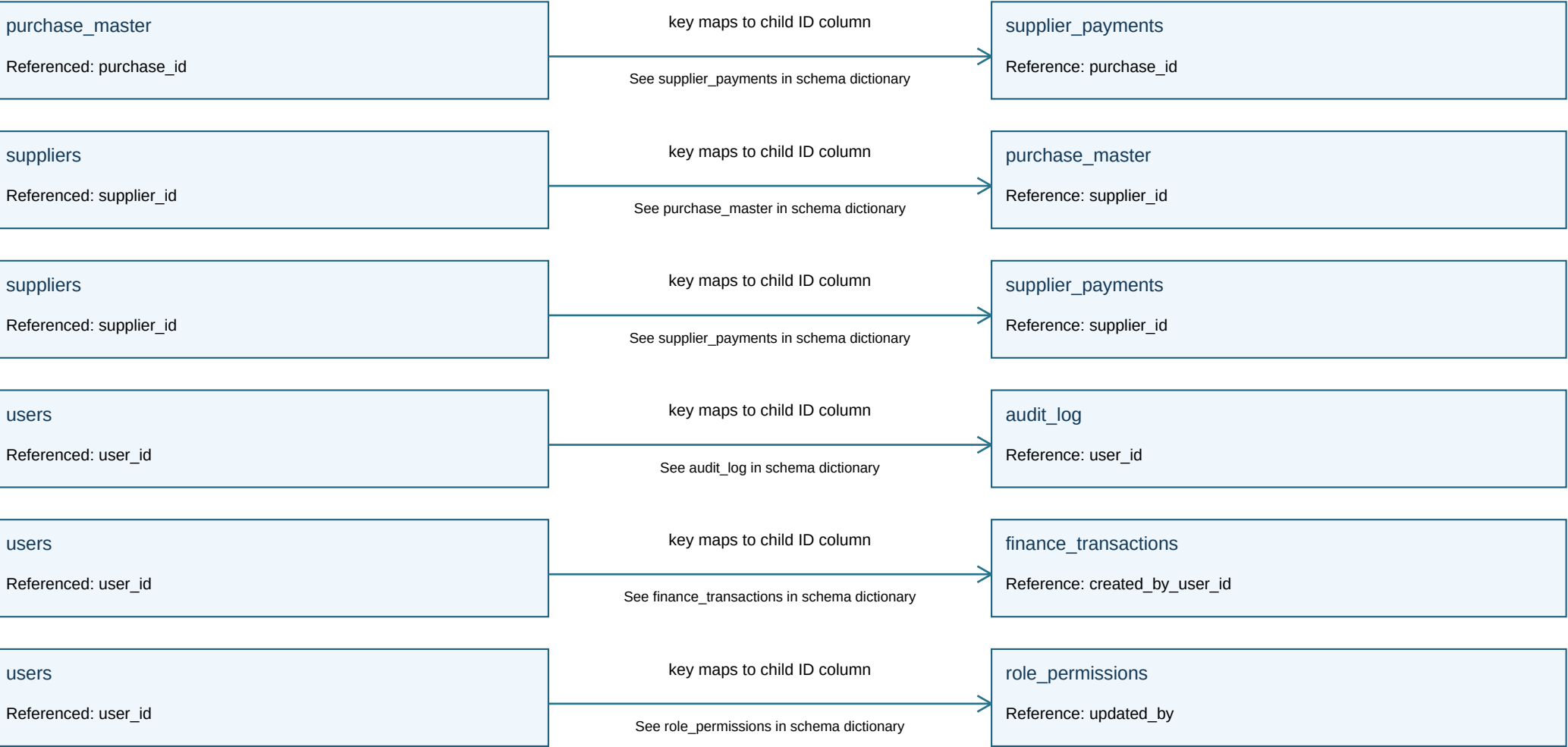
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Masters, purchasing, inventory and administration relationships

Declared FK | parent referenced key -> child reference | rows 15-20 of 20

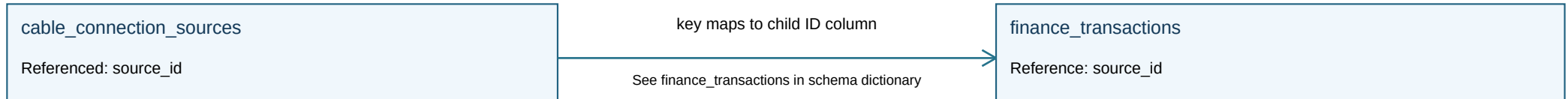
Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Masters, purchasing, inventory and administration relationships

Inferred ID link | parent referenced key -> child reference | rows 1-1 of 1

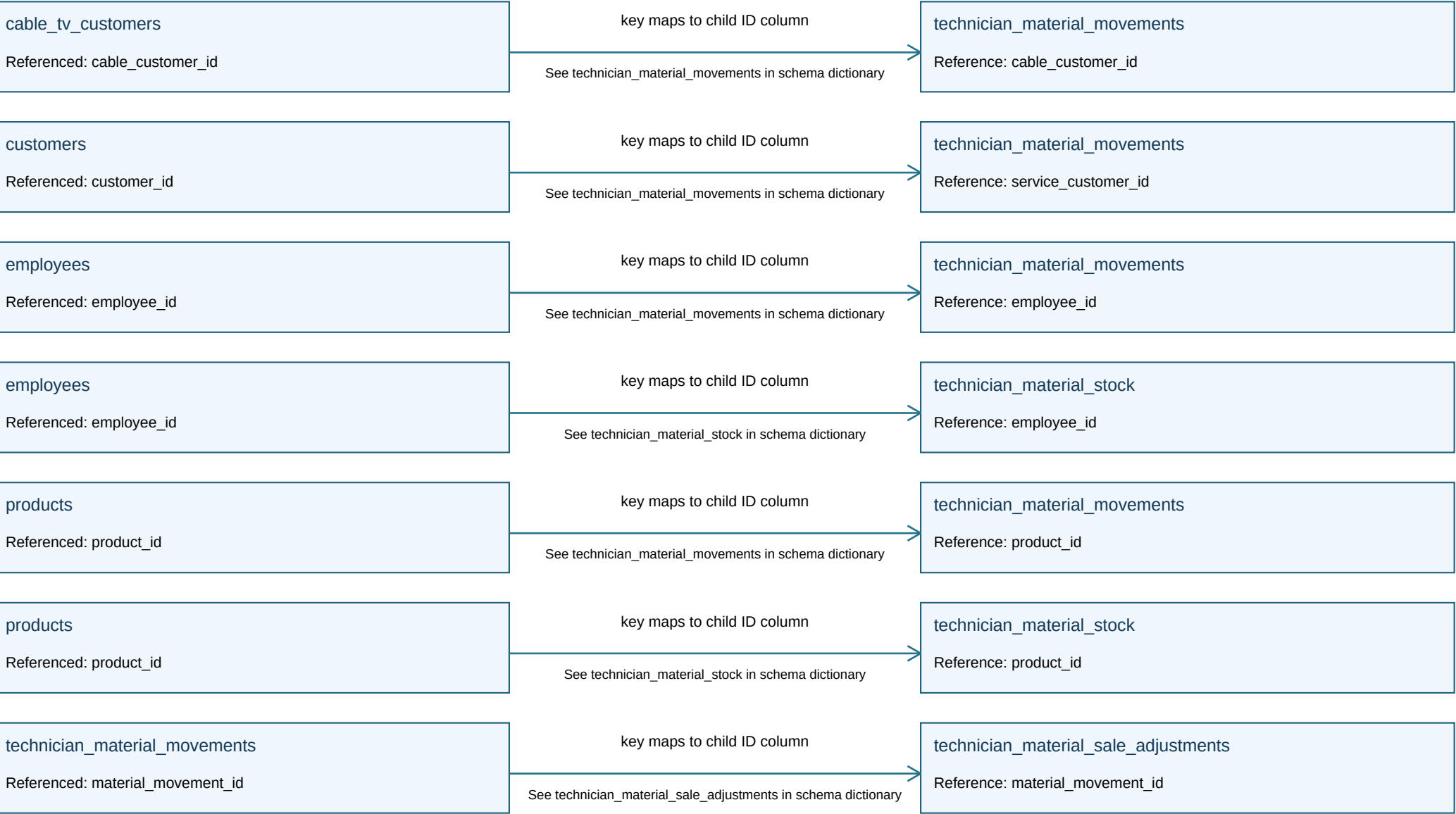
Dashed arrows are inferred ID-name candidates; validate joins and live DDL before relying on them.



Technician materials and sales relationships

Declared FK | parent referenced key -> child reference | rows 1-7 of 8

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.



Technician materials and sales relationships

Declared FK | parent referenced key -> child reference | rows 8-8 of 8

Declared in repository DDL; actual deployment and parent-key uniqueness require live verification.

